

TRACKING AND RECONSTRUCTION OF BI-MANUAL OBJECT HANDLING TASK USING RGB-D SENSING

by

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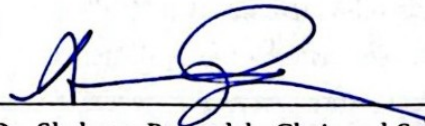
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Abstract

Bimanual object handling is a fundamental human skill, and replaying such motions supports imitation learning, avatar animation, and clinical assessment. Conventional motion-capture systems rely on multi-camera rigs that require careful calibration and expert operation, while wearable systems impose hardware on the subject. This thesis demonstrates that a single consumer-grade RGB-D camera, combined with three printed ArUco markers, can track a person's upper body and a hand-held object during a desk-based task and reconstruct both within a metric scene suitable for digital-twin playback.

The core challenge lies in sensing reliability. The landmark detector may output points even when the corresponding body part is occluded or absent. The object and the opposite hand frequently hide the wrist; the hands themselves obscure the hips, causing the depth map to default to the desk surface. The landmark set does not uniquely determine wrist orientation, and each frame must be solved independently as it arrives.

The proposed pipeline contains two branches. The "landmark branch" converts RGB-D body landmarks into a torso pose and four joint angles per arm. The "marker branch" establishes a world coordinate frame on the desk and tracks the object within it. The branches share only the captured frames and the world frame. Object pose information provides an additional geometric constraint when a wrist is lost, and the elbow pose follows from two-link inverse kinematics. Outlier landmarks are rejected, remembered depth substitutes for occluded hips, and all reconstructed quantities are labelled as measured, held, or constrained before entering the rendered scene.

The evaluation tracks a cube moving across a desk and along a rail, both single-handed and during a handover. Tape-measured distances between marker-defined waypoints range from 3.5 to 37.5 cm, while reconstructed lengths differ by 0.53 to 2.83 cm depending on detection settings. Desk-level segments reconstruct slightly short, whereas later segments reconstruct long. Because the route is not registered into the world frame, point-to-path distances are unavailable. Under single-hand grip, the reconstructed wrist-to-marker separation remains within roughly 2 cm of its tape measurement. Object

constraints assist recovery during one natural occlusion interval but not another. The causal pipeline replays the motion at the recorded rate of approximately 29 fps, producing smooth output playback; however, a full live-camera session was not validated.

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Writing the acknowledgements has, unexpectedly, been one of the hardest parts of finishing this thesis.

The results of an experiment may be reproduced, but the uncertainty and frustration behind them cannot. Looking back, I still remember the nights I spent alone working through derivations by hand, again and again. Those difficult moments eventually became part of the work presented here.

I used to think that growing meant becoming better at solving problems on my own. Only now do I realize how many people supported me along the way.

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"Sometimes life knows what is best for us better than we ourselves know or think."

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Glossary

Selected terms and abbreviations used in this thesis.

Term	Meaning
camera frame {Camera}, C	the aligned colour optical frame: x right, y down, z forward
display frame {Unity}, U	left-handed axes formed by swapping the second and third world axes; the desk-marker origin is retained
marker, fiducial marker	a printed square pattern for measuring pose; ArUco markers are fixed to the wall, desk and object
object frame {Object}, O	centred on the object marker; its third axis is normal to the marker plane
p95	95th percentile: the value at or below which 95 percent of samples fall
PSF1, PSR2, PSB2, PSB3, PSI2	pose-stream record identifiers: colour/depth frames (PSF1), person (PSR2), object (PSB2), calibration (PSB3), combined pose (PSI2); digits identify layout versions
PUMA convention	Programmable Universal Machine for Assembly: the robot drawing convention with revolute joints shown as cylinders
RGB-D camera, sensor	captures red, green and blue colour (RGB) plus depth (D); the sensor here is the Intel RealSense D435
scene frame {Scene}	the final Unity frame after gravity alignment and floor placement; y is vertical, with the origin on the drawn floor
T-pose	arms extended horizontally to the sides; all arm angles are zero
world frame {World}, W	fixed to the desk marker during calibration; the reference for object pose
y-up camera frame {Camera'}, C'	{Camera} with y flipped upward; the optical origin is unchanged

Chapter 1: Introduction

This thesis examines the extent to which a person and a manipulated object can be reconstructed from one consumer-grade colour and depth (RGB-D) camera. It also examines the contribution of object tracking when the body landmarks fail, and the limits that remain from the sensing, the carried state and the single viewpoint. A fixed sensor observes the torso, both arms and a marked cube while a single subject performs a desk task in three recordings. None of the recordings establishes accuracy for general bimanual manipulation.

1.1 Motivation

Imitation learning in robotics uses recorded human demonstrations of coordinated two-handed tasks [1], [2]. Virtual reality (VR) and augmented reality animate avatars from skeletal pose data together with object poses [3], [4].

An avatar replay of a patient's upper-body motion could help a physician assess reaching and hand transfers in Parkinson's disease, stroke, or age-related loss of coordination [5], [6]. A deployment can keep colour and depth frames on the capture device and share only the reconstructed motion.

Metric three-dimensional (3D) measurement requires depth sensing or synchronized camera viewpoints [7]. Multi-camera motion capture needs calibrated arrays with expert operation [8]. Wearables need per-user attachment and calibration, and they raise movement and hygiene concerns [9]. A consumer-grade RGB-D camera captures registered colour and metric depth without hardware on the subject [10].

Hands can occlude each other or the object, disrupting landmark detection [11], [12] and depth readings [13]. Calibration must place every segment and object in one frame [14], and torso and hand motion call for different filtering [15]. With limits on transistor scaling [16], this systems engineering thesis examines the capabilities that algorithm design and system organization can achieve on existing hardware.

Earlier work in the Networked Robotics and Sensing Laboratory at Simon Fraser University developed a hierarchical hand model from RGB-D landmarks [17] and Gaussian Bayesian Network joint estimation for noisy or occluded measurements [18].

This thesis extends the tracked anatomy to the torso and both arms, with concurrent object tracking and Unity visualization [19].

1.2 Problem Statement and Literature Review

That laboratory work covers the hand, but not the torso, the arms or a held object, so the review below follows the three problems that remain. Section 1.2.1 covers tracking the hands and the object, Section 1.2.2 the coupling of two hands on one object, and Section 1.2.3 the replay of the motion graphically. Table 1.1 summarizes the limitation each reviewed approach family carries in this setting.

1.2.1 Hand and Object Tracking Approaches

Early hand tracking systems encoded finger configuration in the sensor signal, through instrumented colour gloves [20] and, in earlier laboratory work, marker-based visual tracking with a vibrotactile glove [21]. Consumer depth cameras then shifted the field to markerless tracking: Particle Swarm Optimization over a geometric hand model [22], an articulated Iterative Closest Point solver [23], and a sphere-based model fitted to depth point clouds [24].

Deep learning approaches followed the model-fitting trackers and include multi-view convolutional networks over depth images [25], transformer-based mesh recovery from monocular RGB [26], and graph-guided state space models [27]. Google's MediaPipe framework [28], [29] provides lightweight landmark detection [30], but its appearance-derived depth is uncertain during motion, contact, and occlusion [3]. This thesis instead back-projects the landmarks using calibrated sensor depth [17], [13].

A held object carries no landmarks, so its pose needs a different signal. For object tracking, fiducial markers give reliable pose estimation without appearance models or training data. ArUco markers balance robustness and efficiency for single-camera setups [31], and marker size and viewing distance set the pose error [32]. Each marker encodes an integer identity, so one detection gives both the object identity and its pose. Occlusion between the hand and the object remains the most persistent failure mode, met either by inpainting the occluded hand before pose estimation [11] or by modelling hand and object jointly through mutual attention [12].

1.2.2 Bimanual Manipulation in Motion Analysis

Occlusion between one hand and the object carries over to the two-handed case, where each hand can also hide the other. Bimanual manipulation goes beyond two hand trackers in parallel. The hands overlap in the image and are kinematically coupled through the object they hold, so an error in one hand spreads into the reconstructed scene [33].

Symmetric and asymmetric coordination give different patterns of landmark dropout [34]. Bimanual datasets show the difficulty of the capture problem: InterHand2.6M used between 80 and 140 synchronized cameras [35]; OakInk2 used multi-camera rigs with wearable sensors [8]. RGB2Hands recovers two interacting hands from a single RGB video [36] with a learned model that is hard to interpret physically.

1.2.3 Graphical Reconstruction in Robotics and VR

Unity [19] has been used for monocular motion replay through kinematic retargeting [37] and Kinect-based rehabilitation feedback [38]. RePose streams MediaPipe-derived joint angles to a Unity avatar [39], the architecture adopted here. RealSense and Unity use different handedness and vertical axes, so the streamed poses need an explicit coordinate conversion [14].

A single fixed change of coordinates moves landmark positions between the two environments. The avatar, however, is driven joint by joint, so the segment rotation angles must be resolved from the measured landmark positions. The depth noise underlying those positions is characterized as Gaussian, consistent with [40], [41].

Approach family	Representative work	Sensing	Limitation for this setting
Instrumented gloves and wearables	[20], [21]	worn sensors	hardware on the user; per-user calibration; hygiene
Model-based depth tracking	[22], [23], [24]	one depth camera	single hand; explicit geometric models; per-user initialization
Learning-based pose estimation	[25], [26], [27]	depth or RGB	large training sets; no biomechanical constraints
Landmark detection with depth fusion	[30], [3], [13], [17]	RGB-D	single hand, or body landmarks without a manipulated object in the same frame
Fiducial object tracking	[31], [32]	RGB	tracks marked objects only; no body tracking
Hand-object occlusion handling	[11], [12]	RGB, depth	single hand and object; learned models
Bimanual capture datasets	[35], [8]	80 to 140 synchronized cameras [35]; multi-camera rig with wearables [8]	laboratory infrastructure far beyond clinical settings
Two-hand reconstruction	[36]	monocular RGB	no object tracking; no metric depth
Avatar replay in Unity	[37], [38], [39]	monocular or Kinect	single tracked stream; no concurrent object in a shared frame

Table 1.1. Summary of the reviewed literature: each approach family, its representative work, its sensing arrangement, and its limitation for single-sensor bimanual capture.

1.2.4 Gap Analysis and Motivation for This Work

Recovery must account for the different kinematics of the torso and arm segments [33]. The reviewed bimanual method omits object tracking [36], the hand-object annotation method of [42] covers one hand, and the sensor-to-Unity conversion is documented for a single hand [14]. No reviewed system reconstructs the upper body together with a tracked object from a single RGB-D viewpoint, or reports the recovery from that viewpoint when

the landmarks fail. This thesis investigates that question in the stages of Section 1.3, building on the laboratory work of [17] and [18].

1.3 Research Objectives

The central question divides into five stages, and the objectives below are those stages in the order the thesis builds them.

Objective 1 is the measurement layer: a calibrated RGB-D pipeline on the Intel RealSense D435 that aligns the depth and colour streams, establishes the camera intrinsics, and places the body landmarks and the marked-object poses in one metric world frame.

Objective 2 reconstructs the person from that layer. Metric 3D positions for the torso, shoulders, elbows and wrists come from MediaPipe Pose detections fused with registered depth, and give the selected degrees of freedom of the upper body. The wrist stays a tracked point.

Objective 3 observes the object independently in the same scene. The scene calibration of Chapter 2 fixes the world frame on the desk marker for both branches. The ArUco-marked object is tracked there, and the two records are synchronized by frame index. The branch filters and streams the object orientation, and no second measurement chain checks it.

Objective 4 uses that observation as an additional geometric constraint where the landmarks fail. The holding wrist comes from the tracked object pose and the hand-object offset, and its elbow from two-link inverse kinematics. The recovered positions are compared with the unmasked reconstruction under synthetic masking and with manual wrist labels on the natural failures. Chapter 10 discusses lightweight learned estimation in the recovery layer as future work, a compact temporal model that operates alongside the explicit kinematic chain rather than replacing it.

Objective 5 integrates the stages and evaluates the result. The person and object records drive one Unity reconstruction. Chapter 7 evaluates it offline on the rail task one-handed and with a handover, and on natural failure windows from the two-hand loop recording. Chapter 8 tests the same architecture for causal operation, and

Chapter 9 discusses the limitations shown by the results. No recording establishes accuracy for general bimanual manipulation.

1.4 Organization of the Thesis

The remaining chapters build the system, evaluate it, and discuss the results; the appendices support the chapters with detail.

- Chapter 2 builds the measurement layer, Chapter 3 the kinematic model, Chapter 4 the object branch, Chapter 5 the recovery layer and Chapter 6 the Unity integration.
- Chapter 7 evaluates the system offline against a physical route not yet registered into the world frame. Chapter 8 tests real-time feasibility as a causal replay, with no live-camera session validated.
- Chapter 9 discusses the results and their limitations. Chapter 10 concludes and outlines future work.
- Appendix A covers the recording procedure, B the hardware and software platform, C pose landmark detection, D depth sensing, E fiducial marker detection, F candidate filters, G hip depth preparation, and H solver verification.

Chapter 2: Experimental Setup and Measurements

Chapter 1 set out to establish the extent to which a person and a manipulated object can be reconstructed from one consumer-grade RGB-D viewpoint. This chapter builds the measurement layer for that task, from the calibrated scene to the filtered landmark and marker observations.

2.1 The Recording Setup

The recording took place in a controlled lab to limit lighting and background interference. The subject moves a cube along the desk and a wooden rail, with one hand in the rail recording of this chapter and with a handover between the hands in the handover recording of Chapter 7. Chapter 7 compares the reconstructed separations between the endpoints of that route with physical references measured on the setup with a tape, and it draws the reconstructed trajectories. The physical route is not registered into the calibrated world frame, so no physical path is drawn. Figure 2.1 shows the setup, and Table 2.1 lists its objects.

The task runs in five phases, and the later chapters refer to them by these names. The hand acquires the cube at its initial position in the object acquisition, and the carry translates the cube from that position toward the rail. The lift raises the cube from the desk level to the rail level, the slide translates it along the rail, and in the release the hand leaves the cube at the far end. The handover recording has the same physical phases, and its slide also contains a right-hand segment, a handover interval and a left-hand segment.

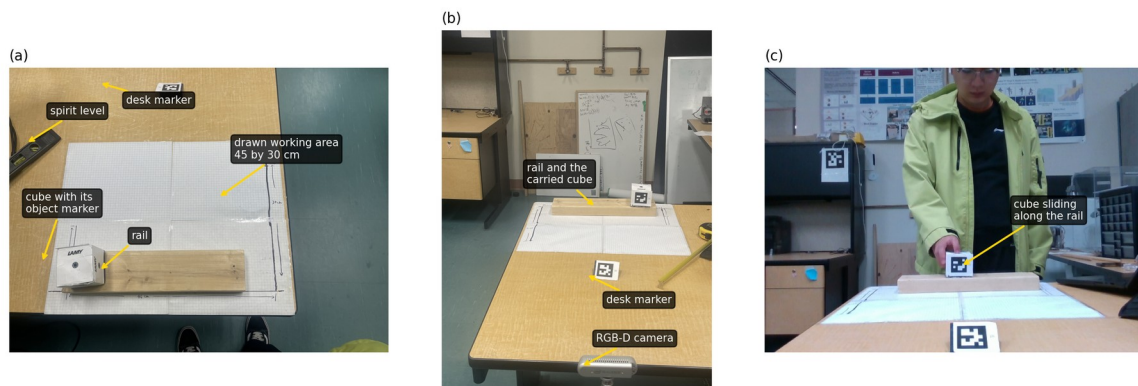


Figure 2.1. The recording setup. (a) The working area on grid paper, with the rail, cube and desk marker. (b) The view from behind the camera. (c) A colour frame.

Object	Role	ArUco identifier	Dictionary	Printed size (mm)	Physical dimensions
Wall marker	Static; gravity reference	0	5 by 5	150	-
Desk marker	Static; world origin	2	5 by 5	45	-
Object marker	Dynamic; tracked cube	1	5 by 5	45	Cube: 70 mm per side
Wooden rail	Reference object	-	-	-	387 by 88 by 38 mm, lying flat

Table 2.1. The objects in the scene. All markers are from the OpenCV 5 by 5 dictionary, and the size is the side of the printed black square. The translation a planar marker solver returns scales with the size it is given, so every object-derived distance in this thesis rests on it.

Figure 2.2 labels the parts of the scene. The camera is an Intel RealSense D435 on a fixed support at the edge of the desk, about 1.3 metres from the subject. The subject stays within 2 metres of it for the whole experiment, because the depth error of the sensor grows with the square of the distance [40], [41], [43].



Figure 2.2. The experimental setup seen from the RGB-D camera, with the subject, the rail, the carried cube, and the three ArUco markers labelled.

The recording is saved as a bag file, the format of the Intel RealSense software. Each frame holds a colour image, a depth image aligned to it, and a shared frame index that pairs the person and object measurements of one moment. Appendix A describes the recording and the file layout.

Figure 2.3 follows the recording through two parallel pipelines. Pipeline A combines landmark detection, depth, filtering and the body solve of Chapters 3 and 5. Pipeline B measures the scene and object poses of Sections 2.2 and 2.4 and Chapter 4, which also support the recovery of Chapter 5. Chapter 6 pairs the two outputs at a common render time for Unity. Appendix B lists the workstation.

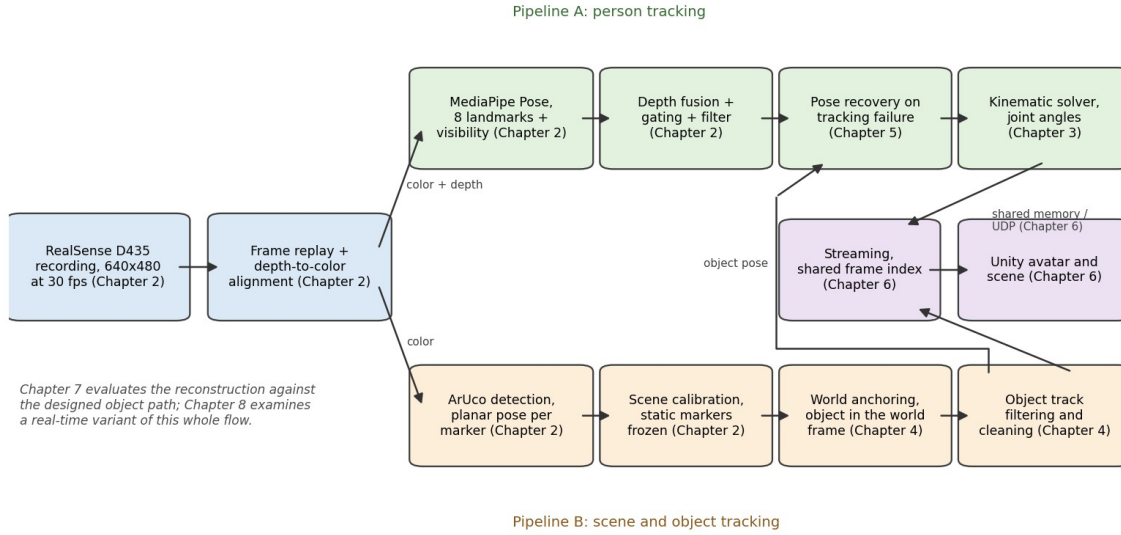


Figure 2.3. Overall data flow. Pipeline A (top) tracks the person, Pipeline B (bottom) the scene and object, and each block names the chapter that develops it.

2.2 Coordinate Frames and Scene Calibration

Each position and orientation is expressed in a coordinate frame, an origin with three perpendicular axes. The transformations between them follow standard robotics notation [44]. Every frame of this thesis carries a name. A name in braces is that frame's label in the superscript and subscript slots of a transformation.

2.2.1 The Frames

The raw camera frame {Camera} is the frame of the colour camera to which each depth image is aligned. Its origin is at the colour camera lens, the x axis points to the right of the image, the y axis points down, and the z axis points forward into the scene [10], which makes it right-handed. The depth of a point is its z coordinate in that frame. Every raw measurement of both pipelines starts there, and Figure 2.4(a) draws it on the sensor as the letter C.

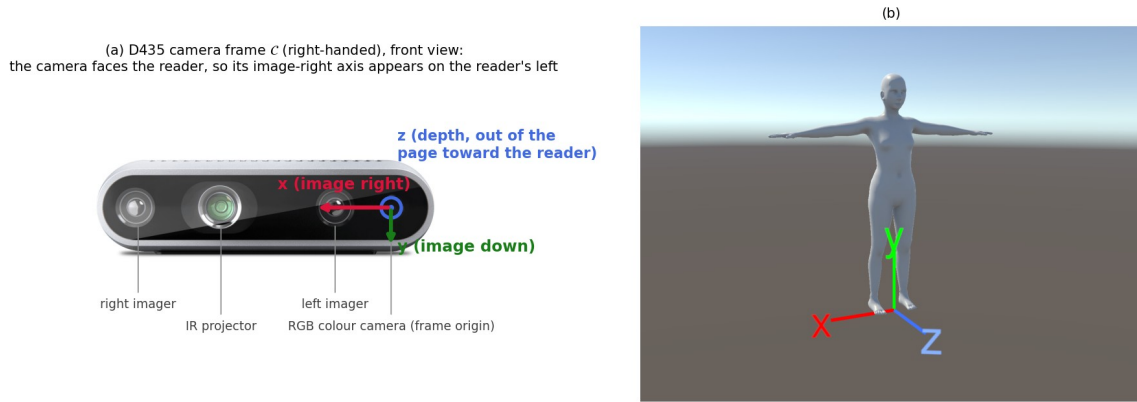


Figure 2.4. (a) The sensor frame C on a D435 photograph (Intel [45]; axes added). (b) The coordinate system of the Unity scene, left-handed, y up, origin on the floor.

Every printed marker of Table 2.1 has a right-handed frame of its own. Its origin is at the centre of the printed square, its x and y axes lie in the printed plane and its z axis points out of the printed face. The world frame is the desk marker's, fixed to the desk. Every object pose ends up in it, and Chapter 6 brings the reconstructed body into it as well. The wall marker's frame sits behind the working area and is never used as an origin. Section 2.2.2 states the assumptions under which its up axis gives the direction of gravity, and the calibration uses that marker for this purpose only. The object frame rides the marker on the cube.

The y -up camera frame $\{\text{Camera}'\}$ shares the optical centre of $\{\text{Camera}\}$ and differs only by flipping the y axis upward, which makes it left-handed. It is the coordinate convention used for body kinematics. Chapter 3 builds the body model in it as a chain of frames, each named after the landmark at its origin, which also gives the side. The root frame sits at the right hip landmark L24 and carries the orientation of the torso. A shoulder frame sits at each shoulder landmark, L12 on the right and L11 on the left, and inherits the orientation of the root. An elbow frame sits at each elbow landmark, L14 on the right and L13 on the left, and carries the basis of the root turned by the solved shoulder rotation.

Chapter 6 first converts the calibrated world coordinates to the left-handed display convention $\{\text{Unity}\}$. This convention retains the desk-marker origin; its second axis follows the desk-marker normal and is not generally vertical. Gravity alignment then rotates these coordinates, and a recording-specific translation places the drawn floor at zero height. The result is the final $\{\text{Scene}\}$ frame of Figure 2.4(b), and Section 6.4

combines those two operations into one transformation without defining an intermediate frame.

The pose of a frame B seen in a frame A is written A_BT , a 4 by 4 matrix that holds a rotation and a translation, and Chapter 3 gives its form in equation (3.2). The same matrix carries a point of B into A, so that $\begin{bmatrix} {}^AP \\ 1 \end{bmatrix} = {}^A_BT \begin{bmatrix} {}^BP \\ 1 \end{bmatrix}$. A pair of them composes when the subscript of the first matches the superscript of the second, and the shared name drops out of the product. Figure 2.5 draws the frames of the scene, with an arrow for each transformation. Arrows point from the reference frame to the described frame, and the labelled transformation maps coordinates in the opposite direction.

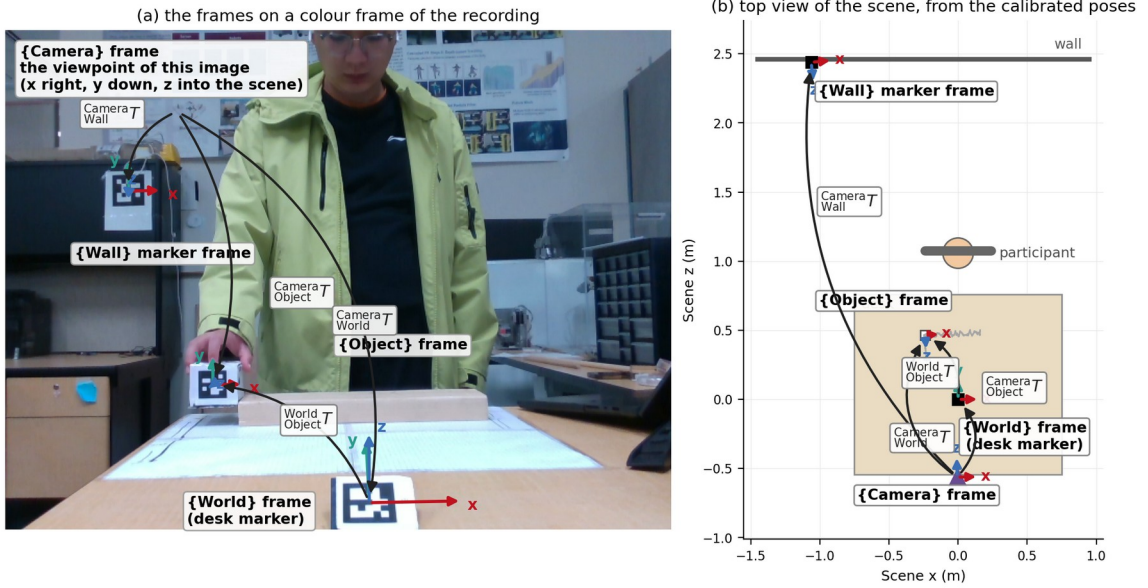


Figure 2.5. The scene frames. (a) A colour frame carrying {Wall}, {World} and {Object}, with an arrow for each transformation. (b) A top view of {Scene} and the object track.

The flip and the swap both reverse handedness. The flip F takes the right-handed sensor frame to the left-handed y-up camera frame, and as a matrix it is $\text{diag}(1, -1, 1)$. The swap S takes the right-handed world frame to the left-handed Unity frame by exchanging

the second and third coordinates, so it is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Each has determinant -1 , so neither

is a rotation, and each is its own inverse. Where an orientation crosses one of them,

Chapter 6 prints the change of basis as a conjugation rather than as a multiplication on one side.

2.2.2 Scene Calibration

The desk and wall markers are assumed fixed. Their poses are calibrated at the start and held constant throughout the recording.

Calibration averages the first ten detections of each static marker. Translation uses the arithmetic mean. The element-wise rotation mean generally loses orthonormality, so it is projected onto the nearest proper rotation [46], giving the chordal mean. Let $\bar{M}$ be the element-wise mean of the rotation blocks of ${}^A_B T$ across the window. Its singular value decomposition is $\bar{M} = U \Sigma V^T$. The projection is

$${}^A_B R = U \operatorname{diag}(1, 1, \det(U V^T)) V^T, \quad (2.1)$$

where the determinant factor excludes reflections and the result is orthonormal. The calibration runs it on ${}^{Camera}_{World} \bar{M}$ for the desk marker and on ${}^{Camera}_{Wall} \bar{M}$ for the wall marker, which gives ${}^{Camera}_{World} R$ and ${}^{Camera}_{Wall} R$. Chapter 4 uses the same projection to smooth the object rotations ${}^{World}_{Object} R$.

Freezing the anchor prevents later desk-marker noise from rotating the reconstructed scene. An anchor orientation error displaces each point in proportion to its distance from the anchor. The frozen calibration can still be biased, but it adds no per-frame anchor wobble.

Calibration also measures the geometry of the scene around the markers, on two physical assumptions. The wall carrying the wall marker is plumb, and the marker is mounted with its printed up axis vertical. Under these assumptions, the up axis in the plane of the calibrated wall marker gives the direction of gravity. A plumb wall alone fixes only the normal of the marker plane, and the marker could still be turned within that plane. The upright mounting was not measured, so an error in the angle at which the marker hangs enters every gravity-referenced height and tilt of this thesis.

The axes of the world frame are not lined up with gravity because the desk marker card sits on a small stand that tilts it toward the camera. Gravity is therefore taken from the wall marker.

A coarser estimate of the same direction comes from a plane fitted to the depth pixels in a ring around that card, once the deprojection of Section 2.3 has turned those pixels into metric points. Its normal fixes the sign of the gravity direction and settles the two-solution ambiguity of the planar pose estimation of Section 2.4. The fitted plane also supplies a point on the tabletop, which fixes the horizontal tabletop model whose normal is the calibrated gravity direction.

The pipeline hands this scene calibration to the reconstruction once, before any per-frame streaming begins, and Chapter 6 builds the Unity scene from it.

2.3 Pose Landmarks From MediaPipe

Pipeline A starts with MediaPipe Pose. It is an open-source detector [30], and it runs in real time on ordinary hardware. For each colour frame, it first locates the person and then regresses 33 body landmarks. Landmark indices are fixed. The pipeline can therefore select the eight required upper-body points directly: shoulders L11 and L12, elbows L13 and L14, wrists L15 and L16, and hips L23 and L24. Every result includes image coordinates, an appearance-derived depth estimate, and a visibility score between 0 and 1. Appendix C gives the detector architecture, the steps that extract the landmarks, and a numerical example.

Because MediaPipe estimates depth from two-dimensional (2D) appearance rather than physical sensor measurements, the pipeline discards those values and uses the depth readings from the aligned RealSense stream instead. It then uses the camera intrinsics to back-project each landmark pixel into metric 3D coordinates (Appendix D). The pipeline treats landmarks with visibility scores below 0.50 as missing data. Figure 2.6 illustrates the detector output.

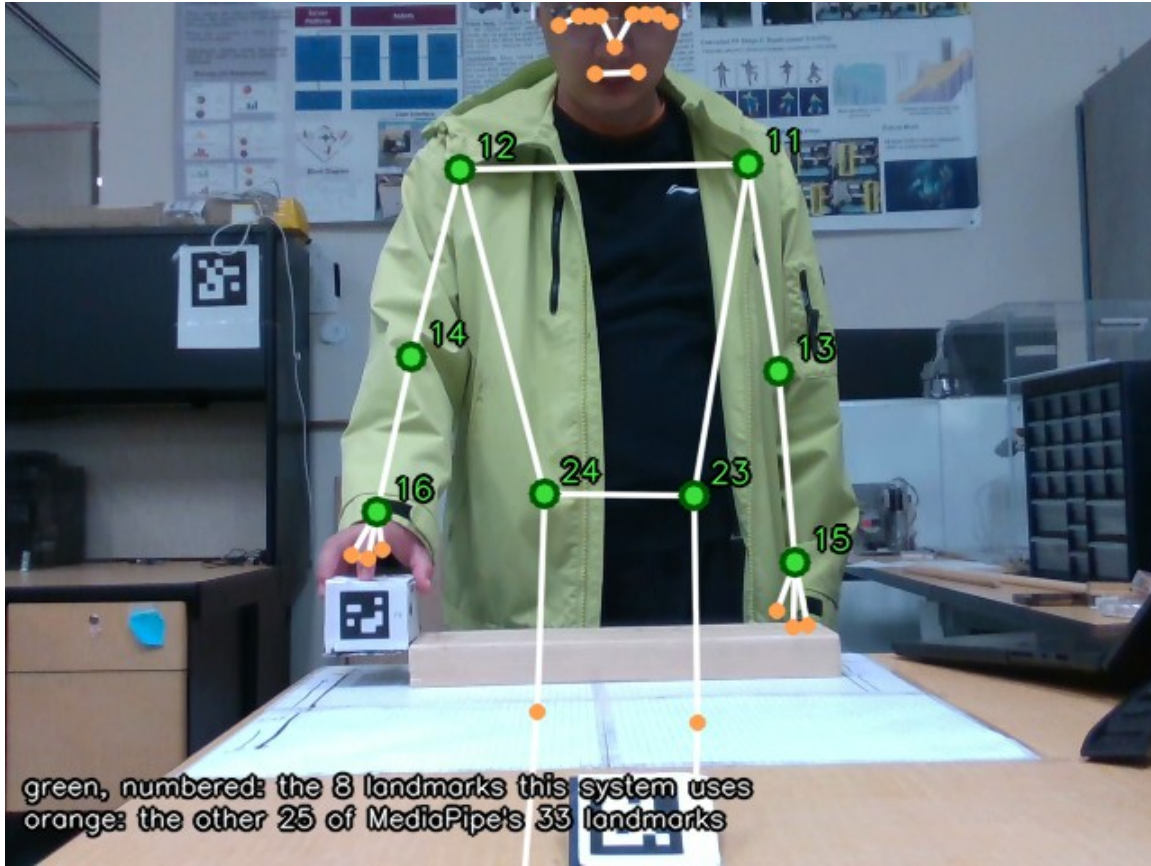


Figure 2.6. MediaPipe Pose output on a mid-task frame. The eight numbered upper-body landmarks are the ones this system uses. The visibility gate removes the leg landmarks the desk hides.

2.4 Object Pose From ArUco Markers

Pipeline B gets its measurements from ArUco fiducial markers [31], which have a black square border around a unique binary identifier (ID) grid. Once supplied with the marker's physical side length, it solves the planar Perspective-n-Point problem [47] to resolve both position and orientation in a single pass. The result is the pose of the marker in the sensor frame. For the object marker that pose is ${}^{Camera}_{Object}T$, and it is measured on every frame.

The wall marker (ID 0, 150 mm) is mounted on a plumb wall. It is printed larger only because it sits farthest from the sensor. Appendix E covers corner detection, planar ambiguity, and rotation matrix conversion.

2.5 Measurement Filtering

The raw 3D landmark trajectories carry three kinds of error:

Case A, spikes: an isolated sample jumps far off the local trajectory for one frame, from a depth speckle or a momentary false detection.

Case B, gaps: a few consecutive samples are missing, because the visibility gate of Section 2.3 blocked them or the depth sample was invalid.

Case C, jitter: every sample carries small broadband noise from detection and depth measurement.

If a filter is strong enough to remove a sharp spike, it also distorts fast motion. No smoothing filter can fill in missing data gaps either. The pipeline therefore handles each case in a separate stage.

A rolling Hampel test handles case A. The Hampel identifier [48] spots outliers by the distance of a point from the local median. It works well on time-series data because a sudden spike will not pull the median off track the way it pulls a mean [49]. The test uses a 7-frame window and a threshold of three times a robust scale. On each coordinate, the pipeline subtracts each sample's own centred rolling median and takes the absolute residual. A second centred 7-frame rolling median over those residuals, multiplied by about 1.48, supplies the scale (Appendix F). A baseline floor of 2 centimetres prevents small sensor noise from triggering the test when the subject is standing still. The pipeline discards any spike the test flags and marks it as missing data, which then passes into case B. Figure 2.7 shows the test flagging a wrist spike that jumped 2.9 centimetres off track in a single frame.

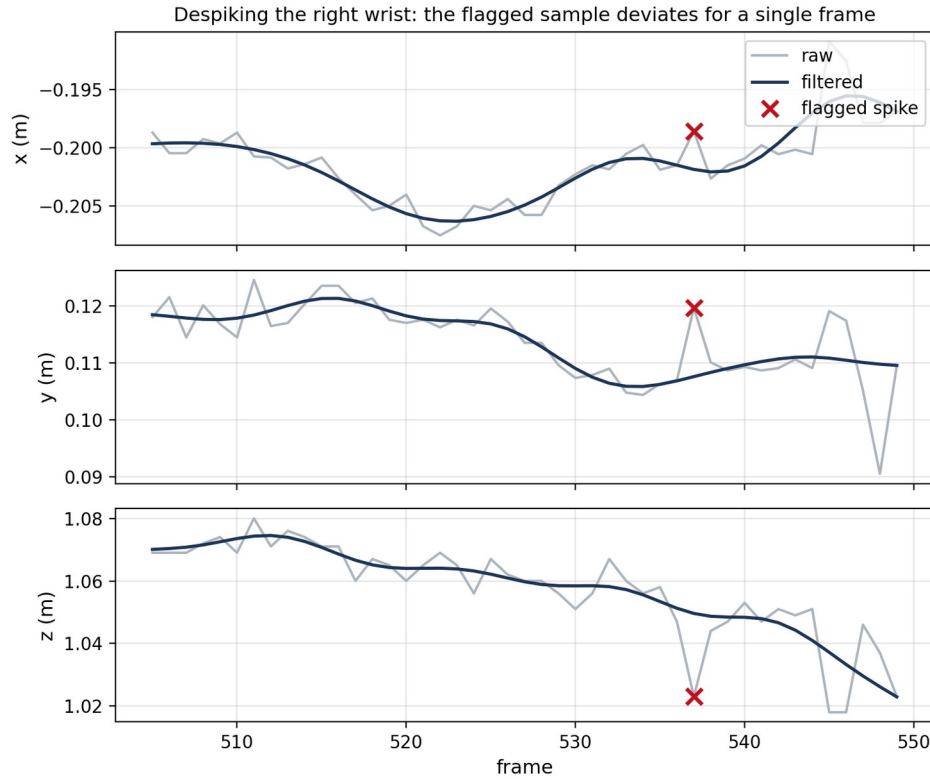


Figure 2.7. Despiking on the right wrist of the recording: per-axis raw and filtered trajectories, with a mark on the sample flagged by the Hampel test.

Linear interpolation handles case B. The interpolation fills interior gaps no longer than 5 frames, a sixth of a second, along a straight line between the two valid endpoints. Longer gaps remain missing, and the pose recovery of Chapter 5 handles them. Figure 2.8 shows bridged wrist dropouts from the recording.

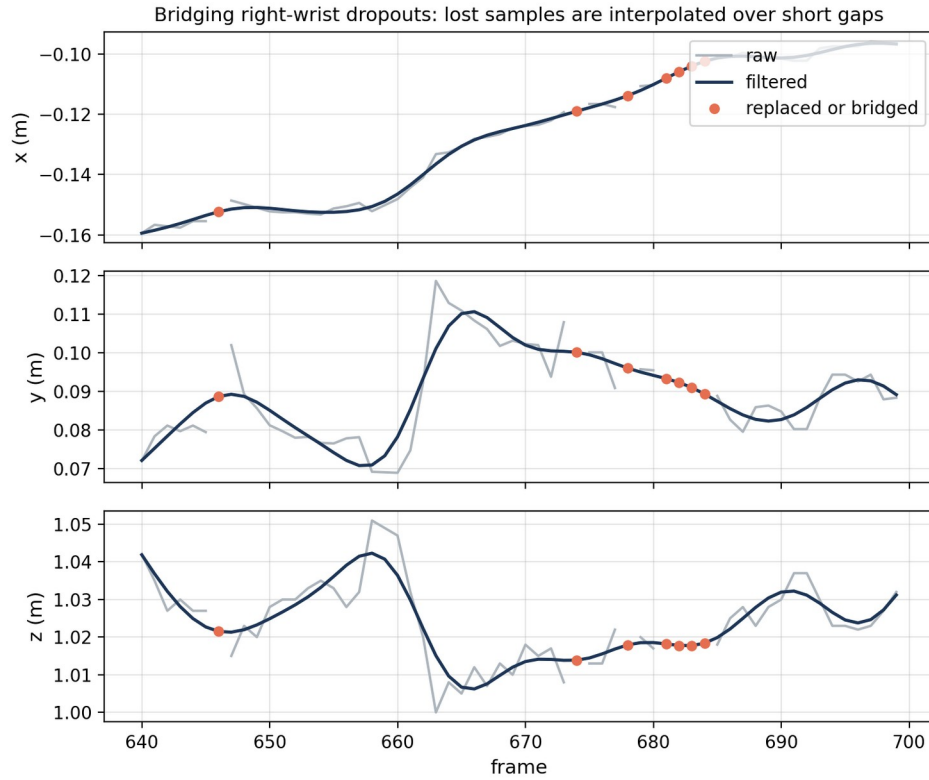


Figure 2.8. Gap bridging on the right wrist of the recording: the short gaps filled by interpolation, with the replaced samples marked.

A zero-phase Butterworth low-pass filter then handles case C. The filter follows the design procedure of [50]. The Butterworth family has a flat passband, so it passes the motion without ripple [51]. A fourth-order filter with a 3 Hz cutoff runs forward and backward over the recording [52]. The two passes cancel the phase delay. The forward and backward low-pass filter reduces jitter but also attenuates motion near its cutoff. The backward pass needs the whole recording, so the filter can only run offline. The real-time alternative is the One Euro filter [53]. Chapter 8 compares the causal filter with the offline one on the same recording (Section 8.3). Appendix F compares the candidate smoothers side by side. Figure 2.9 shows the raw jitter against the filtered track.

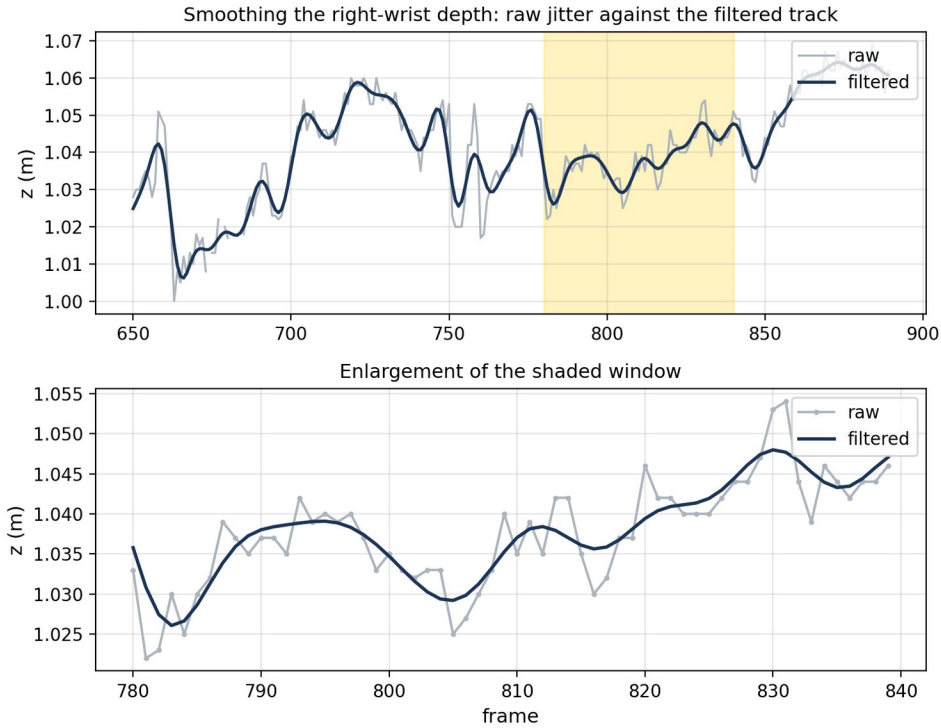


Figure 2.9. Smoothing on the right-wrist depth of the recording: the raw and filtered tracks over an eight-second span, with the shaded window enlarged below.

Every repaired sample keeps a flag, which separates measured data from filled-in data for later stages. Chapter 3 solves the joint angles from these filtered landmarks.

2.6 Hip Depth Preparation

A visible landmark can take the depth of the desk or rail in front of the body. Before solving the body angles, the pipeline checks hip and shoulder widths against calibrated values, then compares the directions of the hip line and shoulder line. It next rejects a torso landmark whose depth is more than 10 centimetres from its remembered value. A final check rejects a hip more than 15 centimetres nearer the camera than the input shoulder midpoint. These checks run in that order, so the replacement described below takes priority over the last check.

Depth memory starts empty. The first sample that survives the checks seeds it; later accepted samples update it with 70 percent previous depth and 30 percent new depth. A repaired sample does not update it, and the memory does not expire. The pipeline places a single rejected point on its retained camera ray at the remembered depth, or at the

shoulder-midpoint depth for the last check. Rebuilding a whole pair instead restores its calibrated width, so its final endpoints can leave their original rays.

Appendix G gives the branching, fallbacks and output states. The preparation assumes usable landmark pixels and, for the shoulder-depth check, limited trunk lean. It does not establish anatomical hip depth. Chapter 5 uses the prepared torso for arm recovery. The plain, direction-memory and object-assisted solves of Chapter 7 all rest on this preparation. The hold-last solve keeps the last accepted joint angles and does not use it. The causal path's angle solve includes this preparation, but its pelvis translation still uses the original hip midpoint.

The measurement layer leaves three things unresolved: the landmarks are metric points, not a kinematic state; the object poses are not yet joined into a track; and nothing links the body to the object. Chapter 3, Chapter 4 and Chapter 5 take each in turn.

Chapter 3: Kinematic Modelling

Chapter 2 delivers eight body landmarks per frame as metric 3D positions in one calibrated scene (Appendix C, Appendix D). Points carry no orientation. This chapter converts them into a kinematic state: a root pose, four angles per arm, and the wrist as a point. The model is closed form: every angle comes from the landmark geometry of the current frame, with no optimization and no state carried between frames.

Figure 3.1 draws the information flow.

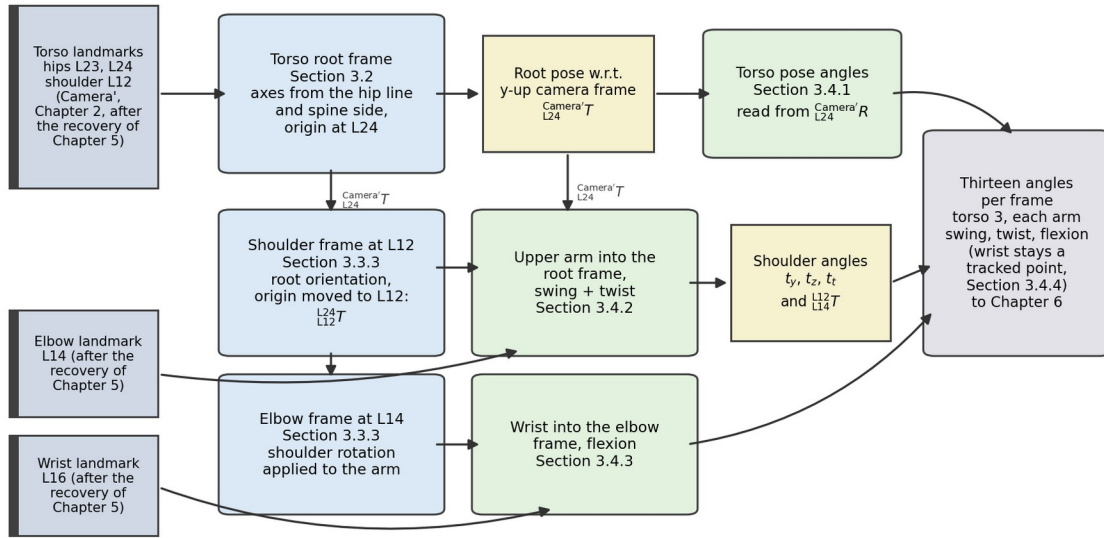


Figure 3.1. The chapter's information flow. Landmarks enter from Chapter 2 on the left, the stored thirteen-angle pose leaves on the right, and the section numbers locate each construction.

3.1 Coordinate Systems and Transformations

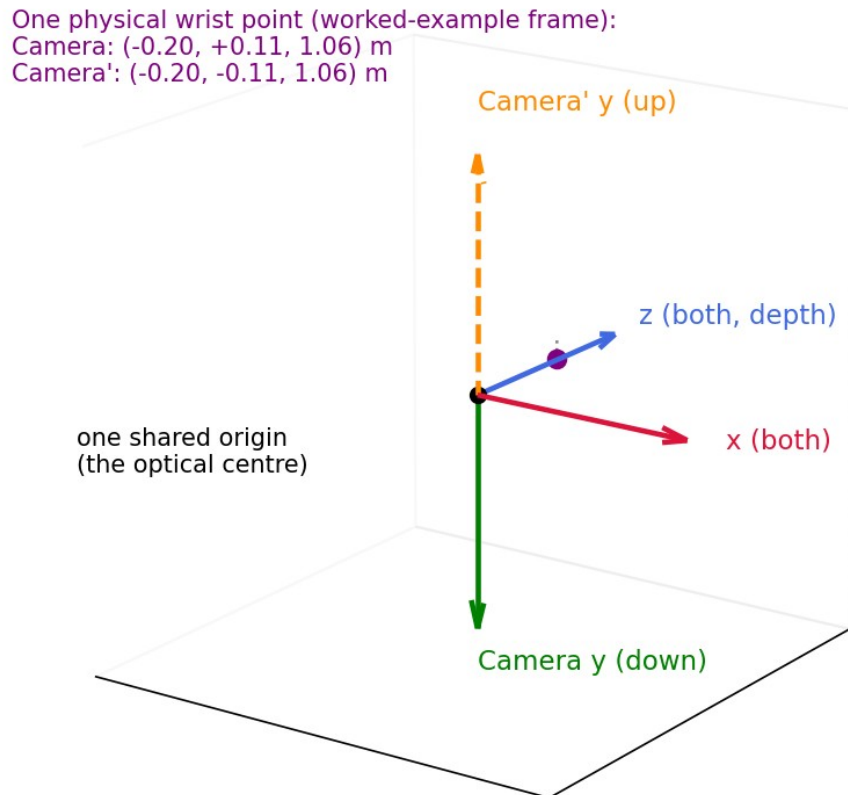
This chapter attaches a coordinate frame to each body segment. Unity uses a left-handed frame with y upward, so the kinematics uses the matching axis convention of {Camera'}.

Converting a position into {Camera'} changes the handedness itself rather than relabelling the axes. If p is a landmark position in the sensor frame, its position q in the y-up camera frame is

$${}^{Camera'}q = F^{Camera}p, F = \text{diag}(1, -1, 1). \quad (3.1)$$

The determinant of F is -1 , the sign of a reflection. F applies to points only, never to rotation matrices, so every rotation in the pipeline comes out proper. Figure 3.2 draws the two frames with the right wrist of Section 3.5. Every landmark position in this chapter is a y-up camera-frame value unless stated otherwise.

Raw camera frame {Camera} and y-up camera frame {Camera'}
Same optical origin; Camera' is not person-centred



$F = \text{diag}(1, -1, 1)$: only y changes sign

Figure 3.2. The raw camera frame {Camera} and y-up camera frame {Camera'} drawn from one optical centre. Only the y axis differs: {Camera} reads y downward and {Camera'} upward.

The frame {Camera'} is not person-centred, and it is not the frame of the finished Unity scene of Figure 2.4(b), which Chapter 6 builds from the calibrated camera pose and the desk marker of Section 2.2.

The left-handed choice changes no physical result. The object track keeps the right-handed world frame instead, and Chapter 6 converts it for Unity with the axis swap of equation (6.1).

Section 2.2 fixes the notation of a rigid body transformation [44]. The reference frame A is a left superscript and the described frame B a left subscript, so ${}^A_B R$ is the orientation of a child frame B in a parent frame A. The columns of the rotation part ${}^A_B R$ are the axis directions of B in the coordinates of A, ${}^A_B R = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix}$. They are perpendicular unit vectors, so the transpose of ${}^A_B R$ is its inverse, which the transformations of this chapter use directly. With the origin of B expressed in A as the translation part ${}^A P_{BORG}$, the two combine into the 4 by 4 homogeneous transformation matrix ${}^A_B T$:

$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = {}^A_B T \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}. \quad (3.2)$$

Running the rotation the other way expresses a measured segment direction in the coordinates of the child frame B:

$${}^B v = {}^B_A R {}^A v, {}^B_A R = \left({}^A_B R \right)^T. \quad (3.3)$$

Throughout, ${}^A P$ is a measured landmark point carrying the frame it is expressed in, and v a segment vector between two points. Equation (3.3) is the rotation part acting alone, so it applies to directions. The chain frames sit at different places, so a landmark moves between them by the full transformation of equation (3.2), whose inverse is

$${}^B_A T = \left({}^A_B T \right)^{-1} = \begin{bmatrix} \left({}^A_B R \right)^T & - \left({}^A_B R \right)^T {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}. \quad (3.4)$$

Figure 3.3(a) draws the chain on the worked-example frame of Section 3.5.

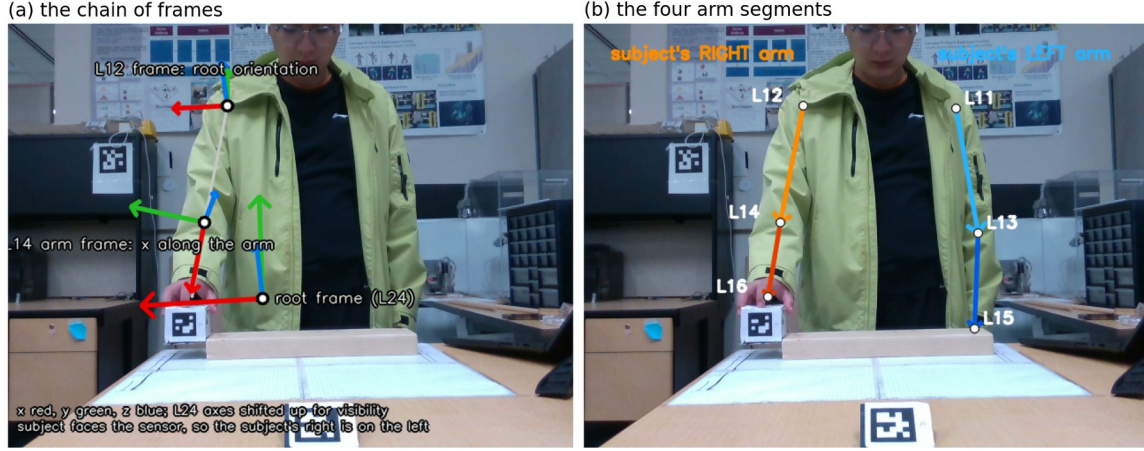


Figure 3.3. The worked-example frame, front view. (a) The chain of frames, the root at right hip L24, shoulder at L12, elbow at L14. (b) The four arm segments.

Each neighbouring pair in Figure 3.3(a) is related by its own homogeneous transformation:

$$\begin{aligned}
 {}^{Camera'}_{L24}T &= \begin{bmatrix} {}^{Camera'}_{L24}R & {}^{Camera'}P_{L24ORG} \\ 0 & 1 \end{bmatrix} \\
 {}^{L24}_{L12}T &= \begin{bmatrix} I & {}^{L24}P_{L12ORG} \\ 0 & 1 \end{bmatrix} \\
 {}^{L12}_{L14}T &= \begin{bmatrix} {}^{L12}_{L14}R & {}^{L12}P_{L14ORG} \\ 0 & 1 \end{bmatrix}.
 \end{aligned} \tag{3.5}$$

A frame that exists once per arm takes its label from its landmark, so the side needs no further subscript: L12 and L14 on the subject's right, L11 and L13 on the left. The root transformation ${}^{Camera'}_{L24}T$ is the pose of the root frame with respect to the y-up camera frame. The two that follow carry the shoulder frame with respect to the root and the elbow frame with respect to the shoulder, with ${}^{L24}_{L11}T$ and ${}^{L11}_{L13}T$ their left counterparts. Section 3.2 builds the root pose, Section 3.3.3 places both origins, and Section 3.4.2 solves the shoulder rotation that fills the last block and mirrors the construction for the left arm. Chaining the three describes the right elbow frame directly in the y-up camera frame,

$${}^{Camera'}_{L14}T = {}^{Camera'}_{L24}T {}^{L24}_{L12}T {}^{L12}_{L14}T. \tag{3.6}$$

Every landmark is converted to the y-up camera frame by equation (3.1), and the inverse transformations of equation (3.4) carry it into the frame where its joint is solved. The shoulder landmark goes into the root frame by ${}^{L24}_{Camera'}T$. The elbow goes into the shoulder frame by ${}^{L12}_{Camera'}T$ for the angles of Section 3.4.2. The wrist goes into the elbow frame by ${}^{L14}_{Camera'}T$ for the flexion of Section 3.4.3. Section 3.5 prints all three as numbers.

3.2 Torso Reference Frame

3.2.1 Landmark Selection

The right hip L24, the left hip L23, and the right shoulder L12, three MediaPipe landmarks [54], define the torso root. Left and right always refer to the subject, who faces the sensor, so the right side appears on the left of the image.

These points move more slowly than the hands and forearms, the object rarely hides them, and they form a broad triangle across the trunk, so a small position error changes the angles little.

With three torso points, the model cannot separate the pelvis from the upper trunk. The torso is therefore one rigid plate with no twist between shoulders and hips, and a forward lean pitches the root frame with it. Chapter 9 lists the simplification among the limitations.

3.2.2 Derivation of the Torso Frame

The flip F of equation (3.1) converts the three positions to the y-up camera frame, giving ${}^{Camera'}P_{23}$, ${}^{Camera'}P_{24}$ and ${}^{Camera'}P_{12}$. Figure 3.4(a) shows the two input vectors on frame 1400 of the two-hand rail recording, and Figure 3.4(b) the finished frame at L24.

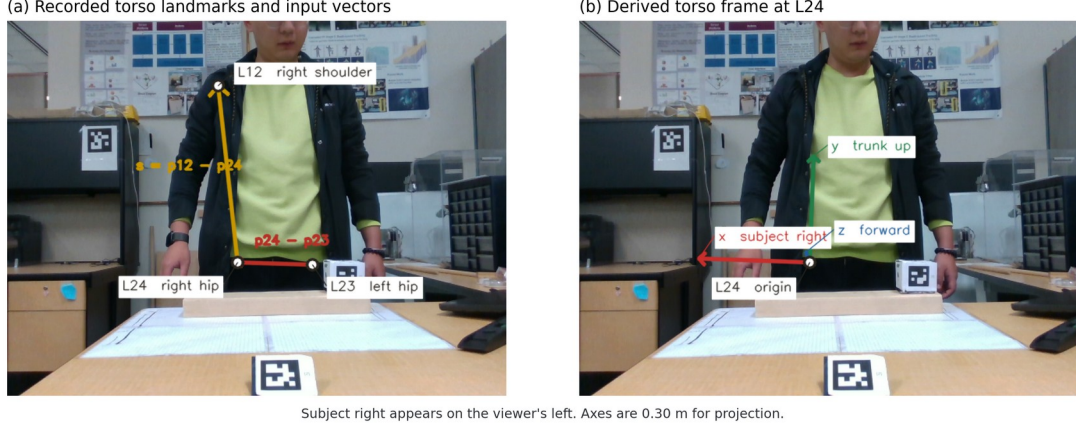


Figure 3.4. Torso-frame construction. (a) The measured shoulder L12 and hips L23 and L24 define the hip vector and the spine-side vector s . (b) The derived axes at L24.

The x axis points along the hip line toward the subject's right, and the hat marks a unit vector.

$${}^{Camera'}\hat{X}_{L24} = \frac{{}^{Camera'}P_{24} - {}^{Camera'}P_{23}}{\left| {}^{Camera'}P_{24} - {}^{Camera'}P_{23} \right|}. \quad (3.7)$$

The second input is the spine-side vector from the right hip up to the right shoulder.

$${}^{Camera'}s = {}^{Camera'}P_{12} - {}^{Camera'}P_{24}. \quad (3.8)$$

This vector is not an axis, because it is generally not perpendicular to the hip line: in the worked-example frame the angle between them is 83.8 degrees. It only selects the torso plane, the one plane through the hip line that also contains ${}^{Camera'}s$.

The z axis is the normalized cross product of the two inputs:

$${}^{Camera'}\hat{Z}_{L24} = \frac{{}^{Camera'}\hat{X}_{L24} \times {}^{Camera'}s}{\left| {}^{Camera'}\hat{X}_{L24} \times {}^{Camera'}s \right|}. \quad (3.9)$$

The order of the factors sets the sign: with ${}^{Camera'}\hat{X}_{L24}$ to the subject's right and ${}^{Camera'}s$ up the trunk, the product points forward, out of the chest toward the camera.

The y axis closes the frame with a second cross product.

$${}^{Camera'}\hat{Y}_{L24} = {}^{Camera'}\hat{Z}_{L24} \times {}^{Camera'}\hat{X}_{L24}. \quad (3.10)$$

The vectors ${}^{Camera'}\hat{Z}_{L24}$ and ${}^{Camera'}\hat{X}_{L24}$ are unit vectors at right angles, so their product needs no normalization. The construction gives an orthonormal frame whenever the two hips are distinct and the shoulder does not lie on the hip line.

The three axes assemble into the root rotation matrix, its columns in the y-up camera-frame order right, up, forward.

$${}^{Camera'}_{L24}R = \begin{bmatrix} {}^{Camera'}\hat{X}_{L24} & {}^{Camera'}\hat{Y}_{L24} & {}^{Camera'}\hat{Z}_{L24} \end{bmatrix}. \quad (3.11)$$

The right hip is the origin of the root frame, ${}^{Camera'}P_{L24ORG} = {}^{Camera'}P_{24}$, so the three axes fill in the first transformation of equation (3.5).

$${}^{Camera'}_{L24}T = \begin{bmatrix} {}^{Camera'}_{L24}R & {}^{Camera'}P_{24} \\ 0 & 1 \end{bmatrix}. \quad (3.12)$$

Figure 3.5 shows the reference configuration that fixes the meaning of the finished matrix: the subject upright, facing the sensor, arms straight out to the sides, the T-pose that is the zero of every arm angle in Section 3.3. Such a subject has the right side toward the camera's left and the chest toward the negative z of the y-up camera frame, giving

${}^{Camera'}_{L24}R$ the columns $(-1, 0, 0)$, $(0, 1, 0)$, $(0, 0, -1)$. These decode in the convention of Section 3.4 to the Euler angles $(0, 180, 0)$.

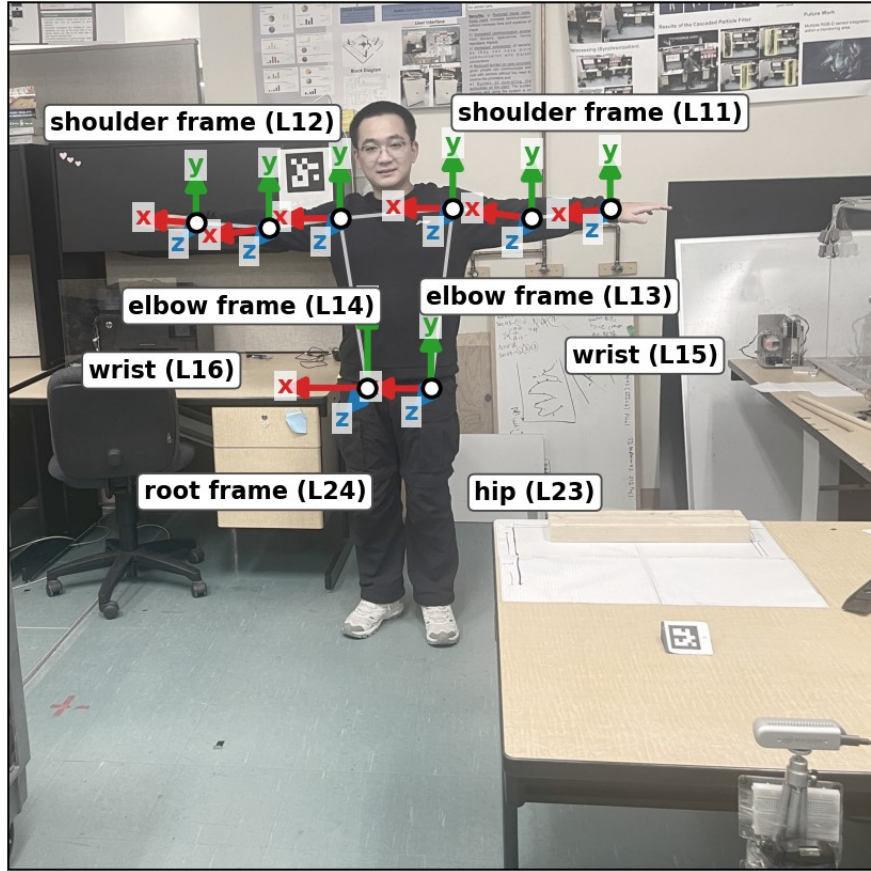


Figure 3.5. The reference configuration: the subject in the T-pose with a frame at each landmark. All frames share the root orientation, right arm along $+x$, left arm along $-x$.

3.3 The Arm Model

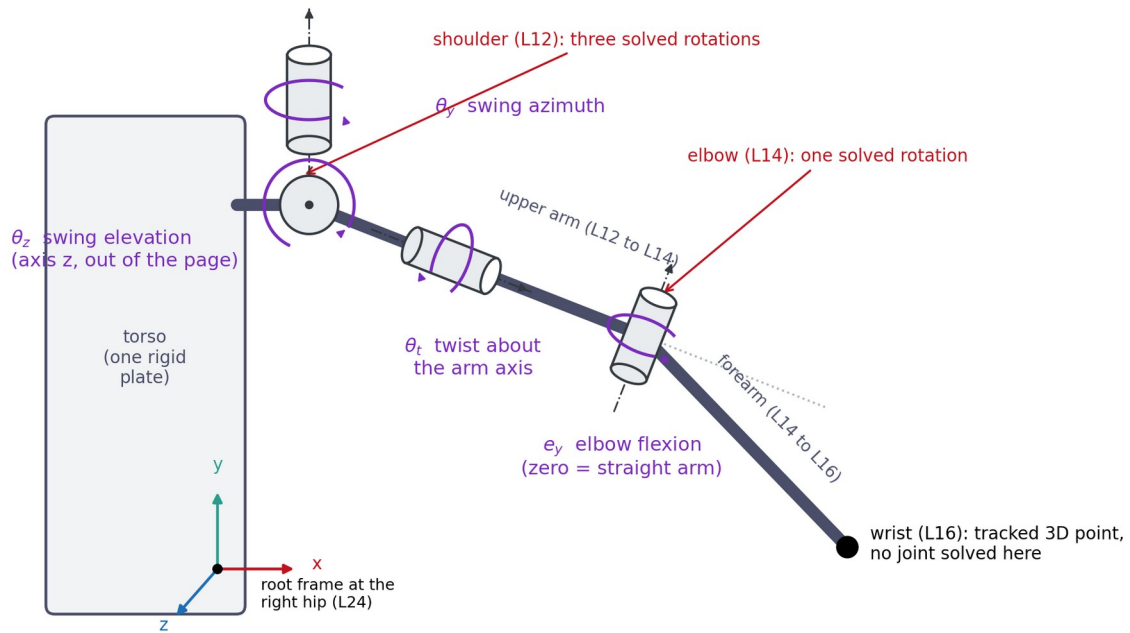
3.3.1 Kinematic Schematic and Solved Angles

A human arm is described in [44] with seven rotational degrees of freedom, three at the shoulder, one at the elbow and three at the wrist. This model solves the first four: the two shoulder swing angles, the shoulder twist and the elbow flexion. The chain ends at the wrist as a tracked point, since the three wrist rotations cannot be observed from the eight body landmarks.

The wrist orientation is not solved. Section 3.4.4 gives the reason no algebra over these landmarks reaches it. Fingers, contact points and grip force lie outside the landmark set, the torso is the single rigid plate of Section 3.2.1, and the legs are not tracked. The recovery of Chapter 5 and the avatar of Chapter 6 use segment lengths to rebuild positions. These are effective lengths, medians of a segment over the landmark track

rather than anatomical bone lengths, and Section 6.3 sets them against the subject's own measurement.

Figure 3.6 places those four joints on the body as a manipulator diagram. The shoulder at L12 is a cluster of three revolute joints with one shared centre, and a single hinge sits at the elbow L14.



Each solved rotation is drawn as a revolute-joint cylinder whose dash-dot line is its rotation axis.
 Four solved angles per arm: shoulder swing θ_y , θ_z , shoulder twist θ_t , elbow flexion e_y .
 Zero reference: the T-pose, with the right arm along the root frame's +x axis (dotted grey line at the elbow).

Figure 3.6. The kinematic schematic of the right arm, a chain of revolute-joint cylinders in the Programmable Universal Machine for Assembly (PUMA) style. Each dash-dot line is a rotation axis.

The azimuth t_y turns the arm about the torso's y axis, the elevation t_z raises it about the z axis, the twist t_t rolls the upper arm about its own length. The flexion e_y bends the forearm relative to the upper arm.

3.3.2 Definition of Shoulder, Elbow, and Wrist

Each arm is a chain of two rigid segments, the upper arm from the shoulder to the elbow and the forearm from the elbow to the wrist. Figure 3.3(b) shows the four segments, the right arm on L12, L14 and L16 and the left arm on L11, L13 and L15.

3.3.3 Establishing Local Arm Frames

Every joint rotation is measured relative to its parent frame, the shoulder relative to the torso and the elbow relative to the rotated upper arm.

The parent frame of the shoulder is L12, the root orientation ${}^{Camera'}_{L24}R$ moved up the trunk to the shoulder landmark. No separate orientation is derived there: a second basis from the shoulder line and the upper arm would add a shoulder-girdle degree of freedom the data cannot support. With the shoulder angles at zero, the right upper arm points along +x. This fills in the second transformation of equation (3.5):

$${}^{L24}_{L12}T = \begin{bmatrix} I & \left({}^{Camera'}_{L24}R \right)^T \left({}^{Camera'}P_{12} - {}^{Camera'}P_{24} \right) \\ 0 & 1 \end{bmatrix}. \quad (3.13)$$

The rotation block is the identity, ${}^{L24}_{L12}R = I$, and the origin

${}^{L24}P_{L12ORG} = \left({}^{Camera'}_{L24}R \right)^T \left({}^{Camera'}P_{12} - {}^{Camera'}P_{24} \right)$ is the measured shoulder landmark in torso coordinates, by the mapping of equation (3.4).

The parent of the elbow is the upper arm, so the elbow frame L14 is the root orientation composed with the solved shoulder rotation, with its origin at the elbow landmark. This follows the PUMA robot model, where each new frame is the previous one turned by the joint rotation [44]. In the T-pose, the forearm continues the arm axis along the local +x.

The third transformation of equation (3.5) is then

$${}^{L12}_{L14}T = \begin{bmatrix} {}^{L12}_{L14}R & \left({}^{Camera'}_{L24}R \right)^T \left({}^{Camera'}P_{14} - {}^{Camera'}P_{12} \right) \\ 0 & 1 \end{bmatrix}, \quad (3.14)$$

with the solved shoulder rotation ${}^{L12}_{L14}R$ of Section 3.4.2 in the rotation block and the elbow landmark in shoulder-frame coordinates, ${}^{L12}P_{L14ORG}$, as its origin. The shoulder frame shares the root orientation, so the same $\left({}^{Camera'}_{L24}R \right)^T$ maps it.

A measured segment reaches its parent frame in two steps. It is formed as a vector between two landmarks, for the right upper arm ${}^{Camera'}v = {}^{Camera'}P_{14} - {}^{Camera'}P_{12}$, and

equation (3.3) pulls it into the parent frame as ${}^{L24}v = \left({}_{L24}^{Camera'} R \right)^T {}^{Camera'} v$. Section 3.4 works backwards from it to the angles.

3.4 Joint Angle Computation

The root orientation is encoded by three Euler angles in the Unity convention (Chapter 6), applied about the parent-frame axes in the order z, then x, then y. The arm angles are joint coordinates; Section 3.4.2 defines their separate composition. In the matrix product the rotation applied first sits nearest the column vector, so the convention reads

$$R = R_y(y) R_x(x) R_z(z), \quad (3.15)$$

where R_x , R_y , R_z are the elementary rotations about the coordinate axes. Multiplied out, with c for cosine and s for sine, the product is the matrix this section reads its angles from:

$$R = \begin{bmatrix} c_y c_z + s_y s_x s_z & -c_y s_z + s_y s_x c_z & s_y c_x \\ c_x s_z & c_x c_z & -s_x \\ -s_y c_z + c_y s_x s_z & s_y s_z + c_y s_x c_z & c_y c_x \end{bmatrix}. \quad (3.16)$$

Write m_{ij} for the entry of a measured numerical matrix m in row i and column j . The three angles follow from its entries:

$$x = \text{asin}(-m_{23}), y = \text{atan2}(m_{13}, m_{33}), z = \text{atan2}(m_{21}, m_{22}). \quad (3.17)$$

The arcsine restricts x to -90 to +90 degrees. At either end of that range ($|m_{23}|=1$) the y and z rotations turn about the same axis and only their combined angle is defined, the configuration called gimbal lock. The z angle is set to zero there by convention, so only the split between the two angles is a choice. The entries m_{13} and m_{33} of the third column both vanish there, so the locked branch reads y from m_{12} and m_{11} instead:

$$\begin{aligned} x = +90^\circ: y &= \text{atan2}(m_{12}, m_{11}), z = 0 \\ x = -90^\circ: y &= \text{atan2}(-m_{12}, m_{11}), z = 0. \end{aligned}$$

3.4.1 Torso Pose

The torso is not a joint. Its pose is the pair ${}^{Camera'}_{L24}R$, ${}^{Camera'}P_{L24ORG}$, and equation (3.17) applied to ${}^{Camera'}_{L24}R$ gives its three angles, as Section 3.5 shows on measured data.

3.4.2 Shoulder Orientation

A shoulder pose must describe the direction of the upper arm and its roll about its own axis, the three angles labelled at the shoulder of Figure 3.6. They separate into swing and twist [55], with the twist innermost:

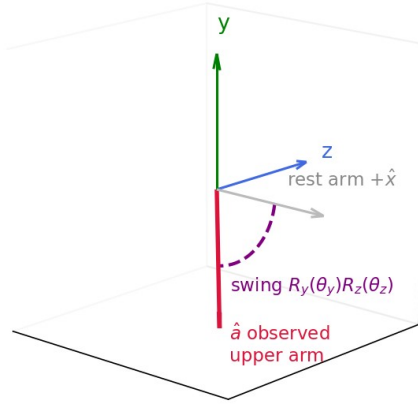
$${}^{L12}_{L14}R = R_y(t_y) R_z(t_z) R_x(t_t), \quad (3.18)$$

where t_y and t_z are the swing angles and t_t the twist about the arm axis +x of the T-pose. A rotation about x leaves the x axis fixed, so with the twist innermost the first column of ${}^{L12}_{L14}R$ is free of the twist factor. This column is the upper arm direction ${}^{L12}\hat{a}$ of the elbow frame described in the shoulder frame,

$${}^{L12}\hat{a} = {}^{L12}\hat{X}_{L14} = {}^{L12}_{L14}R {}^{L14}\hat{X}_{L14} = R_y(t_y) R_z(t_z) {}^{L14}\hat{X}_{L14} = (c_y c_z, s_y c_z, -s_y), \quad (3.19)$$

so the upper-arm direction depends on the swing alone. The twist is invisible in the shoulder-to-elbow vector and has to be measured from a second segment, the forearm. The joint order is not an Euler order, so Chapter 6 rebuilds the rotation from the three angles as three factors in this order. Figure 3.7 shows both halves of the decomposition.

(a) swing aims the arm; $\hat{a} = (c_y c_z, s_z, -s_y c_z)$



(b) twist $R_x(\theta_t)$ rolls about the arm axis

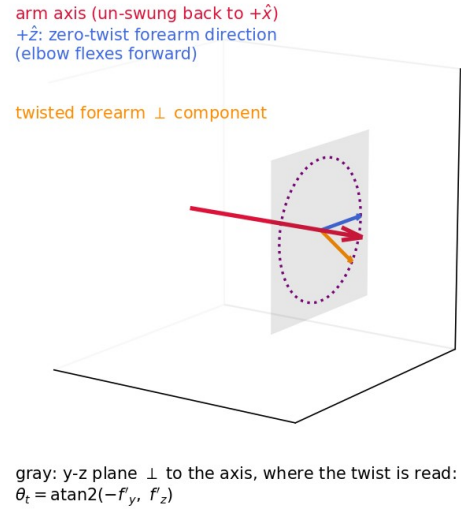


Figure 3.7. The swing-twist decomposition. (a) The swing aims the rest direction $+x$ of the arm into the observed direction $\hat{a}$. (b) The twist rolls about the arm axis.

The swing angles follow from the components of ${}^{L12}\hat{a}$. Its y component is s_z , so

$$t_z = a \sin(a_y), \quad (3.20)$$

the elevation of the arm, positive upward and restricted to -90 to $+90$ degrees. The x and z components share the factor c_z , which cancels in the two-argument arctangent:

$$t_y = \text{atan2}(-a_z, a_x), \quad (3.21)$$

the azimuth of the arm, positive when sweeping backward. It is undefined for the arm pointing straight up or straight down ($t_z = \pm 90^\circ$), where the x and z components both vanish. The convention $t_y = 0$ is applied there, and any axial rotation folds into the twist.

A rotation about the arm axis is only visible in the plane perpendicular to it, so the procedure undoes the swing first, making that plane the y-z plane of the shoulder frame. The matrix product below is a rotation operator given by its axes and its angles, not a frame-relative rotation, so it carries no pair of frame labels. Its input and output are described in the shoulder frame L12:

$${}^{L12}f' = R_z(-t_z) R_y(-t_y) {}^{L12}f, \quad (3.22)$$

where ${}^{L12}f$ is the forearm vector in the parent frame of Section 3.3.3. Once the swing is undone, the nonzero perpendicular component of ${}^{L12}f'$ points along the +z axis of L12 at zero twist. Under a twist t_t , its unit direction in the (y, z) plane is $(-\sin t_t, \cos t_t)$, so the twist reads

$$t_t = \text{atan2}(-f'_y, f'_z). \quad (3.23)$$

The two components the arctangent reads are components of the rotated vector ${}^{L12}f'$ in the shoulder frame L12. The twist is positive for internal rotation, where the forearm rotates downward from forward.

The left arm is solved on mirrored inputs. Both left segment vectors are expressed in the root basis and their x components flipped with $M = \text{diag}(-1, 1, 1)$, the sagittal mirror. The same right-arm solver then runs on the result, so a mirror-symmetric pose gives identical angles on both sides with the same anatomical meaning.

3.4.3 Elbow Flexion and Extension

The elbow is solved in the elbow frame L14 of Section 3.3.3. Its orientation in the y-up camera frame is ${}^{Camera'}_{L14}R = {}^{Camera'}_{L24}R {}^{L24}_{L12}R {}^{L12}_{L14}R$, where the middle factor is the identity of equation (3.13) and the last the solved shoulder rotation of equation (3.18). The unit

forearm direction there is ${}^{L14}\hat{g} = \frac{({}^{Camera'}_{L14}R)^T ({}^{Camera'}P_{16} - {}^{Camera'}P_{14})}{|{}^{Camera'}P_{16} - {}^{Camera'}P_{14}|}$, and its swing is read

from the components of ${}^{L14}\hat{g}$ as at the shoulder:

$$e_z = \text{asin}(g_y), e_y = \text{atan2}(-g_z, g_x). \quad (3.24)$$

Here $e_y = 0$ is a straight arm and $e_y = -90$ is a right angle bend, the hinge angle labelled in Figure 3.6.

The angle e_z is always zero. The shoulder twist aligns the perpendicular component of the forearm with local +z, so the flexion plane is already the x-z plane of the elbow frame by the time the elbow is solved. That closes the dimension count: two measured segment directions give four observable degrees of freedom and the model extracts four

independent angles. The angle e_z stays among the streamed angles for generality but holds no independent information.

A three-landmark arm breaks down in one case. When the arm is straight, the forearm is parallel to the upper arm, its perpendicular component vanishes, and no algebra can recover a roll about an axis from two collinear segments. The solver detects this when the forearm lies along the upper arm axis and the perpendicular part of the forearm vanishes against its length, applies $t_i=0$ so the elbow frame stays defined, and flags the twist as unobservable; later stages hold the last valid value. A nearly straight elbow leaves the flexion well conditioned but makes the twist sensitive to noise, a limitation Chapter 9 discusses.

3.4.4 The Wrist as a Tracked Point

The wrist L16 is not a solved joint. It enters as a tracked 3D point, as Figure 3.6 draws it, and does two jobs: it gives the forearm direction the elbow flexion is read from, and it makes the shoulder twist observable.

A full three-degree-of-freedom wrist orientation would need a landmark beyond the wrist. Forearm pronation, the roll of the forearm about its own axis, does not move the wrist point, so no algebra over these landmarks can observe it. The tracked position is still kept and transmitted, and when the landmark is lost during a grip the recovery in Chapter 5 rebuilds it with an additional geometric constraint from the object pose. A hand model such as [56] would attach at the same point.

3.5 Worked Example

Frame 533 of the rail recording, drawn in Figure 3.3, is taken while the cube slides along the rail in the task of Chapter 2. All eight landmarks are measured on it, none flagged by the despiking stage or filled by the gap-bridging stage of Section 2.5, and no failure detector of Chapter 5 fires on it. Values are computed from the unrounded positions and printed rounded to two decimals, so a step recomputed by hand from the printed inputs can differ by several degrees where the arguments are small. The two-argument arctangent of the printed 0.06 and 0.03 gives 63.4 degrees, while the unrounded values give the 59.9 degrees printed below.

Table 3.1 lists the eight landmark positions in both frames, after the filtering of Section 2.5 and before the hip depth preparation of Section 2.6. The same frame runs through the worked examples of Appendix C, Appendix D and Appendix E.

Landmark	Camera (m)	Camera' (m)
L11 left shoulder	(0.22, -0.34, 1.36)	(0.22, 0.34, 1.36)
L12 right shoulder	(-0.16, -0.33, 1.31)	(-0.16, 0.33, 1.31)
L13 left elbow	(0.28, -0.03, 1.38)	(0.28, 0.03, 1.38)
L14 right elbow	(-0.20, -0.04, 1.19)	(-0.20, 0.04, 1.19)
L15 left wrist	(0.24, 0.20, 1.25)	(0.24, -0.20, 1.25)
L16 right wrist	(-0.20, 0.11, 1.06)	(-0.20, -0.11, 1.06)
L23 left hip	(0.12, 0.18, 0.99)	(0.12, -0.18, 0.99)
L24 right hip	(-0.06, 0.19, 0.99)	(-0.06, -0.19, 0.99)

Table 3.1. The eight landmarks of the worked-example frame in the sensor frame and in the y-up camera frame, printed to two decimals. Each column carries the frame its entries are expressed in, the left superscript of the position symbols used from here on. The conversion flips only the sign of y, and the hips read negative in the y-up camera frame because they sit below the camera height.

The torso construction of Section 3.2 needs three of these points:

$$\begin{aligned}
{}^{Camera'}P_{23} &= (0.12, -0.18, 0.99) \\
{}^{Camera'}P_{24} &= (-0.06, -0.19, 0.99) \\
{}^{Camera'}P_{12} &= (-0.16, 0.33, 1.31).
\end{aligned}$$

Equations (3.7) and (3.8) give the hip line, 0.18 m long, and the spine-side vector, 0.63 m long. The cross products of equations (3.9) and (3.10) close the frame:

$$\begin{aligned}
{}^{Camera'}\hat{X}_{L24} &= (-1.00, -0.05, -0.04) \\
{}^{Camera'}\hat{Z}_{L24} &= (0.01, 0.53, -0.85) \\
{}^{Camera'}\hat{Y}_{L24} &= (-0.06, 0.85, 0.53).
\end{aligned}$$

The x component of ${}^{Camera'}\hat{X}_{L24}$ is close to -1, and ${}^{Camera'}\hat{Z}_{L24}$ is tipped upward by the hip-to-shoulder depth difference. Equation (3.11) assembles the columns:

$${}^{Camera'}_{L24}R = \begin{bmatrix} -1.00 & -0.06 & 0.01 \\ -0.05 & 0.85 & 0.53 \\ -0.04 & 0.53 & -0.85 \end{bmatrix}.$$

Appending the origin ${}^{Camera'}P_{L24ORG} = {}^{Camera'}P_{24}$ gives the root pose of equation (3.12) as numbers:

$${}^{Camera'}_{L24}T = \begin{bmatrix} -1.00 & -0.06 & 0.01 & -0.06 \\ -0.05 & 0.85 & 0.53 & -0.19 \\ -0.04 & 0.53 & -0.85 & 0.99 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Equation (3.13) needs the shoulder landmark in the torso's own coordinates, which equation (3.4) gives as (0.07, 0.63, 0.00). The forward offset is zero, since the spine-side vector lies in the torso plane by construction:

$${}^{L24}_{L12}T = \begin{bmatrix} 1 & 0 & 0 & 0.07 \\ 0 & 1 & 0 & 0.63 \\ 0 & 0 & 1 & 0.00 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

an identity rotation with a pure origin shift. Equation (3.17) decodes the root matrix:

$$\begin{aligned} x &= \text{asin}(-0.53) = -32.1^\circ \\ y &= \text{atan2}(0.01, -0.85) = 179.5^\circ \\ z &= \text{atan2}(-0.05, 0.85) = -3.3^\circ. \end{aligned}$$

Compared with the T-pose (0, 180, 0), the subject faces almost straight at the sensor and the hip line is almost level. The frame is pitched by 32.1 degrees because the hips sit nearer the camera than the shoulders. The two hip landmarks fall on the pixels where the pelvis meets the rail, so the depth sampled there is the depth of the rail, 0.99 m against 1.31 and 1.36 m at the shoulders. Chapter 5 describes this kind of failure. The pipeline replaces the hip depth with the preparation of Section 2.6 before it solves, and the numbers of this section are taken before that replacement. The arm angles below are read against the pitched frame, and these three values are the torso pose of Section 3.4.1.

The y-up camera-frame upper arm vector of the right arm is

$${}^{Camera'}v = {}^{Camera'}P_{14} - {}^{Camera'}P_{12} = (-0.04, -0.29, -0.12),$$

with length 0.32 m. Equation (3.3) pulls it into the parent frame and normalizing gives the local arm direction:

$${}^{L24}\mathbf{v} = \left({}^{Camera'}_{L24}\mathbf{R} \right)^T {}^{Camera'}\mathbf{v} = (0.05, -0.31, -0.05)$$

$${}^{L12}\hat{\mathbf{a}} = (0.17, -0.97, -0.16).$$

The upper arm points slightly outward, strongly downward and slightly behind the pitched torso plane, since +x for this right arm points outward to the subject's right. Figure 3.3(b) shows that pose, the right arm reaching down to the cube. The same two steps on the forearm give

$${}^{L12}\mathbf{f} = \left({}^{Camera'}_{L24}\mathbf{R} \right)^T \left({}^{Camera'}\mathbf{P}_{16} - {}^{Camera'}\mathbf{P}_{14} \right) = (0.01, -0.20, 0.03),$$

with length 0.20 m. Equations (3.20) and (3.21) read the two swing angles of the right arm (Figure 3.6):

$$t_z = \arcsin(-0.97) = -76.6^\circ$$

$$t_y = \arctan 2(0.16, 0.17) = 43.0^\circ.$$

The arm points 76.6 degrees below horizontal, and equation (3.22) undoes this swing on the forearm vector:

$${}^{L12}\mathbf{f}' = R_z(76.6^\circ) R_y(-43.0^\circ) {}^{L12}\mathbf{f} = (0.19, -0.06, 0.03).$$

Its perpendicular part points mostly along negative y. Equation (3.23) gives the twist:

$$t_t = \arctan 2(0.06, 0.03) = 59.9^\circ.$$

The forearm turns inward across the front of the body, an internal rotation. In the elbow frame, the forearm direction is

$${}^{L14}\hat{\mathbf{g}} = (0.94, 0.00, 0.34).$$

Its y component is zero, the $e_z=0$ property of Section 3.4.3 on the measured frame. The flexion of equation (3.24) is

$$e_y = \arctan 2(-0.34, 0.94) = -19.8^\circ.$$

The solved shoulder rotation and the local upper-arm vector ${}^{L24}\mathbf{v}$ complete the last transformation of equation (3.5), the elbow pose of equation (3.14):

$${}_{L14}^{L12}T = \begin{bmatrix} 0.17 & 0.95 & -0.27 & 0.05 \\ -0.97 & 0.12 & -0.20 & -0.31 \\ -0.16 & 0.30 & 0.94 & -0.05 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Its first column is exactly ${}^{L12}\hat{a}$ and its last column is the elbow position in the shoulder frame. The numeric chain closes with two checks. The product of the three matrices of equation (3.6), applied to the origin, $\begin{bmatrix} {}^{Camera'}P_{L14ORG} \\ 1 \end{bmatrix} = {}_{L14}^{Camera'}T(0,0,0,1)^T$, lands on (-0.20, 0.04, 1.19), the measured elbow position of Table 3.1, to machine precision. Equation (3.4) applied three times carries the measured wrist through the inverse chain into the elbow frame, $\begin{bmatrix} {}^{L14}P_{16} \\ 1 \end{bmatrix} = ({}_{L14}^{Camera'}T)^{-1} \begin{bmatrix} {}^{Camera'}P_{16} \\ 1 \end{bmatrix}$. The result is (0.19, 0.00, 0.07), whose length is the forearm length 0.20, and dividing by that length reproduces the ${}^{L14}\hat{g}$ used for the flexion.

The left arm runs through the same mirrored solver and gives a shoulder azimuth of 76.1 degrees, an elevation of -53.5 degrees, a twist of 108.9 degrees and an elbow flexion of -37.3 degrees, the arm hanging at the side in Figure 3.3(b). The stored pose has thirteen angular values: three root Euler angles, two shoulder swing angles, shoulder twist and elbow flexion for each arm, plus the two redundant elbow elevation fields, eleven of them independent.

Appendix H records the verification of the chapter's mathematics against synthetic exact references. A missing or misplaced landmark leaves that arm without a state. Chapter 5 supplies an additional geometric constraint from the object Chapter 4 tracks. Chapter 6 carries the finished state into the reconstruction.

Chapter 4: Object Tracking and Scene Reconstruction

Chapter 2 left the marker branch at its raw product: each detected marker's pose relative to the camera. This chapter builds that pose into an independent branch that describes the carried object in the same calibrated metric scene. Appendix E covers the conversion of a detection into a pose.

4.1 World Anchoring and Camera Placement Invariance

The static calibration of Section 2.2.2 fixed the world frame at the desk marker and the other quantities of the scene. Every pose the detector returns is expressed relative to the camera, which is the wrong frame for a reconstruction: the same cube on the same spot gives a different camera-relative pose when the camera moves to another corner of the room. The object must instead be expressed relative to the desk marker's own frame, the world frame W . Figure 2.5 draws the three frames of the chain: the camera frame C , the world frame W at the desk marker, and the object frame O , which moves with the cube.

Figure 4.1 shows the object marker as the camera sees it at four moments of the task. On each frame the outline and the axes drawn on the cube are the pose the detector returns for its marker in the camera frame. The axes on the desk marker show the per-frame detected estimate of the world frame W , while the reconstruction uses the frozen calibrated anchor.

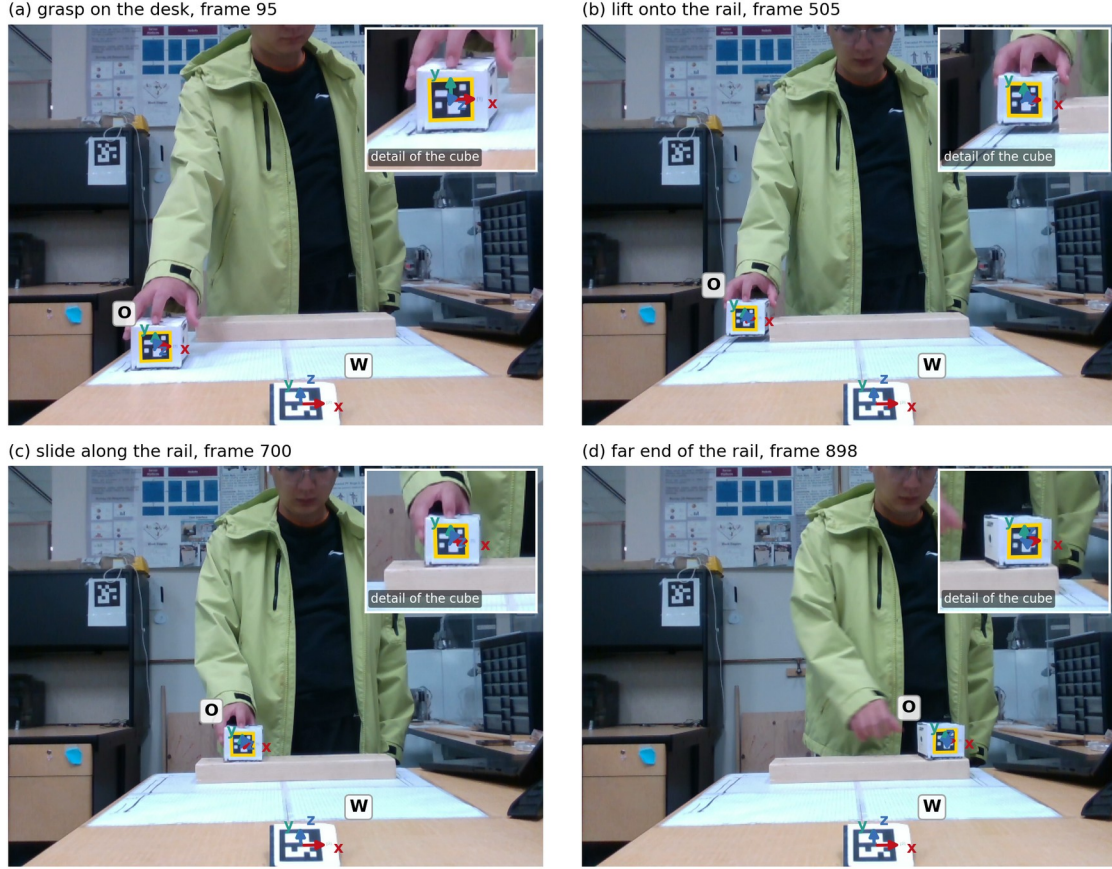


Figure 4.1. The object marker: (a) on the desk, frame 95; (b) lifted, frame 505; (c) sliding, frame 700; (d) the rail end, frame 898. The wall marker is omitted.

The marker branch writes each pose as the 4 by 4 homogeneous transformation of Chapter 3. The calibrated anchor ${}^{Camera}_{World}T$ maps a point of the world frame into camera coordinates, and the per-frame object pose ${}^{Camera}_{Object}T$ maps a point of the object frame into camera coordinates. Both are available on any frame on which the cube is seen. The chain must deliver ${}^{World}_{Object}T$, the object as seen from the world, and only one path connects the two frames:

$${}^{World}_{Object}T = {}^{World}_{Camera}T {}^{Camera}_{Object}T. \quad (4.1)$$

The first factor is the inverse of the calibrated anchor, ${}^{World}_{Camera}T = \left({}^{Camera}_{World}T \right)^{-1}$, which equation (3.4) gives as a transpose and one matrix-vector product. Its translation column is the camera's own position in the world frame, so the camera pose in the reconstructed scene

comes from the anchoring without a separate measurement. Chapter 6 places the sensor in the Unity scene at that pose.

The chain makes the reconstruction independent of the location where the camera is set up. Both measured transformations change when the camera moves, but the camera frame enters equation (4.1) twice, as the described frame of the inverted anchor and as the reference frame of the object pose. Those two labels are adjacent in the product and cancel, so the remainder depends only on the cube's position relative to the desk. They cancel only if the anchor and the object pose are measured from the same camera position. On a recording the anchor is the frozen one of Section 2.2.2. The reconstruction is therefore invariant to a camera placed differently before the calibration window. A camera moved after that window breaks the invariance. The camera stands on a fixed support at the desk edge for the whole session. The world frame and the Unity frame are defined in Section 2.2.1, and Chapter 6 converts between them before anything is drawn.

4.2 Filtering and Cleaning the Object Track

The track that Section 4.1 delivers holds one entry per frame in the world frame, with a flag that records the marker's detection status. It passes a signal filter, the Pipeline B counterpart of the landmark filtering of Section 2.5, and then a single cleaning step against marker failure.

The filter runs the three stages of Section 2.5 over the track, with rotation added to each and with its own parameters, among them a bridging limit of eight frames rather than five. A spiked pose solution is wrong as a whole, so the despiking test on position drops its rotation along with its position. Bridging turns along the shortest path between the endpoint rotations. A run at either end of the recording holds the first or the last valid pose, and an interior run longer than the bridging limit holds the valid pose before it. The smoothing stage runs a Savitzky-Golay filter on position [57]. On rotation the filter takes a rolling chordal mean, equation (2.1) applied to the world-frame object rotations over a short window.

The detection flag passes through every stage unchanged. A row the marker was never seen on stays marked as not detected, and a despiked sample keeps its incoming flag

while carrying the bridged pose. The raw track is kept unmodified in its own output.

Figure 4.2 shows the track before and after the filter.

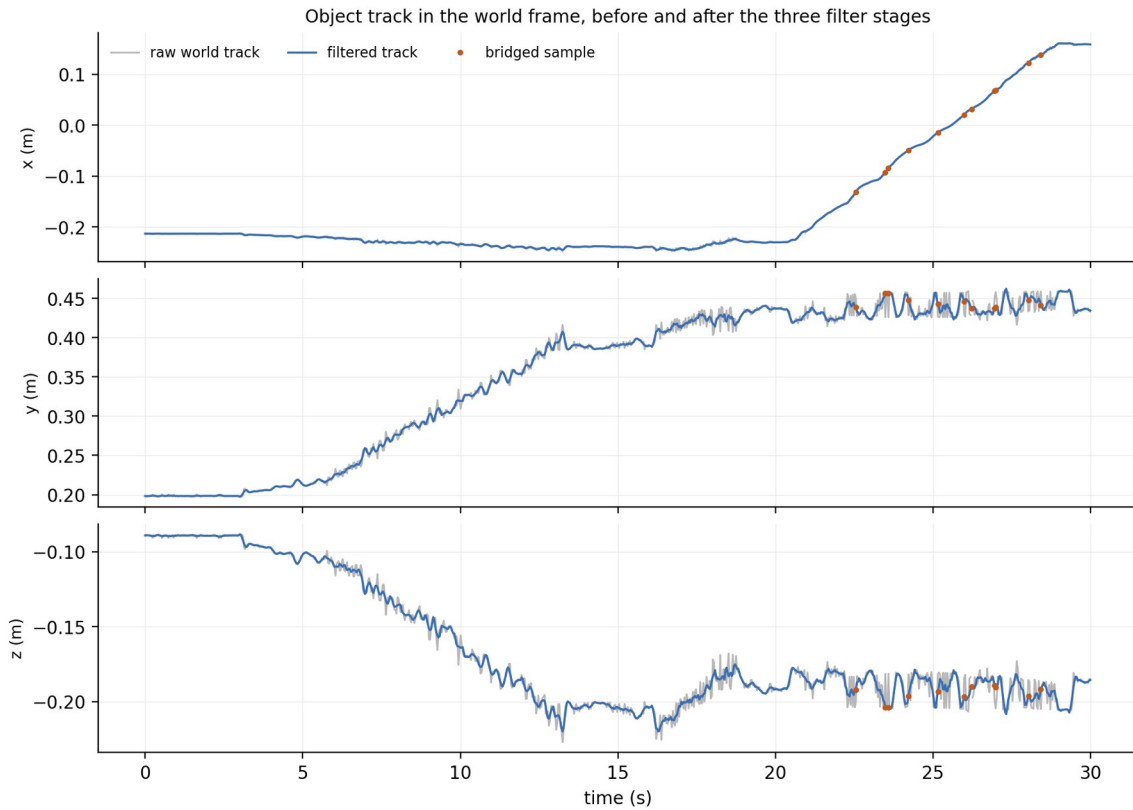


Figure 4.2. The object marker origin in the world frame, one panel per axis: grey detected, blue filtered, orange the eleven bridged rows, ten despiked and one undetected.

On the recording the hand never covers the marker, and the detector returns a pose on 899 of the 900 frames. The filter bridges that one undetected row along with the ten the despiked test emptied. The held case of the filter, which the recording does not show, has a physical cause. A hand closing over the marker leaves the detector with nothing for longer than the bridging limit, so the track stays where the cube was last seen. Chapter 9 shows such a loss on the loop recording.

The branch ends with two cases, and neither clears the detection flag. A detected row beside a held run is smoothed together with that run, so its pose can drift from the detection on its own frame while it still carries the detected flag. A partly covered marker, which the recording does not show either, may decode into a pose that is not right.

The cleaning step at the end of the branch answers both cases. No later stage carries marker logic of its own: the recovery of Chapter 5, the streaming of Chapter 6, and the evaluation of Chapter 7 read the same cleaned track, and Section 8.2 gives the timing of the cleaning step in the live path.

The rule is physical, carries no tuning to a particular recording, and walks the track forward. It keeps a sample only if the marker was detected, the pose is finite, and the step from the last kept sample is no larger than the task's slow movement can produce in the time between the two:

$$\begin{aligned} \left\| {}^{Unity}P_{Object\ ORG}(i) - {}^{Unity}P_{Object\ ORG}(last) \right\| &\leq v_{max}(t_i - t_{last}) \\ angle \left({}^{MappedMarker(last)}_{MappedMarker(i)} R \right) &\leq w_{max}(t_i - t_{last}). \end{aligned} \quad (4.2)$$

Equation (4.2) uses the Unity frame of Section 2.2.1, because the rule reads the Unity copy of each row. That frame exchanges two coordinates of the world frame, which changes no distance and no relative angle, so the rule bounds the same two quantities in either frame. In that frame the object's own axes are the mapped marker axes, which Chapter 6 obtains by applying that exchange to both sides of the object rotation. The rule forms the second bound from the two orientations of the Unity copy,

$\left({}^{Unity}R_{MappedMarker(last)} \right)^T {}^{Unity}R_{MappedMarker(i)}$, in which the reference frame cancels and leaves the relative rotation between the two samples.

The two bounds are rates rather than fixed steps: v_{max} bounds the linear step and w_{max} the angular step, each an allowance per unit of elapsed time between the candidate and the last kept sample. A real occlusion, during which the cube does travel, therefore leaves the rule ready to accept the marker when it is detected again. Measuring against the last kept sample rather than the previous row keeps one bad pose from dragging the track behind it. Both bounds carry a margin of about a factor of two over the largest step the clean parts of the recording show, so the rule is a guard.

A sample that fails the test has its pose columns blanked and its detection flag cleared, and so does every row the marker was never detected on. A later stage reads filtered poses on the kept rows and blanks elsewhere, and on a blanked row it holds the last kept pose and flags it as held.

4.3 Worked Example

The transformation chain runs on frame 533 in the scene image of Figure 2.5, the worked-example frame of Chapter 3, where the subject holds the cube against the rail and its marker is cleanly detected. The cleaning step runs on frame 786, the one frame of the recording on which the marker is not detected.

Vectors and matrices are printed to two decimals. Averaging the ten desk rotations of the calibration window element by element gives

$${}_{World}^{Camera}\bar{M} = \begin{bmatrix} 1.00 & 0.00 & -0.02 \\ -0.02 & -0.48 & -0.88 \\ -0.01 & 0.88 & -0.48 \end{bmatrix},$$

which is not a rotation matrix, since its determinant is not exactly one and its columns are only nearly perpendicular. Equation (2.1) projects it back onto the nearest rotation, and at the precision printed here no entry changes, because the ten input rotations disagree so little. With the mean of the ten translations, the projected rotation forms the frozen anchor ${}_{World}^{Camera}T$. That anchor puts the calibrated desk marker 55.3 cm in front of the camera and 18.7 cm below it.

The same calibration gives the scene description of Section 2.2.2. Gravity, taken from the wall marker, is the unit direction ${}^{World}\hat{g} = (-0.02, 0.53, 0.85)$, at 32.0 degrees to the world z axis, which is the tilt of the desk marker card on its stand. The world origin sits 3.8 mm below the horizontal tabletop model of Section 2.2.2.

Transposing the anchor rotation gives ${}_{Camera}^{World}R$, and applying it to the negated translation gives the inverted anchoring transformation of equation (3.4),

$${}^{World}P_{Camera\ ORG} = - {}_{Camera}^{World}R {}^{Camera}P_{World\ ORG} = (-0.01, -0.39, 0.43) m,$$

which puts the camera 58.4 cm from the world origin and 15.4 cm above the same tabletop model.

The chain of equation (4.1) runs on the scene image. The object marker is detected there, and its origin in the camera frame is

$$^{Camera}P_{Object\ ORG} = (-0.21, 0.16, 1.02) m.$$

The cube is therefore 106 cm from the camera, on the rail in front of the subject. Multiplying the inverted anchoring transformation by that object transformation gives one expression for each block: the transposed anchor rotation multiplies the object rotation and the object translation, and the translation block also includes the camera position:

$$^{World}P_{Object\ ORG} = ^{World}R_{Object}^{Camera} ^{Camera}P_{Object\ ORG} + ^{World}P_{Camera\ ORG}.$$

Adding the camera position to the transformed object translation gives the world translation, with the rotation product beside it:

$$^{World}P_{Object\ ORG} = (-0.24, 0.43, -0.19) m$$

$$^{World}R_{Object} = \begin{bmatrix} 1.00 & 0.01 & 0.07 \\ 0.05 & 0.52 & -0.85 \\ -0.05 & 0.86 & 0.52 \end{bmatrix}.$$

Projected onto $^{World}\hat{g}$, the marker centre is 6.3 cm above the tabletop model. The cube is held against the rail at that instant, and the rail height measured with a tape puts the cube centre 7.3 cm above the tabletop. The comparison uses marker-centre height in place of cube-centre height for this mounting and orientation. The sources of the discrepancy have not been separated.

For the cleaning step, the object marker is seen on frames 785 and 787, at world x 3.0 cm and 3.4 cm, and the detector returns nothing on frame 786 between them. The gap is one frame, inside the bridging limit of eight, so the filter bridges it. The bridged position sits 1.0 mm from the mean of the two detections beside it.

The rule then walks the filtered track. Frame 786 is 3.3 mm and 1.2 degrees from the last kept sample, frame 785, well inside the allowance. The marker was never seen on it, so the row is blanked. Frame 787, two frame intervals later, is 4.9 mm from frame 785 and is kept. The rule rejects no detected sample anywhere in the recording.

The frozen scene fixes the desk, gravity, tabletop, camera and cube edge of Table 2.1 in one frame. The cleaned track gives a measured cube pose, a marked gap where the

marker was missed, and the marker identity on each row. A grip ties the branches together in Chapter 5, constraining the carrying arm. Chapter 6 streams the position and orientation. Section 7.2 evaluates the track against the physical route; a line fit to the slide samples describes spread, not accuracy. No second measurement chain checks the streamed orientation.

Chapter 5: Pose Recovery During Tracking Failure

Chapter 3 solves the joint angles from a fully measured chain, and Chapter 4 tracks the carried object in the same scene. Neither handles a frame with a missing wrist or a wrist whose depth belongs to the cube. When body landmarks become unreliable, object observations provide an additional geometric constraint on the region the arm can occupy. This layer applies that constraint between Chapter 2 and the solve of Chapter 3, treating occlusion as missing data on a torso that is assumed to be measured (Section 5.1).

The arm chain has calibrated upper-arm length L_1 and forearm length L_2 , and a missing landmark is rebuilt from the measured landmarks and the carried state in three cases.

- In Case A a missing wrist is placed at the forearm length from a measured elbow, along the remembered forearm direction (Section 5.3).
- In Case B the wrist of a holding hand is placed from the measured object pose of Chapter 4 and the hand-object offset h (Section 5.4).
- In Case C a missing elbow is placed by two-link inverse kinematics on the two calibrated lengths, from a measured shoulder and a measured or rebuilt wrist (Section 5.5).

Where no case anchors a landmark, the joint holds its last angle. The solve produces seven joint groups, the root and three per arm, and each leaves the layer in one of three states. A group is measured when every landmark it needs was measured and unflagged on this frame, held when it keeps the value it last had, and constrained when it was solved from a rebuilt landmark or with its change limited.

The states travel with the angles into Chapters 6 and 7. Figure 5.1 draws this flow, from the Chapter 2 landmarks and the Chapter 4 object pose to the solve.

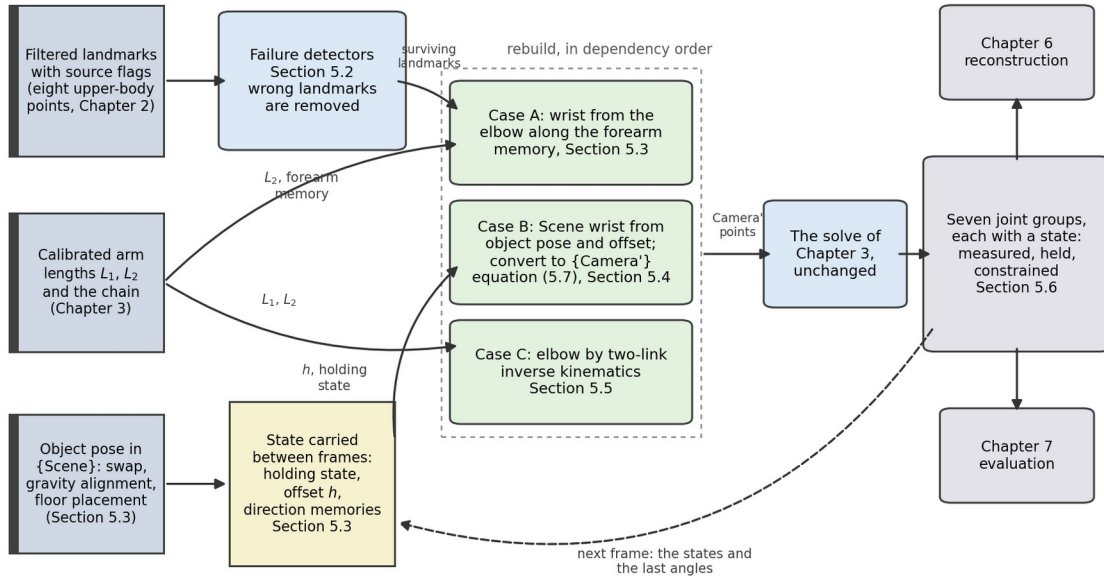


Figure 5.1. Information flow of the recovery layer in the Scene formulation of Section 5.3. Inputs from Chapters 2 to 4, the detectors, the three rebuild cases and the solve.

5.1 Assumptions and Parameters

The recovery rests on six assumptions, and Chapter 9 takes up the consequences of their failure. The recording is mostly clean: every landmark is measured on most frames, and a failure is a short window the recovery fills. The torso is measured on every frame, so the two hips and the two shoulders are taken as correct, landmark 24 above all, since it is the origin of the root frame of Chapter 3. A lost root frame is outside the scope of this thesis.

The torso assumption needs one preparation of the recording. The hip landmarks take the depth of the rail in front of the pelvis, and the preparation of Section 2.6 replaces that depth outside this layer. The plain, direction-memory and object-assisted solves of Chapter 7 all rest on this preparation, and the hold-last solve, which keeps the last accepted joint angles, does not.

The grasp is rigid within one grip episode: the wrist keeps a fixed position in the mapped marker frame $\{\text{MappedMarker}\}$ that Chapter 6 establishes, and a regrasp starts a new episode and a new offset. The object marker stays visible while the hand is hidden, since Case B needs a measured object pose on the frame it repairs. Where the marker is lost too, Case A or the held angle takes over. The upper arm and the forearm keep their

calibrated lengths through the recording. The subject moves slowly, as the task of Chapter 2 is defined, so a segment direction changes little between frames and the direction memories of Section 5.3 can rely on it.

The recovery needs parameters of three kinds. The calibrated constants are the upper-arm and forearm lengths, taken from the clean frames at the start of the recording or from an earlier calibration. The fixed mapping into {Camera'} of Section 5.3 is a calibrated constant as well. The design values were fixed before the recording. They are the detector thresholds, gains, radii, runs, bounds and limits, each defined with its rule in Sections 5.2 to 5.5. The states are estimated on clean frames and carried forward: the hand-object offset per grip episode, the holding state of each hand, and the direction memories of the two arms.

5.2 Landmark Tracking Failures

A landmark can be wrong in two ways, and only one of them is announced. The source flag of Section 2.3 reports a usable point, a visibility score below the gate of that section, or a depth window with no valid depth sample (Appendix D). The last two are declared failures, already treated as missing. The second way is silent: the pixel is correct and the visibility score is high, but the depth behind it belongs to a hand, to the cube, or to the desk. The point is then lifted tens of centimetres from the body, and the visibility score cannot show it.

Geometry shows it. Over a few frames the body is rigid: a limb segment cannot change length, the trunk cannot fold, and a landmark cannot jump across the scene. Each statement becomes a test per frame, and a frame that fails one carries a wrong measurement whatever the source flag says. The object supplies a sixth test: a hand that holds the cube cannot have its wrist far from it. Figure 5.2 draws the six tests.

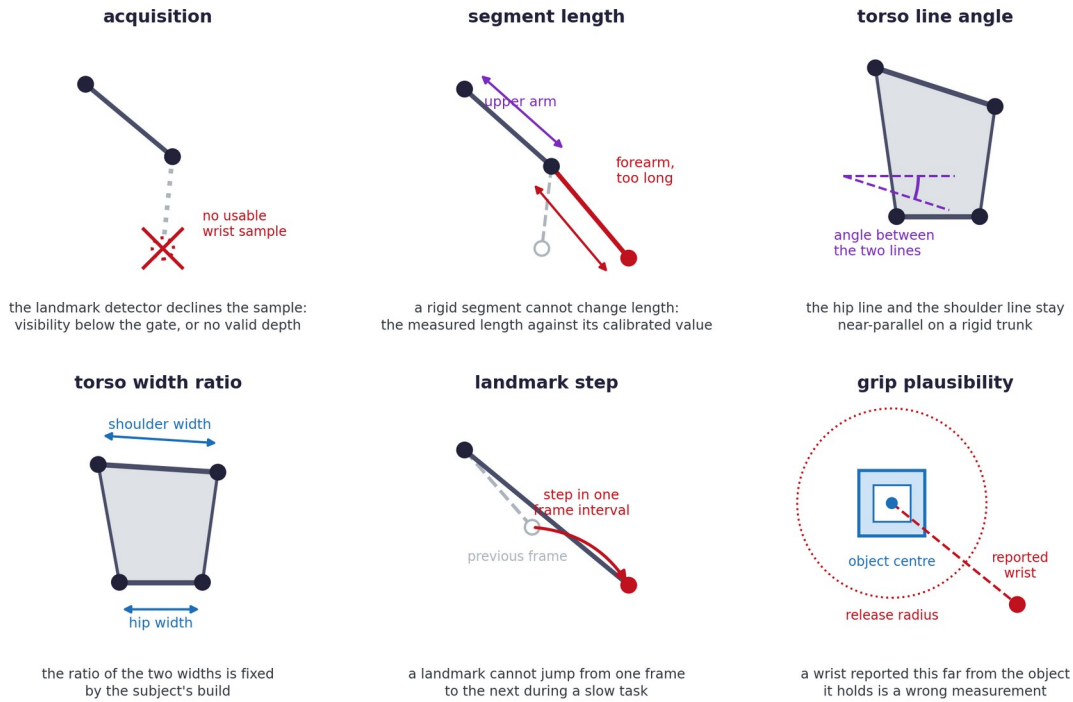


Figure 5.2. The six failure detectors, one panel each: acquisition, segment-length and line-angle in the upper row, width-ratio, landmark-step and grip-plausibility in the lower. Red marks the firing condition.

The acquisition detector is a report rather than a test: it fires on the source flags above. The four geometry detectors turn those statements into numbers, the trunk statement giving two. The segment-length detector compares the upper arm and forearm with the median of that segment over the clean frames, and fires at a departure of 35 percent. The line-angle detector measures the angle between the hip line and the shoulder line, close to parallel while the trunk is rigid, and fires above 22 degrees. The width-ratio detector compares the ratio of shoulder width to hip width with its clean median, and fires at a departure of 20 percent. The landmark-step detector measures the distance each landmark moved between two consecutive usable frames, and fires above 5 centimetres.

The grip-plausibility detector uses the object. A wrist reported more than 35 centimetres, the release radius of Section 5.3, from the object centre while that hand holds the cube is a wrong measurement and not a release. That bound is a design choice for the cube and grip of this task, and its suitability elsewhere has not been established. The four geometry thresholds are properties of a human body and of a slow task, each set at the 99th percentile of the clean frames of a separate reference recording plus a safety margin. That

recording is clean for the right arm and the torso, while its left arm has a defect that fires the segment-length detector. Widening the tolerance until it stops firing would calibrate the threshold to a defect, so it stays where the clean statistics put it.

The per-frame results are combined into three failure masks, one per arm and one for the torso, each set whenever any of its detectors fires and then cleaned twice. Gaps of at most five frames are closed, and runs shorter than five frames are dropped. The grip-plausibility test runs afterwards, because it needs the holding state of Section 5.3, and whatever it flags is added to the arm mask.

A flagged arm has its elbow and wrist landmarks removed before the solve, so the detectors turn a wrong measurement into a missing one. The layer reports the torso mask and rebuilds no torso landmark, since under the assumption of Section 5.1 the torso is measured on every frame, after the hip depth preparation named there. The two torso detectors only show the frames in which that assumption fails.

5.3 Grip Episodes and the Hand-Object Offset

This chapter writes the three landmarks of one arm as p_{sh} , p_{el} and p_{wr} for its shoulder, elbow and wrist. On the right side these are the points Chapter 3 numbers p_{12} , p_{14} and p_{16} , and the left arm carries the same names. Every symbol is defined where it first appears, with its frame given there. In the displayed mathematics a left superscript names the frame a quantity is expressed in, and a left subscript the frame a rotation or a transformation describes.

While a hand grips the cube and does not slide on it, the wrist is a fixed point of the object. The wrist is not on the marker: it sits on the side of the cube, at an offset that changes with every grasp and that nothing in the setup measures directly. The layer estimates it on the frames where both the wrist and the marker are measured, and carries it into the frames where the wrist is lost. Chapter 7 evaluates a wrist reconstructed this way against the unmasked reconstruction on frames where the wrist was measured and then withheld (Section 7.4.1), and against manually labelled wrist positions in the natural failure windows (Section 7.4.2).

On a frame with a measured object pose, the layer receives an origin $^{Scene}o$ and an orientation $^{Scene}_{MappedMarker}R$. Chapter 4 supplies that pose in {World}, and Sections 6.2 and 6.4 derive the chain that carries it into the display frame {Scene}. The chain begins with the swap S , which also turns the object's own marker frame into the left-handed mapped marker frame {MappedMarker} about the same marker centre. It closes with the proper gravity-alignment operator G and the floor translation t_f of height d , which Section 6.4 fixes from the recording's calibrated tabletop position and the drawn desk height. The same chain, applied after the flip F of equation (3.1) and the calibrated sensor pose of Chapter 4, gives the fixed transformation $^{Scene}_{Camera'}T$ from {Camera'} to {Scene}. Its inverse is $^{Camera'}_{Scene}T$. Writing $^{MappedMarker}h$ for the position of the wrist in the mapped marker axes, the wrist of a holding hand satisfies

$$^{Scene}p_{wr} = ^{Scene}o + ^{Scene}_{MappedMarker}R ^{MappedMarker}h, \quad (5.1)$$

where $^{MappedMarker}h$ is constant for as long as the grasp is rigid.

The layer computes the offset directly on a frame where both the wrist and the object are measured, with the wrist carried into {Scene} by $^{Scene}_{Camera'}T$. The displacement from the origin to the wrist is pulled into the mapped marker axes by the transpose, the operation of equation (3.3),

$$^{MappedMarker}h_k = \left(^{Scene}_{MappedMarker}R_k \right)^T \left(^{Scene}p_{wr,k} - ^{Scene}o_k \right), \quad (5.2)$$

where k is the frame index. The same floor translation occurs in both positions in equation (5.2) and cancels from their difference. It cancels a second time in the inverse transformation, so the implementation, which forms the offset before the display adds that translation, delivers the identical Camera' point to the solver. Figure 5.3 draws the two equations, on a clean frame and on a later frame where the object has moved and turned and the wrist is hidden.

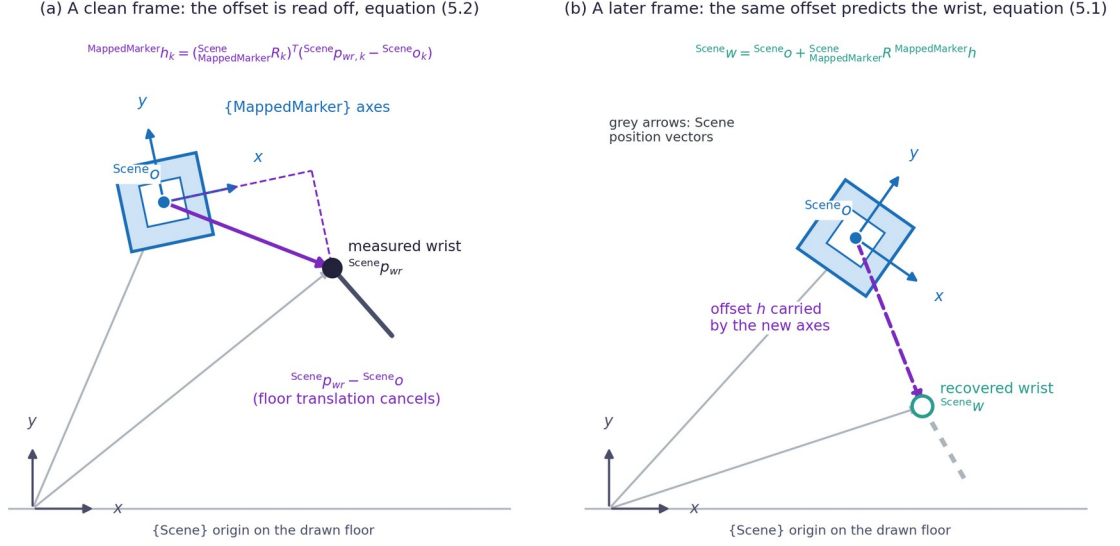


Figure 5.3. The hand-object offset. (a) A clean frame, with the wrist and marker origin measured, equation (5.2). (b) The offset carried by the later object pose, equation (5.1).

Each clean frame contributes one observation $^{MappedMarker}h_k$ of the same constant. A least squares fit of a single $^{MappedMarker}h$ to N clean frames minimizes the total squared distance between the measured wrists and the wrists equation (5.1) predicts. Because $^{Scene}_{MappedMarker}R$ is orthonormal and leaves lengths unchanged, that objective has three equal forms:

$$\begin{aligned} & \sum_k \left| ^{Scene}p_{wr,k} - ^{Scene}o_k - ^{Scene}_{MappedMarker}R_k ^{MappedMarker}h \right|^2 \\ & \sum_k \left| ^{Scene}_{MappedMarker}R_k \left(^{MappedMarker}h_k - ^{MappedMarker}h \right) \right|^2 \\ & \sum_k \left| ^{MappedMarker}h_k - ^{MappedMarker}h \right|^2. \end{aligned}$$

The rotation therefore drops out and the fit reduces to the mean of the observations,

$$^{MappedMarker}h = \frac{1}{N} \sum_k ^{MappedMarker}h_k. \quad (5.3)$$

A single offset per recording will not do, because each new grasp puts the hand somewhere else on the cube. Each grasp is one grip episode, the unit the rest of this chapter and Chapters 8 and 9 work in, and the offset belongs to the episode. Within one long episode the hand also shifts a little, so the offset is estimated with a short memory

instead of fitted once. On each frame that contributes a clean observation, the estimate moves towards that observation by a small gain c ,

$$^{MappedMarker}h_{new} = (1 - c)^{MappedMarker}h_{old} + c^{MappedMarker}h_k. \quad (5.4)$$

Equation (5.4) is a first-order recursive low-pass update, a smoother that can run on a stream of samples [50]. It mixes a little of the new observation into the previous estimate, so the estimate follows the current offset and forgets an old one. The gain of 0.02 weights it towards about the last fifty frames.

The estimator has three further rules. A frame with no clean observation leaves the estimate untouched, so through a failure window the value in use is the offset from just before tracking was lost. The estimate restarts with each grip episode. Before five clean observations have initialized the local estimate, the mean of equation (5.3) over that episode stands in. This fallback uses the recording-wide fit if the episode has fewer than fifteen clean observations; once the local estimate is available, it takes priority. Flagged arm frames do not enter the local updates or episode means. The recording-wide fallback is fitted over the frames the instantaneous test of this section accepts, so it is gated by the conditions of that test rather than by the cleaned arm failure mask.

The estimate follows a shift of the grip with a lag set by the gain, so the value frozen at a failure can still carry the offset of the last clean frames. The grasp is also not always as rigid as Section 5.1 assumes. On the loop recording the hand wraps around the cube and shifts on it. Over the frames where the left hand holds the cube, the per-frame offset observations scatter with a standard deviation of 7.0, 2.7 and 6.3 centimetres on the three mapped marker axes. A non-rigid grip and the lag of the estimator are therefore two conditions under which the constraint the object supplies is itself wrong, and Section 7.4.2 reports their cost on the natural failure windows.

Equation (5.1) applies only to a hand that holds the object, so the layer must decide which hand that is. On a frame where everything is measured the decision takes five conditions together. The marker is measured, the object is away from its resting place, and the wrist is measured and unflagged. The wrist lies within the hold radius of the object centre, and the forearm of that arm measures within 30 percent of its own median length. The last

condition rejects frames where the elbow and the wrist take their depth from the same near surface and collapse onto each other. The test needs a measured wrist, so the holding decision is carried as a state per hand, with rules for entering and leaving it.

The hand enters the holding state after the instantaneous test passes on five consecutive frames. It stays in that state through a wrist failure for as long as the object is still being carried. It leaves the state when the object comes to rest, or when a cleanly measured wrist sits farther than the release radius from the object centre, in either case over five consecutive frames. The release radius of 35 centimetres is wider than the hold radius of 25 centimetres, which stops the state flickering at the boundary. The run of frames between entry and exit is one grip episode. Both hands may hold at once, and a handover is two grip episodes that overlap. Figure 5.4 draws the state and its rules.

In the offline preparation the state is set after the run that confirms it: an entry starts from the first frame of its five-frame run, and a confirmed exit clears the five frames of the exit run. The warm-up mean of an episode is likewise taken over the whole episode. Section 8.2 describes the version that decides with a delay instead.

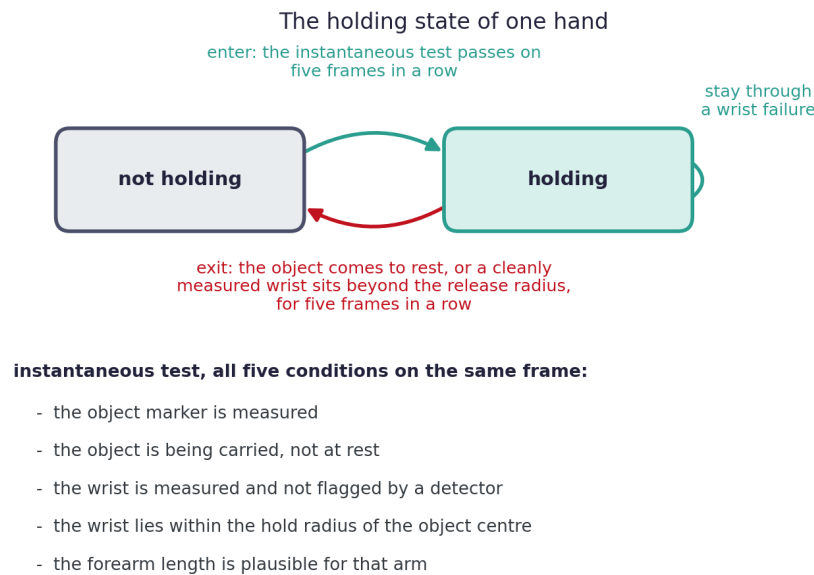


Figure 5.4. The holding state of one hand: the five conditions of the instantaneous test, and the rules for entering, staying in and leaving the state.

The arm that is not holding gains nothing from the object, and its failures fall back on the direction memory. For every body segment the layer keeps a running average of the unit direction from the landmark nearer the trunk to the one farther from it. Writing $\hat{u}$ for the memory of the upper arm and $\hat{u}_{meas}$ for the direction measured on the current frame, the update on a frame where both ends are measured is

$${}^{Camera'}\hat{u}_{new} = \frac{(1 - \alpha){}^{Camera'}\hat{u}_{old} + \alpha {}^{Camera'}\hat{u}_{meas}}{\left| (1 - \alpha){}^{Camera'}\hat{u}_{old} + \alpha {}^{Camera'}\hat{u}_{meas} \right|}, \quad (5.5)$$

with the gain $\alpha = 0.30$, which weights the memory towards the last few frames. The forearm memory $\hat{f}$ follows equation (5.5) as well. The denominator scales the result back to unit length. The memory is updated only from measured landmarks, so nothing it holds was ever reconstructed.

Case A follows from this memory. A missing wrist whose elbow is measured is placed at the calibrated forearm length from the elbow, along the remembered forearm direction:

$${}^{Camera'}p_{wr} = {}^{Camera'}p_{el} + L_2 {}^{Camera'}\hat{f}. \quad (5.6)$$

A missing elbow whose shoulder is measured is rebuilt by equation (5.6) with L_1 and $\hat{u}$, and Section 5.5 covers the case where the wrist is known and the elbow is placed from it instead. Direction memory rebuilds are allowed for at most forty-five consecutive frames on which the affected landmark farther from the trunk is unmeasured. The rebuild that Section 5.5 uses near full reach is one of them. The counter tracks that landmark, rather than the age of every segment-direction update. The two-link solve of Section 5.5 uses the remembered upper-arm direction without that counter check. When the counter prevents a rebuild, the affected joints hold their last angle, so a gap that nothing anchors is carried in angle space.

5.4 Wrist Recovery From the Object Pose

Case B applies to the holding hand. On a failure frame of a holding hand, the lost wrist is replaced by the point that equation (5.1) predicts from the current object pose and the current offset:

$${}^{Scene}w = {}^{Scene}o + {}^{Scene}_{MappedMarker}R {}^{MappedMarker}h. \quad (5.7)$$

The estimate ${}^{Scene}w$ is a function of a same-frame measurement, and only the offset is carried from the past, on the short memory of Section 5.3. A wrist placed from its last direction drifts the longer the window lasts, while the object-derived wrist follows the cube and does not age. Case B therefore takes priority over Case A whenever the hand holds the cube and the marker is measured.

Equation (5.7) does not by itself recover the wrist. It supplies one additional geometric constraint, from a measurement the failing branch does not have. The arm that leaves the layer depends on the current offset, the calibrated lengths, the chain constraints, the two-link geometry of Section 5.5 and the state carried from earlier frames.

An offset longer than 30 centimetres does not describe a hand gripping a cube of a few centimetres, so the layer discards an estimate built on it and the frame falls back to Case A. The grip-plausibility detector of Section 5.2 removes a measured wrist that sits too far from the object its hand holds, and the object estimate takes over. A release is recognized only by the exit rule of the holding state.

The wrist estimate of equation (5.7) is formed in {Scene}, including the gravity alignment and floor placement of Section 6.4, while the solve works in the y-up camera frame {Camera'} of Chapter 3. The fixed rigid body transformation ${}^{Camera'}_{Scene}T$ of Section 5.3 therefore carries the estimate into that frame before the solve. It undoes the full scene mapping and changes no length and no angle.

5.5 Elbow Recovery by Two-Link Inverse Kinematics

Case C starts from a shoulder and a wrist. The upper arm and the forearm are two rigid links of known length, so placing the elbow is the two-link inverse kinematics problem of a planar manipulator lifted into three dimensions [44]. The two lengths are the calibrated L_1 and L_2 of Section 5.3. The construction below is written with the measured wrist ${}^{Camera'}p_{wr}$. On a failure frame of a holding hand the object-derived estimate ${}^{Camera'}w$ of equation (5.7) takes its place, once carried into the y-up camera frame, and nothing else changes. The elbow is any point that satisfies both

$$\left| {}^{Camera'}p_{el} - {}^{Camera'}p_{sh} \right| = L_1, \left| {}^{Camera'}p_{el} - {}^{Camera'}p_{wr} \right| = L_2. \quad (5.8)$$

The two conditions place the elbow on a sphere of radius L_1 about the shoulder and on a sphere of radius L_2 about the wrist. The spheres meet in a circle when their centres are closer than the sum of the radii and farther apart than the difference, and that circle is the set of all elbows consistent with the two endpoints.

Let b be the vector from the shoulder to the wrist, r its length, and $\hat{b}$ the unit vector along it. The triangle at the shoulder, the elbow and the wrist has sides L_1 , L_2 and r , all known. The law of cosines fixes its angle j at the shoulder, between the upper arm and the shoulder-to-wrist line:

$$\cos j = \frac{L_1^2 + r^2 - L_2^2}{2L_1r}. \quad (5.9)$$

The perpendicular from the elbow to the shoulder-to-wrist line has its foot at a distance $L_1 \cos j$ from the shoulder, and length $L_1 \sin j$. Neither changes when the triangle turns about that line, so the foot is the centre p_c of the circle and the perpendicular is its radius r_c :

$${}^{Camera'}p_c = {}^{Camera'}p_{sh} + L_1 \cos j \hat{b}, r_c = L_1 \sin j. \quad (5.10)$$

The circle lies in the plane through p_c perpendicular to $\hat{b}$. Let n_1 and n_2 be any two unit vectors that are perpendicular to each other and to $\hat{b}$, and let s be the turn about the shoulder-to-wrist line that picks out a point of the circle. Every point of the circle is then

$${}^{Camera'}p_{el} = {}^{Camera'}p_c + r_c (\cos s {}^{Camera'}n_1 + \sin s {}^{Camera'}n_2). \quad (5.11)$$

The angle s is the one degree of freedom the two endpoints do not determine, the swivel of the arm about the shoulder-to-wrist line, and Chapter 3 meets it as the shoulder twist that the elbow position cannot determine.

The direction memory of Section 5.3 picks the point, and here it is read without the age check of that section. It supplies the remembered upper-arm direction $\hat{u}$, and the elbow is

placed at the point of the circle nearest that direction. That point follows by removing from the memory its component along the shoulder-to-wrist line, which leaves the perpendicular part $u_{\perp}$, and scaling that part to the radius:

$${}^{Camera'}u_{\perp} = {}^{Camera'}\hat{u} - \left({}^{Camera'}\hat{u} \cdot {}^{Camera'}\hat{b} \right) {}^{Camera'}\hat{b}, \quad {}^{Camera'}p_{el} = {}^{Camera'}p_c + r_c \frac{{}^{Camera'}u_{\perp}}{|{}^{Camera'}u_{\perp}|}. \quad (5.12)$$

Equation (5.12) finds, in closed form, the point p_{el} on the circle closest to $p_{sh} + L_1 \hat{u}$, the elbow the remembered direction would give if it were reachable.

The memory is unusable when the remembered direction lies along the shoulder-to-wrist line, or when no measured frame has been seen yet. A downward direction then replaces it, since a relaxed elbow hangs downward, and the depth axis of the y-up camera frame is the third candidate. These two fallback directions are perpendicular, so one always gives a point on the circle. Figure 5.5 draws the construction.

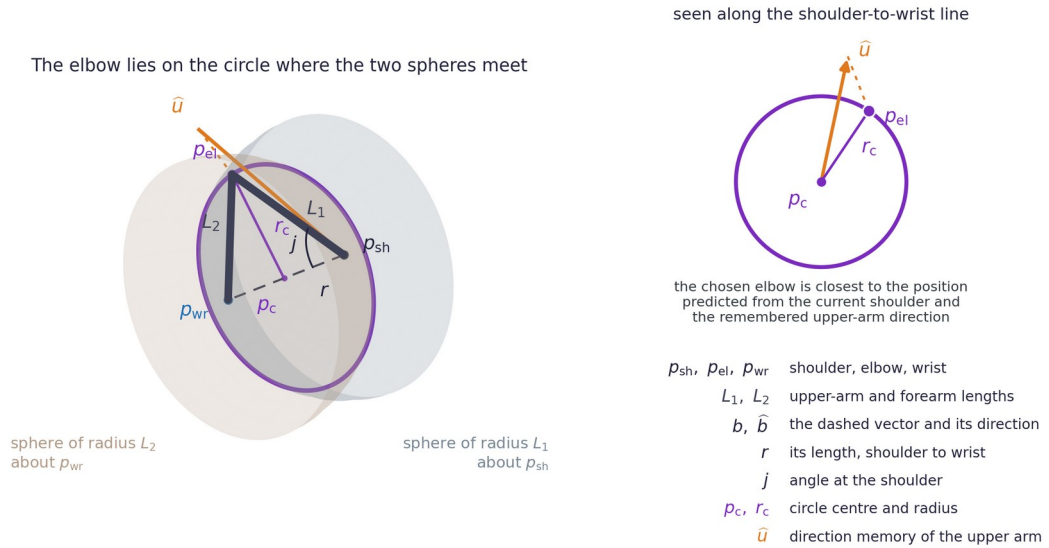


Figure 5.5. The elbow reconstruction. Spheres of radius L_1 and L_2 meet in the circle of equation (5.11). The direction memory, in orange, selects the point in the inset.

The construction is exact only while the shoulder-to-wrist distance lies between the difference and the sum of the two link lengths. Outside that range no elbow satisfies both conditions of equation (5.8), and the layer clips the cosine of equation (5.9) into its valid range instead. A wrist estimate beyond the reach of the two links clips the cosine at one,

the circle radius of equation (5.10) falls to zero, and the elbow collapses onto the shoulder-to-wrist line as a straight arm pointing at the object. A wrist closer to the shoulder than the difference of the lengths pushes the cosine above one as well, since the upper arm is the longer link, and clips at the same end. In both cases the elbow keeps the upper-arm length, while the forearm is left longer or shorter than its calibrated value.

A single configuration has no answer: a wrist estimate that coincides with the shoulder leaves the shoulder-to-wrist line without a direction, so the two-link solve returns nothing. The elbow is then placed from the shoulder along the upper-arm memory, and the groups that use it leave the layer constrained, not held. Only without a fresh memory does the arm fall back on holding its last angles.

Equation (5.9) is ill conditioned near a straight arm: the sensitivity of j to r grows without bound as r approaches $L_1 + L_2$. The elbow is therefore taken from the direction memory above 98 percent of the full reach, with the wrist still anchored at the object-derived estimate, so on those frames the forearm is not in general at its calibrated length either. The two-link solution stays as the fallback where no fresh memory exists.

The layer applies two rules to the angles after the solve. The shoulder twist is read from the part of the forearm perpendicular to the upper arm, and that part vanishes as the elbow straightens, as Section 3.4.3 sets out. A twist solved from measured landmarks keeps its previous value below 15 degrees of elbow flexion, and its group leaves the layer held. The hold is released only above 25 degrees. The hold does not reach a twist solved on a rebuilt arm: that group has already left the layer constrained, and below 15 degrees of flexion its twist is bounded only by the rate limit that follows.

The layer then limits each arm angle to 15 degrees of change between consecutive frames. While the limit is active the group is marked constrained until the limited and the solved value agree again. The limit applies to the arm angles and not to the root.

5.6 Worked Example: Frames 549 and 560

The recovery now runs on one real frame, as Section 3.5 did for the solve. The frame comes from the rail recording, under a second after frame 533 of that section, and it is the one frame of the worked examples on which something is lost, so the recovery has work

to show. From frame 550 the landmark detector stops reporting a usable right wrist for about three seconds while the hand slides the cube along the rail, and the acquisition detector of Section 5.2 fires on the right arm. The example takes frame 560, ten frames in, with the right elbow and wrist removed together and both rebuilt, as Section 5.2 sets out. It solves that frame from the state left by frame 549, the last clean wrist frame before the window. The torso landmarks and the right shoulder are reported, with the hip depth replaced by the preparation of Section 2.6. The right hand holds the cube and the marker is measured, so Case B places the wrist and Case C places the elbow. Figure 5.6 puts the two frames side by side.

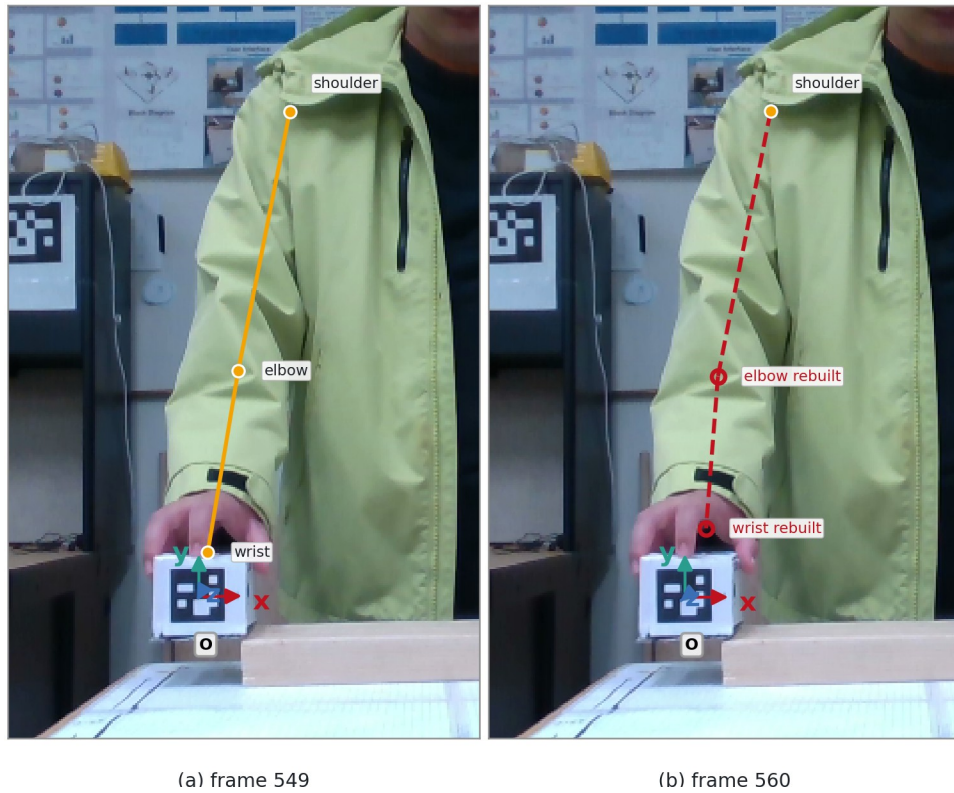


Figure 5.6. (a) Frame 549, the last clean frame of the right arm. (b) Frame 560, solved from that state, with the elbow and wrist rebuilt.

The layer receives the right shoulder in the y-up camera frame and the two lengths of the right arm calibrated on the clean frames of this recording. The Chapter 4 object pose is expressed in {Scene} by the swap, gravity alignment and floor placement of Sections 6.2 and 6.4, with $d = 71.6$ cm on this recording:

$$\begin{aligned}
{}^{Camera'}p_{sh} &= (-0.16, 0.34, 1.32) \\
L_1 &= 31.7 \text{ cm}, L_2 = 20.5 \text{ cm} \\
{}^{Scene}o &= (-0.22, 0.79, 0.45) \\
{}^{Scene}_{MappedMarker}R &= \begin{bmatrix} 1.00 & 0.03 & 0.04 \\ -0.04 & -0.09 & 0.99 \\ 0.03 & -1.00 & -0.09 \end{bmatrix},
\end{aligned}$$

The current offset is the recursive estimate of equation (5.4). Its last update came on frame 549, the last clean wrist frame before the window, and it is frozen from there on:

$${}^{MappedMarker}h = (0.01, -0.04, 0.05), |{}^{MappedMarker}h| = 0.07 \text{ m}.$$

The offset is inside the bound of Section 5.4. Equation (5.7) turns it into Scene axes and adds it to the object origin:

$$\begin{aligned}
{}^{Scene}w &= {}^{Scene}o + {}^{Scene}_{MappedMarker}R {}^{MappedMarker}h = (0.01, 0.06, 0.04) \\
&\quad + {}^{Scene}_{MappedMarker}R {}^{MappedMarker}h = (-0.21, 0.85, 0.48).
\end{aligned}$$

The wrist lands just above the marker origin.

The inverse mapping derived in Section 5.3 gives the fixed transformation ${}^{Camera'}_{Scene}T$ from {Scene} to the y-up camera frame {Camera'}. For this recording it is

$${}^{Camera'}_{Scene}T = \begin{bmatrix} 1.00 & -0.04 & 0.02 & 0.05 \\ 0.04 & 1.00 & -0.05 & -0.90 \\ -0.02 & 0.05 & 1.00 & 0.51 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Its rotation block is close to the identity, since the two sets of axes are nearly parallel here, and its last column is ${}^{Camera'}P_{SceneORG}$, the Scene origin on the drawn floor expressed in the y-up camera frame. Applied to ${}^{Scene}w$, it gives the wrist estimate the solve receives:

$$\begin{aligned}
\begin{bmatrix} {}^{Camera'}w \\ 1 \end{bmatrix} &= {}^{Camera'}_{Scene}T \begin{bmatrix} {}^{Scene}w \\ 1 \end{bmatrix}. \\
{}^{Camera'}w &= (-0.19, -0.09, 1.04).
\end{aligned}$$

The estimate lies about three centimetres from the wrist measured on frame 549.

Case C now has a shoulder and a wrist. The vector between them, its length and its unit vector are

$$\begin{aligned} {}^{Camera'}\mathbf{b} &= {}^{Camera'}\mathbf{w} - {}^{Camera'}\mathbf{p}_{sh} = (-0.02, -0.43, -0.27) \\ r &= 50.7 \text{ cm}, \quad {}^{Camera'}\hat{\mathbf{b}} = (-0.04, -0.84, -0.54). \end{aligned}$$

The wrist lies at 97 percent of the full reach of 52.2 cm, under the 98 percent guard of Section 5.5, so the two-link solve runs on a nearly straight arm. The law of cosines of equation (5.9) gives the angle at the shoulder:

$$\cos j = \frac{31.7^2 + 50.7^2 - 20.5^2}{2 \cdot 31.7 \cdot 50.7} = 0.98, \quad j = 11.0^\circ,$$

Equation (5.10) then places the centre of the circle on the shoulder-to-wrist line and fixes its radius:

$$\begin{aligned} {}^{Camera'}\mathbf{p}_c &= {}^{Camera'}\mathbf{p}_{sh} + L_1 \cos j \cdot {}^{Camera'}\hat{\mathbf{b}} = (-0.18, 0.07, 1.15) \\ r_c &= L_1 \sin j = 0.06 \text{ m}. \end{aligned}$$

The circle of possible elbows is small because the arm is nearly straight, so the free turn about the shoulder-to-wrist line can move the elbow by about twelve centimetres.

The direction memory of the upper arm was last updated on frame 549. Equation (5.12) removes from it the component along the shoulder-to-wrist line:

$$\begin{aligned} {}^{Camera'}\hat{\mathbf{u}} &= (-0.11, -0.91, -0.39), \quad {}^{Camera'}\hat{\mathbf{u}} \cdot {}^{Camera'}\hat{\mathbf{b}} = 0.98 \\ {}^{Camera'}\mathbf{u}_\perp &= {}^{Camera'}\hat{\mathbf{u}} - ({}^{Camera'}\hat{\mathbf{u}} \cdot {}^{Camera'}\hat{\mathbf{b}}) {}^{Camera'}\hat{\mathbf{b}} = (-0.07, -0.09, 0.14), \quad |{}^{Camera'}\mathbf{u}_\perp| = 0.18. \end{aligned}$$

The memory points almost along the shoulder-to-wrist line, and the projection leaves less than a fifth of its length. Scaling that part to the radius gives the elbow:

$${}^{Camera'}\mathbf{p}_{el} = {}^{Camera'}\mathbf{p}_c + r_c \frac{{}^{Camera'}\mathbf{u}_\perp}{|{}^{Camera'}\mathbf{u}_\perp|} = (-0.20, 0.04, 1.20).$$

Both conditions of equation (5.8) hold on the result: the elbow lies 31.7 cm from the shoulder and 20.5 cm from the wrist. It lies six millimetres from the elbow measured on frame 549. The placed upper arm is 0.7 degrees from the remembered direction, so the elbow has moved as little as the new wrist allows. The solve of Chapter 3 then runs on

the assembled points and reads an elbow flexion of -28.1 degrees in the convention of Section 3.4.3. The three right-arm groups leave the layer in the constrained state, since two of their landmarks were rebuilt, and so does the root, whose hips carry a replaced depth.

The layer hands the solve of Chapter 3 an assembled set of landmarks on every frame, with each joint group tagged measured, held or constrained. Chapter 6 transmits these states. The value of the constraint depends on the object being visible and on the conditions Section 5.4 lists. Section 7.4 evaluates it, and Chapter 9 reads the two outcomes. Chapter 8 describes the causal version of the layer.

Chapter 6: System Integration and Graphical Reconstruction

Chapters 2 to 5 leave two measurement branches in one calibrated scene. The landmark branch of Chapters 3 and 5 delivers, for every frame, a pelvis position in the y-up camera frame {Camera'}, thirteen joint angles, and the state the recovery layer gave each joint group, measured, held or constrained. The marker branch of Chapters 2 and 4 delivers a cleaned track of the carried object in the world frame. This chapter renders both in one Unity scene.

6.1 Communication Architecture

Unity performs no measurements and runs no solver. The receiver composes the transforms of Sections 6.2 to 6.4 from the streamed values, so Unity holds no second implementation of the kinematic model of Chapter 3.

Figure 6.1 shows the processes at run time.

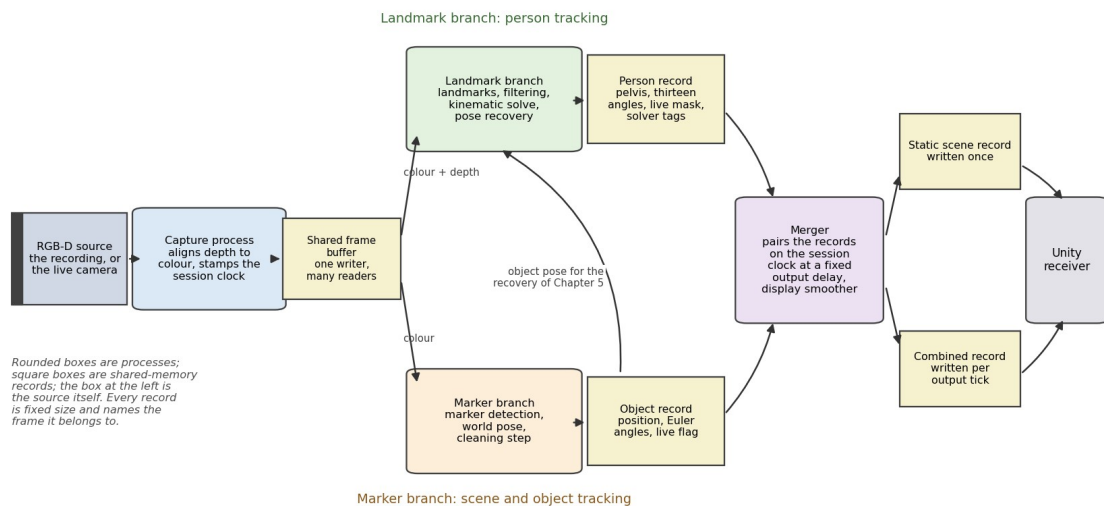


Figure 6.1. The integration architecture: the capture process, the two branches, the link from the landmark branch to the object record, and the merger that feeds the display.

The two branches run as independent processes. The marker branch imports nothing from the landmark branch. The landmark branch accesses the marker branch only through the link, which reuses the marker branch's swap and Euler routines to read the object record.

The frame buffer carries the images from the capture process to the two branches; Unity never reads it. The four records of Table 6.1 carry the computed numbers between the branches, the merger and Unity, which reads the scene record once and the combined record on every rendered frame.

6.1.1 The Frame Buffer: Capture Process to the Branches

Only the capture process opens the source, since two processes cannot share one camera. It aligns the depth frame to the colour frame once and stamps each frame with a session timestamp from the sensor's hardware timestamps. It then publishes the pair into a shared frame buffer with one writer and any number of readers.

The buffer is a ring of eight slots, one aligned colour and depth pair in each (Figure 6.2). The session timestamp is the hardware timestamp less the clock origin held in the header. The capture process numbers the frames as it publishes them, and that frame index travels in every record that describes the frame. Frame n lands in slot $n \bmod 8$, so a reader has eight frames, about a quarter of a second at the sensor's 30 frames per second, before the writer comes round to that slot again.

A sequence counter on each slot separates a settled slot from one being written. The capture process never waits for a branch, so a branch that falls behind loses whole frames rather than reading a half-written one.

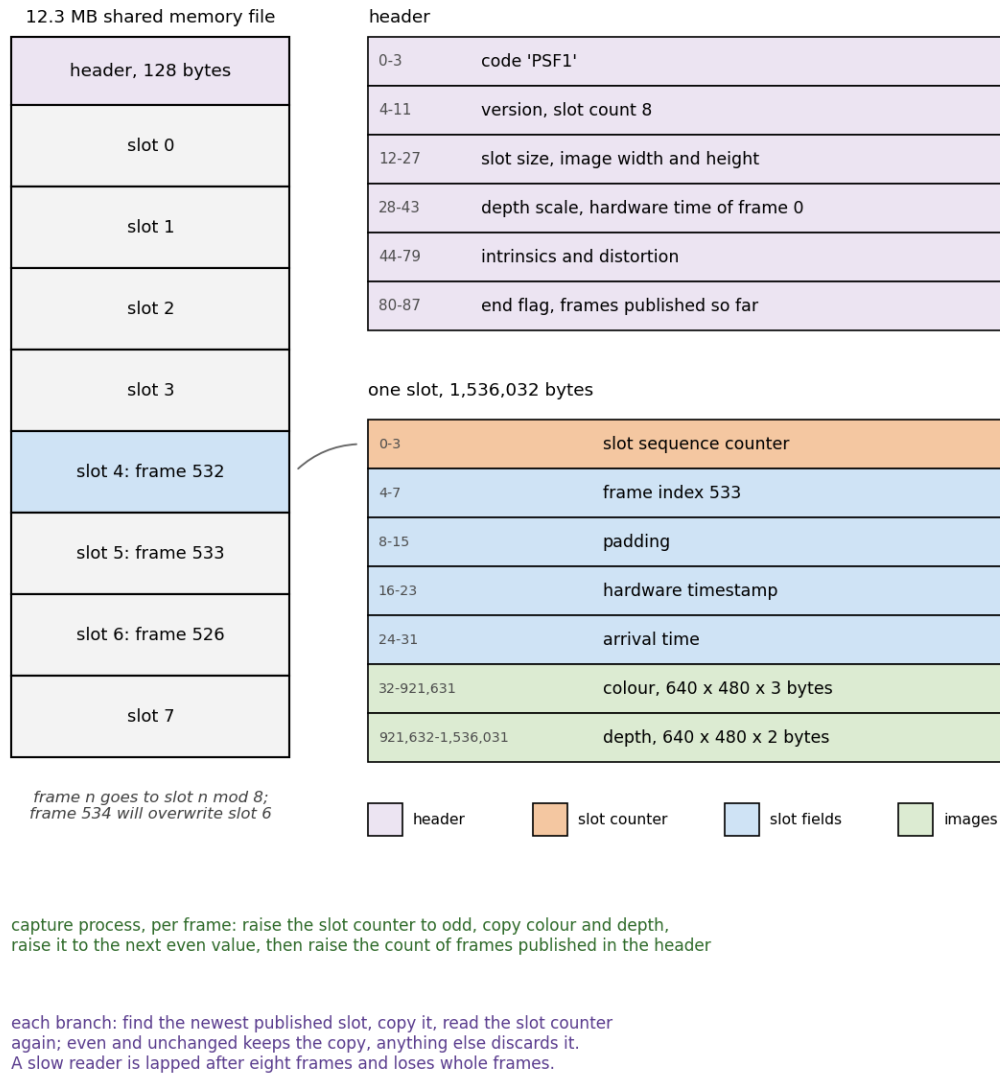


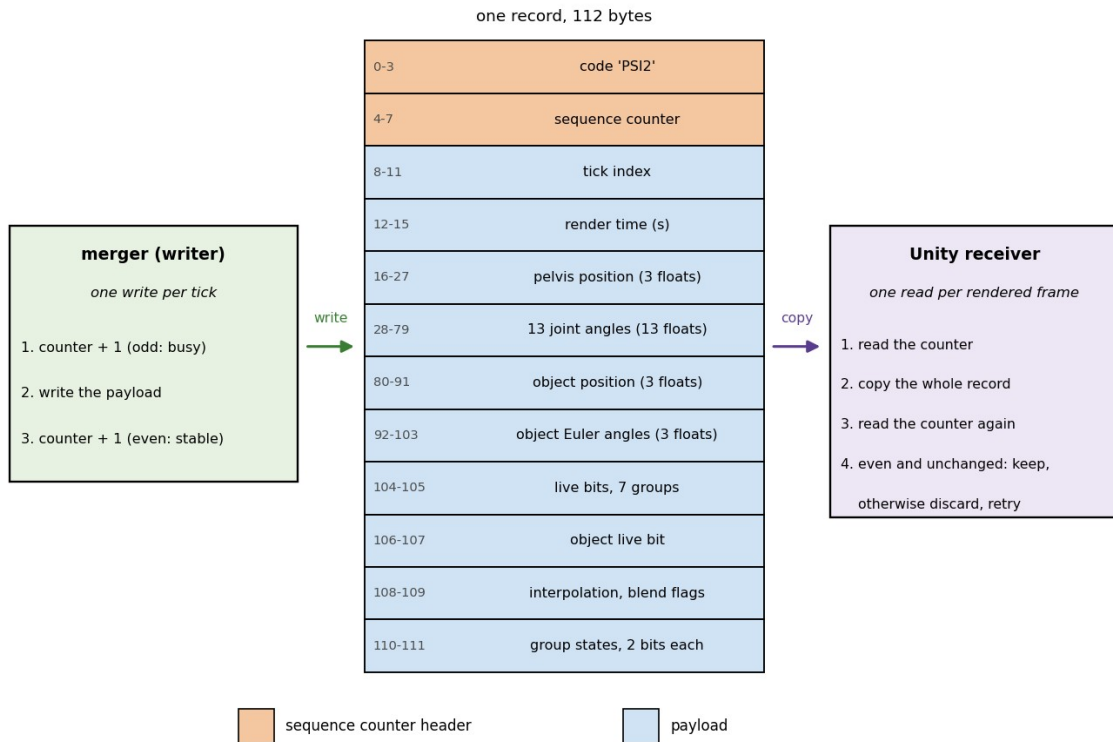
Figure 6.2. The frame buffer: the header, the eight slots, the byte layout of one slot, and the write and read steps. Frame 533 of the rail recording, traced in Section 6.5, sits in slot 5, since frame n goes to slot n modulo 8.

Both branches see the same aligned frame, so a shared frame index means the same captured instant on both sides. The capture process also selects the source, a recording or the camera, so no later stage needs to know which it is. Section 8.5 lists the remaining requirements of a live session.

6.1.2 The Records: Branches to the Merger and Unity

The person, object, scene and combined records are blocks of shared memory of one shape: a four-character code, a sequence counter, the frame or tick described, and the

payload in a fixed byte order. Each carries the output of one stage to the next under the counter rule of Section 6.1.1. The person record is 84 bytes, the object record 44 and the scene record 100. Table 6.1 sets out each record's contents, its writer and its frequency, and Figure 6.3 gives the byte layout of the combined record.



the person record (84 bytes), the object record (44) and the scene record (100) follow the same pattern, code, counter, frame or tick, payload; every writer and reader runs the two protocols shown here

Figure 6.3. The combined record as a memory block: 112 bytes with their offsets, the merger's write protocol on the left and the receiver's read protocol on the right.

Record	Code	Written by	Rate	Contents
frame buffer	PSF1	capture process	per frame	slot: aligned colour and depth images, frame index, timestamps; header: image size, intrinsics, clock origin, frames published
person record	PSR2	landmark branch	per frame	frame index, session timestamp, pelvis in {Camera'}, thirteen angles, live bit and state per group
object record	PSB2	marker branch	per frame	frame index, session timestamp, object pose in display axes, live bit down when unmeasured
scene record	PSB3	merger	once	frozen scene description, Section 2.2.2, display axes
combined record	PSI2	merger	per output tick	tick index, render time, pelvis, thirteen angles, object pose, per-group and object live bits, interpolation flags, group states

Table 6.1. The shared memory blocks. Every block is fixed size and begins with its four-character code. The records carry the sequence counter in their header, the frame buffer in each slot.

The live bit of a joint group indicates the measured status of that group on this frame. Its state is measured, held or constrained in the sense of Chapter 5.

The link crosses in one direction only: the landmark branch reads the object record, because the wrist recovery of Chapter 5 needs the object pose. Bringing that pose back into the camera frame undoes the axis swap of Section 6.2 into the world frame, then applies the calibrated pose of the world frame in the camera frame from Section 2.2.2. The link rejects a record whose frame index sits more than three frames from the frame being solved.

The merger keeps a short buffer of recent samples from each output stream and emits 30 ticks per second. Each tick renders the world at a render time on the session clock two frame intervals behind the tick, not at the newest sample. That offset is the configured buffering and render delay described in Section 8.2. By that render time both branches usually hold a sample to pair. Only samples whose live bit is up enter the object buffer.

Each stream leaves the merger in one of four states: measured, interpolated, held or blended. A stream is measured when a sample lands on the render time and interpolated when the render time falls between two samples no more than three frame intervals apart. The merger holds it across a longer gap, and blends the held pose into the next sample over two ticks. The merger interpolates the angles component by component, and the object pose in display axes by a straight line in position with a shortest-arc path in orientation.

The record's flags mark a stream as interpolated or blended, and distinguish an interpolation that bridged a dropout from one that only resampled between two adjacent frames. A held person shows in its group states. A held object lowers the object live bit of the combined record once its last sample is more than half a second older than the render time. The group states travel through the same buffer, ordered measured, held, constrained. On an interpolated tick each group takes the higher of the two states in that order, and on a held tick the merger raises every group to at least held.

6.2 Coordinate System Conversion

The two branches reach the display through four frames, drawn in Figure 6.4.

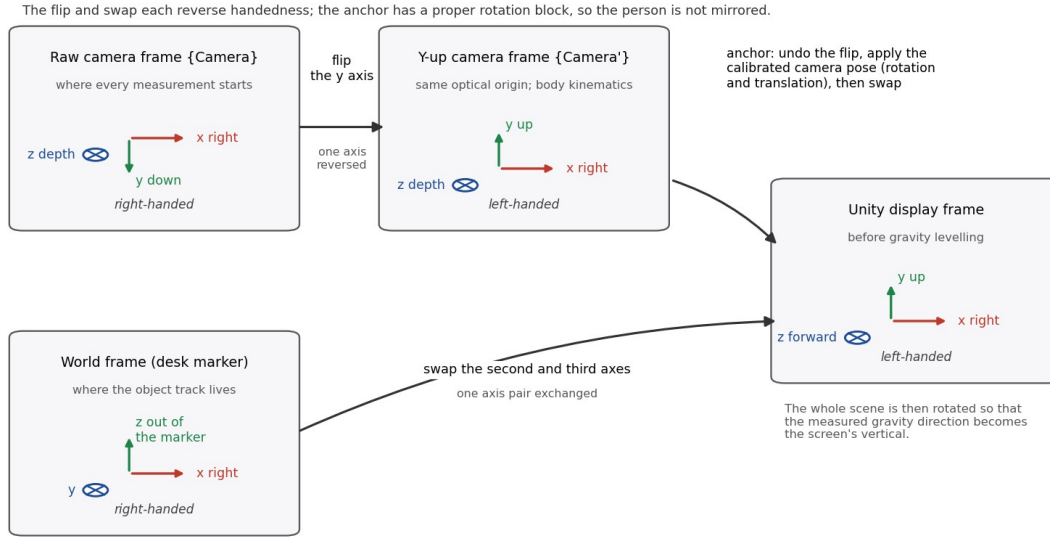


Figure 6.4. The four frames of the coordinate conversion before gravity alignment, with the path of each branch to the display. Section 6.4 completes the mapping to the scene frame.

The landmark branch works in the y-up camera frame {Camera'}, which shares the optical centre of {Camera} and flips only its y axis through F of Section 3.1. The marker branch works in the desk-anchored world frame of Section 2.2.1. Unity uses left-handed coordinates [58]. The swap S exchanges the second and third world coordinates into the display frame {Unity}; its second axis follows the desk marker normal until gravity alignment is applied,

$$\begin{aligned}
 {}^{Unity}P &= S {}^{World}P \\
 {}^{MappedMarker}R &= S {}^{World}R S \\
 S &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.
 \end{aligned} \tag{6.1}$$

Here S changes both the reference axes from {World} to {Unity} and the marker axes from {Object} to {MappedMarker}. Changing both bases requires conjugation. Changing only the reference basis requires left multiplication, as in the subsequent levelling and rig mapping. The swap is its own inverse, so the link of Section 6.1.2 undoes it by applying it again. Its determinant is minus one, as is that of the flip F, so each changes the handedness of the frame.

The two results do not land in the same place, because a y-up camera-frame point is expressed relative to the camera and a world point relative to the desk marker. The calibration of Section 2.2.2 froze the camera's own pose in the world, which Section 4.1 obtains by inverting the anchoring transformation, and that pose closes the gap. The inverse flip, equal to F because F is self-inverse, returns a y-up camera-frame point q to raw sensor coordinates. The calibrated camera orientation rotates it into the world, and the swap carries it into the display frame. The swapped camera position supplies the offset. Collecting the three linear maps into one matrix, and the whole map into one homogeneous transformation,

$$\begin{aligned} \text{Unity } P &= \text{Unity}_{Camera'} R \text{ Camera}' q + \text{Unity } P_{CameraORG}, \text{ Camera}' R = S \text{ World}_{Camera} R F^{-1}, \\ \begin{bmatrix} \text{Unity } P \\ 1 \end{bmatrix} &= \text{Unity}_{Camera'} T \begin{bmatrix} \text{Camera}' q \\ 1 \end{bmatrix}, \text{ Unity}_{Camera'} T = \begin{bmatrix} \text{Unity}_{Camera'} R & \text{Unity } P_{CameraORG} \\ 0 & 1 \end{bmatrix}, \end{aligned} \quad (6.2)$$

where $\text{World}_{Camera} R$ and $\text{World } P_{CameraORG}$ are the rotation block and the translation column of $\text{World}_{Camera} T$, the calibrated camera pose in the world of Section 4.1. $\text{Unity}_{Camera'} T$ is the same map as one homogeneous transformation matrix, the product of the 4 by 4 forms of the swap, $\text{World}_{Camera} T$ and the flip. The determinant of $\text{Unity}_{Camera'} R$ is minus one times plus one times minus one, so the two handedness changes cancel and the person arrives in {Unity} without a mirror image. The product combines two handedness-changing basis maps with the calibrated rigid transform [44]. The y-up camera frame and the display axes are both left-handed, so the algebra of the reference carries over, although its frames are right-handed and these are not.

The composition splits into a factor the display already has and a constant. Inserting the swap twice in the middle changes nothing. Regrouping then gives the anchor rotation as the calibrated camera orientation conjugated into display axes, times the fixed pair formed by the swap of equation (6.1) and the flip of equation (3.1). That pair is a rotation of minus ninety degrees about the x axis,

$$\begin{aligned}
{}^{Unity}_{Camera}R &= S_{Camera}^{World} R F^{-1} = S_{Camera}^{World} R (S S) F^{-1} = (S_{Camera}^{World} R S) (S F^{-1}), \\
{}^{Unity}_{MappedCamera}R &= S_{Camera}^{World} R S, \quad {}^{MappedCamera}_{Camera}R = S F^{-1}, \\
S F^{-1} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = R_x(-90 \deg).
\end{aligned} \tag{6.3}$$

The first factor is the orientation the display already uses to place the sensor body in the scene. It relates the display axes to the mapped camera frame, the sensor node's own system in the display, with the optical axis upward and the downward image direction forward. The receiver creates one static anchor node at the calibrated camera pose composed with the constant rotation. Every quantity from the person record passes through this node, positions as points and rotations by multiplying on the left. Conjugating a body rotation by the anchor would cancel the camera orientation out of the pose and turn the figure the wrong way.

The root orientation and the object pose travel as Euler angles in the fixed order the animation system uses [58], which is the order Section 3.4 decodes a rotation matrix into. The receiver rebuilds those two rotations from the three numbers. The arm angles travel as the joint coordinates of Section 3.4, which Section 6.3 reassembles.

6.3 Avatar Reconstruction in Unity

The avatar is a humanoid character supplied by the laboratory. The receiver drives five of its bones. The spine bone at the base of the trunk takes the root rotation and the position, and the upper arm and forearm of each side take the arm rotations. Everything else, the legs included, keeps its rest pose, since the eight tracked landmarks measure no leg.

The thirteen streamed angles reassemble five rotations. The root Euler triple rebuilds the trunk rotation directly. The shoulder swing pair and twist of Section 3.4 follow as axis-angle factors about the up, forward and right axes in that order. The elbow adds two angles about up and forward. The left arm repeats the pattern with every component except the twist reversed in sign, the sagittal mirror of Chapter 3.

Each reconstructed rotation relates two frames of the chain of Chapter 3: the root frame at landmark L24 in the y-up camera frame, the upper arm frame at landmark L14 in the root

frame, and the forearm frame in the upper arm frame. The forearm frame has its origin at the wrist landmark L16 and its first axis along the forearm. Chapter 3 uses the elbow rotation as an operator on a vector while here it relates two frames, because the notation follows the way a rotation is used. The spine bone takes the root rotation alone, and the chain composition adds one factor per joint,

$$\begin{aligned} \text{Camera}'_{L14} R &= \text{Camera}'_{L24} R_{L14}^{L24} R, \\ \text{Camera}'_{L16} R &= \text{Camera}'_{L24} R_{L14}^{L24} R_{L16}^{L14} R. \end{aligned} \quad (6.4)$$

The shoulder frame at landmark L12 inherits the orientation of the root frame, so the shoulder factor is equally the upper arm frame taken in that shoulder frame.

The rest pose is the first correction, since the receiver composes the chain product against it. Each driven bone carries two frames: the segment frame that the chain of equation (6.4) ends on, and the bone frame, the axes the rig's author gave that bone. Nothing is expressed in the bone frame; it is named because the rotation written to the bone relates it to the scene frame of Section 6.4. That rotation, taken one bone at a time, is

$$\text{Scene Bone} R = \text{Scene Camera} R \text{Camera}'_{\text{Segment}} R_{\text{Bone}}^{\text{Segment}} R. \quad (6.5)$$

The leading factor includes the proper gravity alignment operator G of Section 6.4:

$$\text{Scene Camera}' R = G_{\text{Camera}'}^{\text{Unity}} R.$$

The receiver reads this global anchor rotation. The bone and segment axes are unchanged by the anchor, so the chain is left-multiplied rather than conjugated. The floor translation changes joint positions but not this orientation.

In equation (6.5), the leading factor is the orientation of the y-up camera frame in the scene frame. The trailing factor is the bone's rest rotation, captured when the scene starts, which absorbs the axis conventions built into the rig as a constant. Reading it as the relation between the segment frame and the bone's own axes rests on an assumption this thesis states rather than establishes.

The avatar is taken to be authored so that with every driven bone at its identity rotation it stands upright, its right side along the scene's first axis and its chest along the third. It is

taken to be placed with no rotation, so the model's axes coincide with the scene axes when the scene starts. The model's zero configuration is then the reference configuration of Section 3.2.2, the T-pose that is the zero of every arm angle. The composition is correct only under that alignment. The setup therefore forces the T-pose first, rotating each arm segment straight out to the side, and captures the five rest rotations once the arms are aligned and the rig sized.

Proportions are the second correction. The landmark branch sends joint angles and one point, so the rendered hand ends up wherever the rig's bone lengths put it. The avatar setup scales the character uniformly to 170 centimetres. It then sizes the rig from five lengths supplied for the recording being replayed, the four arm segments and the trunk. The capture writes those five values to a file before the scene starts, and the receiver reads them there when the scene starts. Each arm length is the median of that segment over the landmark track of that recording, which is the length the recovery of Chapter 5 solved on. The avatar therefore reproduces the geometry the solver used.

The rail recording gives a right upper arm of 31.7 centimetres and a right forearm of 20.5, with a left upper arm of 31.0 and a left forearm of 25.7. The handover recording gives 24.1 and 23.3 centimetres on the right, with 25.2 and 22.3 on the left.

The trunk length is supplied in the same way, because it runs from the pelvis point the record carries to the mid-shoulder. This thesis defines that length by two rules, and each capture states which one it used. The rail and handover captures take the median distance from the recovery pelvis of Chapter 5 to the measured mid-shoulder, over the frames whose torso solve did not fail, on the filtered landmarks. That rule gives 57.5 centimetres over 620 frames on the rail recording and 51.7 centimetres over 1173 frames on the handover recording. That pelvis is the corrected point of the hip depth preparation of Section 2.6, which sits deeper than the raw hips.

The loop capture was not repeated and keeps its own five lengths. Its arms are 25.6 and 25.2 centimetres on the right and 26.2 and 24.7 on the left, again the medians of that recording's landmark track. Its trunk is 47.9 centimetres by the second rule, the distance from the raw hip midpoint to the mid-shoulder over the whole track.

Section 5.6 prints the rail lengths in the worked example of the recovery. Every length named here is a landmark median, so the recordings differ mainly in the split of the arm between the upper arm and the forearm. The totals of the right arm are 50.8 centimetres on the loop recording and 52.2 on the rail recording.

A direct measurement of the subject gives about 25 centimetres for both the upper arm and the forearm. That measurement is an independent anatomical comparison. It sets no length of the avatar and no length of the solver. The loop medians lie near it and the rail medians do not, which is consistent with the elbow landmark sitting farther along the arm on the rail recording. Scaling the trunk matters because the trunk moves the shoulder joints. Uniform scaling leaves every rotation unchanged.

A display smoother on the angles and the pelvis is the third correction; the processing side applies it to the angles and the pelvis as they enter the output record. The solver of Chapters 3 and 5 changes state in short ramps because its rate limits cap the distance a joint may move between frames. A linear low-pass alone cannot remove a ramp, only delay it, so the smoother pairs a first-order low-pass with a cap on the angle change per output tick. It changes only the displayed values; the processing side writes the tables Chapters 7 and 8 analyse before the smoother runs.

Each tick the receiver maps the streamed pelvis point from {Camera'} into {Scene} through the anchor, including the levelling and floor translation of Section 6.4. It translates the root so that the spine bone lands on that point, and the root rotation orients the trunk about it.

A root, shoulder swing or elbow group not measured on a tick has a small sphere at its joint, at the pelvis, at the shoulder and at the elbow in turn, while the twist group has none. The sphere sits a little in front of the body along the line of sight of the camera that renders it. Red marks a held value and blue a constrained one, the states of Section 6.1.2. A smaller amber sphere at the pelvis shows a pose interpolated across a gap in the stream or ramping back after a long gap.

The captures of Figure 6.5 come from the receiver that applies both start-up corrections; it reads only the live bits, so every sphere there is red. The receiver that reads the merger's record draws all three colours. It applies the uniform height scale but neither the

forced T-pose nor the supplied segment lengths, and it captures the rest rotations of equation (6.5) from the rig as authored.

6.4 Scene Reconstruction in Unity

The receiver builds the room from the static scene record, which holds the frozen scene description of Section 2.2.2, already in display axes by the swap of Section 6.2.

Levelling comes first. The world frame's third axis is the desk marker's face normal, and the card sits on a leaning stand, so raw world axes would draw a tilted room. The receiver creates a parent node for the scene geometry, object and person anchor, and rotates it by the shortest rotation that carries the gravity direction of Section 2.2.2 onto the screen's vertical. The object pose is applied beneath the node as a local pose, and the rig receives global poses through the anchor.

Write this gravity alignment operation as G , a proper rotation matrix. It acts on the swapped coordinates from {Unity}, preserving lengths and handedness. The receiver then adds the translation t_f to raise the reconstruction until the drawn floor sits at zero height.

Together these operations define the final frame {Scene}, the coordinate system Unity itself draws in. Its second axis follows the calibrated upward gravity direction, and its origin lies on the drawn floor directly below the desk marker origin. The translation raises that marker origin by the recording-specific distance d . The full transformation from {Unity} to {Scene}, and its application to a world point after the swap, are

$$\begin{aligned} \begin{matrix} Scene \\ Unity \end{matrix} T &= \begin{bmatrix} G & t_f \\ 0 & 1 \end{bmatrix}, t_f = \begin{bmatrix} 0 \\ d \\ 0 \end{bmatrix} \\ \begin{matrix} Scene \end{matrix} P &= G S^{World} P + t_f. \end{aligned} \tag{6.6}$$

The offset d belongs to the recording and not to the system. It is the height of the drawn desk top above the floor, the desk thickness plus the leg height, less the small offset between the horizontal tabletop model and the origin of the world frame. That gives 71.6 centimetres on the rail recording and 70.9 centimetres on the handover recording. The final coordinate system drawn in Figure 2.4(b) is the scene frame.

The wall is a thin slab in the plane of the calibrated wall marker, and the desk a horizontal slab on the tabletop model of Section 2.2.2. That model passes through a point on the tabletop derived from depth, with its normal set by calibrated gravity. No measurement sets the desk width, the slab thicknesses, the legs or the floor. The rail is not part of the scene record, so the receiver does not draw it.

A plate printed with the marker pattern sits at each calibrated marker pose. The receiver draws the desk card over the desk slab. Its calibrated centre lies 0.4 centimetres below the horizontal tabletop model, so the slab would otherwise cut the tilted card in half. The depth image puts that centre 1 to 2 centimetres above the desk around it, a difference within the pose error of a 45 millimetre marker seen from 58 centimetres. The object is a cube of the measured edge length, half an edge behind its own plate along the face normal. It turns red for a held pose once the object live bit drops, and amber for a pose interpolated across a dropout or blended back. A model of the sensor sits at the calibrated camera pose, and a camera attached to it renders an inset at the calibrated vertical field of view for comparison with the recorded colour frame.

Figure 6.5 shows the reconstruction on two frames of the rail recording, rendered from that camera at the sensor's pose. The root has a sphere at the pelvis on both frames, because the depth pixels of the hips land on the rail in front of the subject and the hip depth preparation of Section 2.6 corrects them before the solve. The idle left arm hangs at the subject's side, partly out of the frame, so its groups are not measured and it has one at its shoulder and one at its elbow.

On frame 114 the right arm reaches forward and down to the cube on the desk, and the rendered hand lands beside the cube, 13.3 centimetres from the measured wrist. The elbow is bent by 14.4 degrees there, so the rule of Section 5.5 holds the shoulder twist. The rendered difference includes the solver state and the placement of the shoulder, and this capture does not isolate their individual contributions. The rig carries the segment lengths of the recording being replayed, so no length of another recording enters this view. On frame 700 the elbow is bent by 31.0 degrees, the twist is measured, and the hand sits on the cube, 6.6 centimetres from the measured wrist.

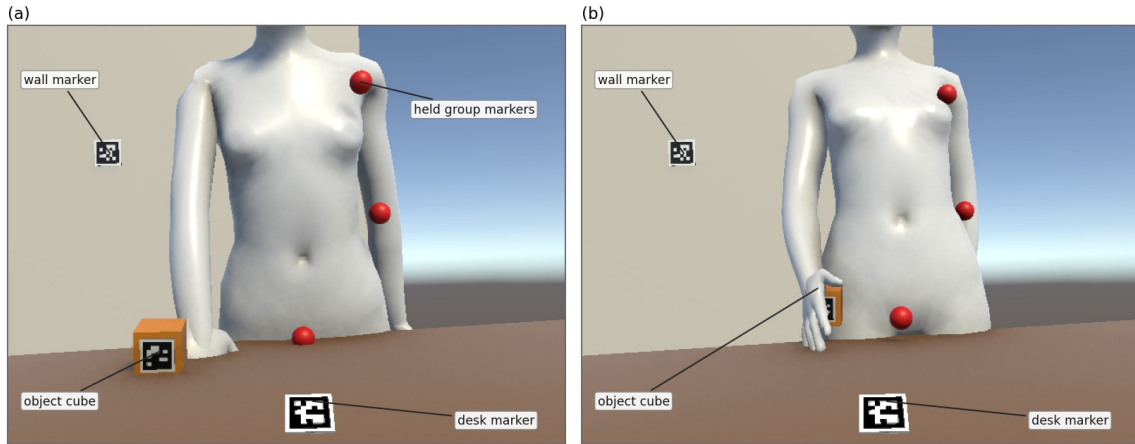


Figure 6.5. The reconstruction from the virtual sensor: (a) frame 114, the hand at the cube; (b) frame 700, sliding it along the rail. Section 2.1 defines the task phases.

6.5 Worked Example

This section takes one frame of the rail recording, so that the records of Section 6.1 and the maps of Sections 6.2 to 6.4 appear with their numbers, as Section 5.6 did for the recovery. The frame is 533, replayed through the causal structure of Section 8.2 at the recording's own pace (Section 8.4), at a moment when the cube slides along the rail. Table 6.2 collects the values.

The capture process published the frame at 17.78 seconds on the session clock into slot 5 of the frame buffer, 533 modulo 8. Both branches took their copy of slot 5.

The landmark branch then wrote the person record: frame 533, the pelvis and the thirteen angles. Its live bits are down for the root and the right shoulder twist. Its states say the root is constrained and the twist held. The record for frame 532 carried the same two states. The solve did draw on the object record on this frame: the right hand was inside a grip episode, so the link of Section 6.1.2 offered the solve a right wrist position derived from the cube pose. The left hand was offered none.

The marker branch wrote the object record with its live bit up, as it had for frame 532: the marker was detected on both frames, so no dropout runs into this one. Table 6.2 lists the processing time stored in each branch's record for this frame. Section 8.4 reports processing times per frame over the recording.

The merger's tick at 17.82 seconds on the session clock, with frame 534 just published, rendered the world as of 17.75 seconds, two frame intervals earlier. The person samples of frames 532 and 533 lay on either side of that render time, 5 milliseconds before it and 29 milliseconds after it, so the merger interpolated between them. The object stream had both of those frames as well, so the merger interpolated the object pose between the same two and raised no dropout flag. Into the combined record went both poses, with the root still constrained and the twist still held.

The pelvis in the person record is (3.1, -18.7, 99.0) centimetres in the y-up camera frame. Equation (6.2) carries it into display axes, and the calculation runs here with its numbers. The rotation block and the translation column of the calibrated camera pose of Section 4.1 are

$$\begin{aligned} {}^{World}_{Camera}R &= \begin{bmatrix} 1.00 & -0.02 & -0.01 \\ 0.00 & -0.48 & 0.88 \\ -0.02 & -0.88 & -0.48 \end{bmatrix} \\ {}^{World}P_{CameraORG} &= (-0.01, -0.39, 0.43)m. \end{aligned}$$

The swap, this rotation and the flip multiply into the anchor rotation, and the swap carries the camera position into display axes:

$$\begin{aligned} {}^{Unity}_{Camera}R &= S {}^{World}_{Camera}R F^{-1} = \begin{bmatrix} 1.00 & 0.02 & -0.01 \\ -0.02 & 0.88 & -0.48 \\ 0.00 & 0.48 & 0.88 \end{bmatrix} \\ {}^{Unity}P_{CameraORG} &= S {}^{World}P_{CameraORG} = (-0.01, 0.43, -0.39)m. \end{aligned}$$

The pelvis point of the record then maps as

$$\begin{aligned} {}^{Camera'}q &= (0.03, -0.19, 0.99)m \\ {}^{Unity}P &= {}^{Unity}_{Camera}R {}^{Camera'}q + {}^{Unity}P_{CameraORG} = (0.00, -0.21, 0.38)m. \end{aligned}$$

The matrices and vectors are printed to two decimals. The pelvis sits at (0.3, -21.2, 38.3) centimetres in display axes, before the parent levelling and floor translation of Section 6.4. The three steps below give the same point. Undoing the flip gives the raw camera point (3.1, 18.7, 99.0), within a millimetre of the midpoint of the two hip landmarks, which the hip depth preparation left unchanged on this frame. The calibrated camera orientation and position carry it into the world frame, (0.3, 38.3, -21.2), and the swap

exchanges the last two coordinates into display axes. The camera itself sits at the swapped camera position, $(-1.4, 43.1, -39.3)$ centimetres in the same axes.

The parent node of Section 6.4 applies the full transformation of equation (6.6). Its gravity alignment carries the calibrated gravity direction, $(-0.02, 0.85, 0.53)$ in display axes, onto the screen's vertical, a turn of 32.0 degrees. Its translation uses the offset d at 71.6 centimetres on this recording, so that the drawn floor sits at zero height:

$$G = \begin{bmatrix} 1.00 & 0.02 & 0.01 \\ -0.02 & 0.85 & 0.53 \\ 0.01 & -0.53 & 0.85 \end{bmatrix}$$

$$G^{Unity} P = (0.00, 0.02, 0.44) m$$

$$\begin{bmatrix} Scene \\ P \\ 1 \end{bmatrix} = \begin{matrix} Scene \\ Unity \end{matrix} T \begin{bmatrix} Unity \\ P \\ 1 \end{bmatrix}$$

$$Scene P = G^{Unity} P + t_f = (0.00, 0.74, 0.44) m.$$

The spine bone lands at $(0.1, 74.0, 43.7)$ centimetres in the scene frame, 2.0 centimetres above the drawn desk top at 72.0. In the replay of Section 6.3 the receiver's log places the spine bone on the mapped point to within a millimetre on every frame.

The object record holds the marker origin of the cube already in display axes, $(-24.0, -19.6, 42.7)$ centimetres with Euler angles $(-61.4, 1.5, -6.3)$ degrees. The link of Section 6.1.2 reverses the step above, from the world to the camera. The anchor's rotation block is the transpose of the one above and its translation column is the desk marker seen from the camera:

$$\begin{matrix} Camera \\ World \end{matrix} R = \left(\begin{matrix} World \\ Camera \end{matrix} R \right)^T = \begin{bmatrix} 1.00 & 0.00 & -0.02 \\ -0.02 & -0.48 & -0.88 \\ -0.01 & 0.88 & -0.48 \end{bmatrix}$$

$$Camera P_{WorldORG} = (0.02, 0.19, 0.55) m.$$

The cube's marker origin then returns to the camera frame as

$$World P = S^{Unity} P = S(-0.24, -0.20, 0.43) = (-0.24, 0.43, -0.20) m$$

$$Camera P = \begin{matrix} Camera \\ World \end{matrix} R \begin{matrix} World \\ P \end{matrix} + \begin{matrix} Camera \\ WorldORG \end{matrix} P = (-0.21, 0.16, 1.02) m.$$

The result, $(-21.2, 15.8, 102.4)$ centimetres in the camera frame, is the pose the marker branch reported, with the marker's face normal turned to the camera 4.4 degrees off the

optical axis. The depth image reads 98.6 centimetres at the marker's pixel, 3.8 centimetres nearer than the pose, within the pose error of a 45 millimetre marker at that distance.

The thirteen angles become bone rotations by equation (6.5), composed here from the record's angles as the receiver of Figure 6.5 composes them. The right upper arm takes the root Euler angles $(-13.2, 176.5, -2.4)$ degrees, the shoulder azimuth and elevation $(-44.1, -76.3)$ degrees and the held twist of -0.1 degrees. After the anchor and the levelling they place the bone's axis along $(-0.14, -0.93, -0.34)$, 0.4 degrees from the measured shoulder-to-elbow direction carried through the same maps. The elbow is bent by 17.7 degrees about the up axis and 0.0 about the forward axis. The forearm lands 10.0 degrees from the measured direction while the shoulder twist is held under the rule of Section 5.5.

The receiver that reads the merger's record draws the pelvis sphere blue, because the root's state is constrained, and no sphere for the twist. The cube keeps its plain colour on this tick: its pose lies between two live measurements, so it is neither held nor bridged nor blended back. Neither stream crossed a dropout, so the pelvis carries no amber sphere either. The receiver of Figure 6.5, which reads the live bits alone, would draw the pelvis sphere red.

Stage	Record or map	Frame 533
Capture process	frame buffer, slot 5	17.78 s on the session clock; colour and depth, 1,536,032 bytes
Landmark branch	person record	8.7 ms; pelvis (3.1, -18.7, 99.0) cm; live bits down for the root and the right twist; root constrained, twist held
Marker branch	object record	13.6 ms; (-24.0, -19.6, 42.7) cm, (-61.4, 1.5, -6.3) deg; live; frame 532 live as well
Merger	combined record	tick at 17.82 s, render time 17.75 s; person 532 to 533, object 532 to 533, no dropout bridged
Link	swap undone, anchor applied	cube marker origin (-21.2, 15.8, 102.4) cm in the camera frame
Receiver, position	equations (6.2) and (6.6)	pelvis (0.3, -21.2, 38.3) cm in display axes; (0.1, 74.0, 43.7) cm in the scene frame; 32.0 deg
Receiver of Figure 6.5, rotation	equation (6.5)	right upper arm 0.4 deg from the measured direction; forearm 10.0 deg

Table 6.2. Frame 533 of the rail recording through the causal structure and the receiver. Positions in centimetres; clock times on the session clock, durations in milliseconds.

Both branches now reach one rendered scene, with every joint group tagged measured, held or constrained. The chapter does not establish the reconstruction's accuracy or its real-time operation. Chapter 7 evaluates it offline, against physical references including the tape measurements of Section 7.2, against measured landmarks and manual wrist labels, and against within-recording comparisons. Appendix H holds the solver verification, and Chapter 8 tests the causal operation of the architecture.

Chapter 7: Experimental Evaluation

This chapter reports the offline reconstruction results, the reference behind each comparison, and the claims the numbers do and do not support. The references are the tape-measured object route, the measured landmarks, manual wrist labels and within-recording comparisons, and the physical route is not registered into the world frame.

7.1 Evaluation Method

The evaluation uses the single-hand rail recording, the handover recording on the rail, and the loop recording for natural occlusion. Tape measurements provide the reference for object segment lengths once the waypoints are matched at both ends. Rendered elbow and wrist positions are compared with synchronized MediaPipe landmarks deprojected using aligned depth. Synthetic removal uses valid unmasked kinematic reconstruction as its reference. Natural occlusion uses manual wrist labels, with their observed separation from accepted wrist measurements evaluated on clean frames.

Physical length error is defined as absolute and relative error. Positional differences and trajectory scatter are summarized by the median, the 95th percentile (p95), and the maximum. Each joint and method uses its stated frame set; n is the number of evaluated frames. Percentiles use linear interpolation between ordered observations.

The reference wrist for quantitative reconstruction error is the accepted raw MediaPipe and depth three-dimensional wrist measurement, taken before temporal filtering. The same reference applies to the kinematic-model wrist error, to the rendered Unity rig wrist error, and to any clean-frame baseline that means reconstruction error against a measured wrist. A filtered wrist is used elsewhere for algorithmic purposes; wherever it appears it is named as a filtered measurement and is not the evaluation reference. No comparison table mixes the two references.

The object frame counts reported below use two terms. A detected frame is a frame on which the ArUco extractor returned a marker pose. An accepted object sample is a returned pose that remained usable after the cleaning rules of Section 7.2. The two answer different questions, so each count names the term it reports.

7.2 Object Reconstruction Accuracy

The route has four physical waypoint roles: W1 is the parked start on the desk, W2 is the foot of the lift, W3 is the start of the rail, and W4 is the far endpoint. The segments are W1 to W2, W2 to W3, and W3 to W4. A segment comparison requires the same named point on the object at both physically measured endpoints and at their corresponding reconstructed positions.

Let q_j and q_{j+1} be reconstructed positions at the independently identified waypoint events, and let L_j be the physically measured endpoint separation. Reconstructed length, absolute error, and relative error are defined by

$$\begin{aligned} L_{rec_j} &= \|q_{j+1} - q_j\| \\ A_j &= |L_{rec_j} - L_j| \\ R_j &= 100 \frac{A_j}{L_j} (\%). \end{aligned} \tag{7.1}$$

Here L_j is positive. Reconstructed length is endpoint separation, not accumulated sample-to-sample travel or the longest span of a fitted line. A stationary-frame interval, when available, defines an endpoint by the mean of its accepted reconstructed samples; an identified single-frame event uses that sample.

For reconstructed samples p_i , $i = 1, \dots, n$, within one segment and expressed in {Scene} of Section 6.4, orthogonal least-squares line fitting uses the sample mean c and covariance C :

$$\begin{aligned} c &= \frac{1}{n} \sum_{i=1}^n p_i \\ C &= \frac{1}{n} \sum_{i=1}^n (p_i - c)(p_i - c)^T. \end{aligned} \tag{7.2}$$

Principal component analysis (PCA) gives a unit eigenvector u of C associated with its largest eigenvalue. This direction and the line through c minimize the squared perpendicular distances:

$$\min_{\substack{c,u \\ \|u\|=1}} \sum_{i=1}^n \left\| (I - uu^T)(p_i - c) \right\|^2 \quad (7.3)$$

$$line(s) = c + su.$$

The parameter s ranges over the real numbers and I is the identity matrix. The perpendicular distance of sample p_i is

$$d_i = \left\| (I - uu^T)(p_i - c) \right\|. \quad (7.4)$$

The distances d_i describe scatter of the reconstructed trajectory about its own fitted line. They do not measure physical accuracy. A degenerate sample set with no unique principal direction does not define a fitted segment direction. Physical length error and trajectory scatter are reported separately.

The measured point is the ArUco marker centre on the cube face, at both physically measured endpoints and in the reconstruction. The physical separations are tape measurements made by the author between marker-centre positions, at a resolution of 1 millimetre, and the physical setup is the same in both recordings. The physical route is not registered into the world frame of the desk-marker calibration or into $\{\text{Scene}\}$, so the trajectory plots report reconstructed samples rather than surveyed physical paths or physical waypoint coordinates.

7.2.1 Single-Hand Rail Task

The rail recording holds 900 frames, of which 889 supply accepted object samples of the marker centre. The other 11 frames are excluded because the marker was not detected, because the stored sample was filled by the tracker, or because the recorded position was not finite. No frame is excluded because of the size of any error.

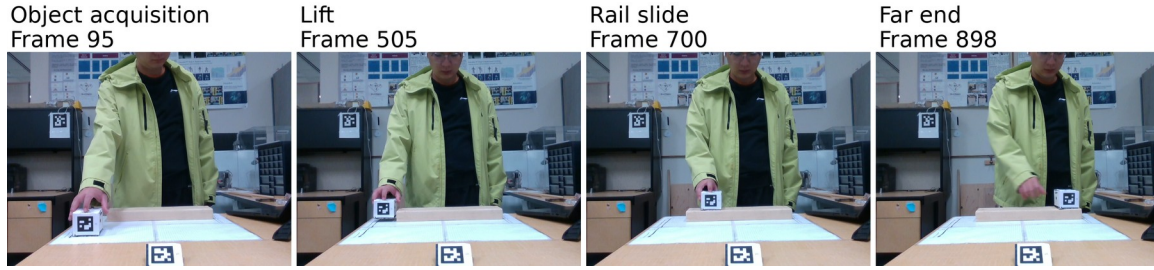
Waypoint timing comes from the marker-centre motion of this recording alone. Each axis velocity is a centred difference over 7 accepted samples on either side. A transition is bracketed at the half-peak of the velocity of the moving axis, and the boundary is the last sample within 5 millimetres of the local pre-transition level of that axis. W1 covers frames 0 to 92, an opening interval of 93 samples whose mean gives the position. W2 is frame 493 and W3 is frame 617, each a single transition sample, because the cube does

not pause on either side of the lift and no dwell exists there to average. W4 covers frames 875 to 899, a closing interval of 25 samples whose mean gives the final position.

Table 7.1 compares the reconstructed segment lengths with the physical reference, using equation (7.1). Figure 7.1 places recorded views of the task above the accepted samples and the four waypoint positions.

Segment	Physical length (cm)	Reconstructed length (cm)	Absolute error (cm)	Relative error (%)
W1 to W2	25.5	24.36	1.14	4.45
W2 to W3	3.5	4.03	0.53	15.25
W3 to W4	37.5	38.59	1.09	2.91

Table 7.1. Rail recording: reconstructed marker-centre segment lengths against the physical reference, with absolute and relative error.



Rail recording, 889 accepted samples

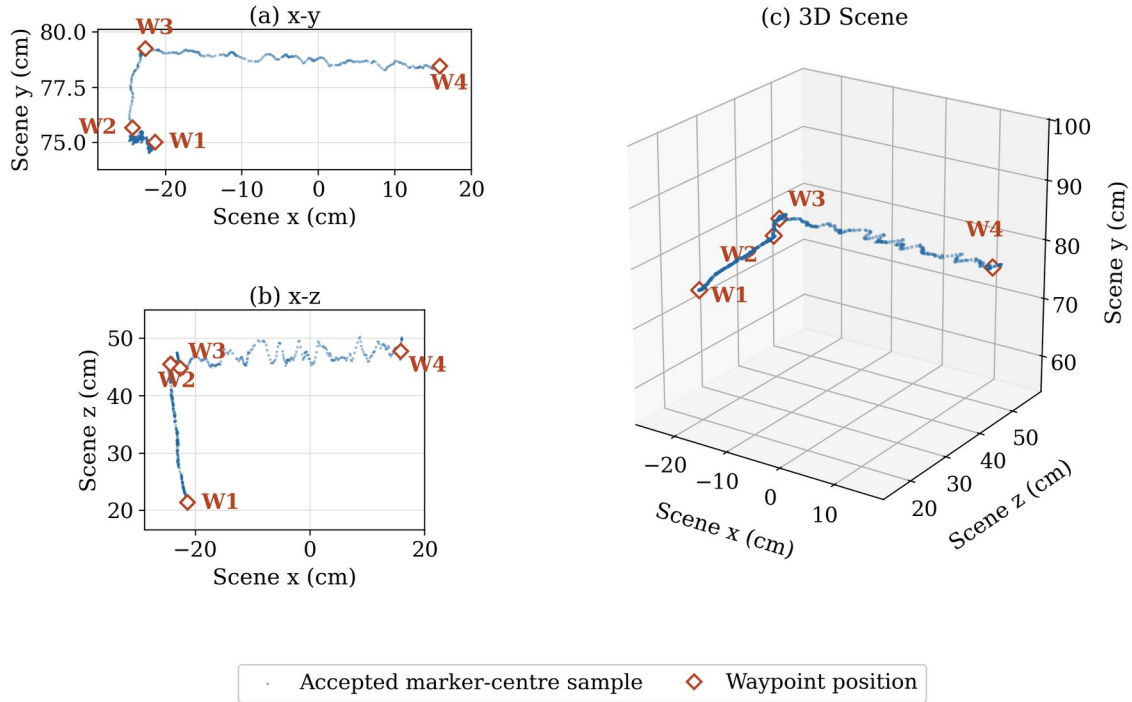


Figure 7.1. Rail recording: video frames, accepted marker-centre samples and the four reconstructed waypoints in $\{\text{Scene}\}$. The x-y view expands the height scale; the others use equal metric scale.

Each reconstructed length is reported with a range obtained by varying the detection parameters and shifting each boundary by two and by five accepted samples. Variants that no longer reproduce the depth, vertical and lateral phase pattern are excluded. The ranges are 23.95 to 25.15 centimetres for W1 to W2, 3.76 to 4.70 centimetres for W2 to W3 and 37.78 to 38.87 centimetres for W3 to W4. The point estimate alone overstates the precision of the result.

W2 and W3 are single accepted samples, so each carries the instantaneous depth error of one frame rather than an averaged value. Within 7 frames on either side the depth

coordinate spreads 8.6 millimetres at W2 and 22.4 millimetres at W3. The criterion based on departure from the pre-transition level also fails to separate either boundary from the half-peak of its axis velocity. The reported sample is therefore the latest the search allows. The transition lies at or before it.

The W4 interval holds 25 samples, about 0.8 seconds. Its depth coordinate spans 38.1 millimetres, and its largest single-sample step is 9.0 millimetres, between frames 882 and 883. The closing interval is therefore not a single settled cluster. The recording ends about ten frames after the cube settles, and no longer settled interval is available to average.

Scatter about the fitted line is a separate within-recording comparison, defined by equations (7.2) to (7.4), between the reconstructed trajectory and a line fitted to the same samples. Over the 248 samples between W3 and W4 the median perpendicular distance is 0.83 centimetres, the p95 is 2.12 centimetres and the maximum is 2.92 centimetres. The fitted line lies 1.3 degrees from horizontal. These distances describe the spread of the reconstructed samples and are not comparable with the errors of Table 7.1.

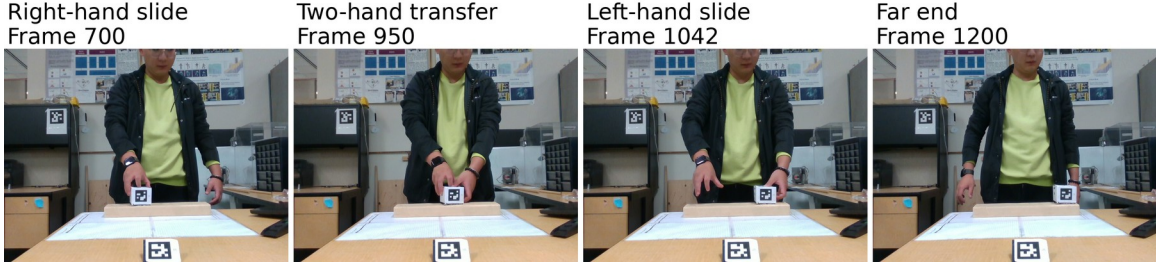
7.2.2 Handover Task

The handover recording holds 1499 frames, of which 1487 supply accepted object samples of the marker centre, and the other 12 are excluded under the same rule. Its waypoints are detected from its own marker-centre motion, with the estimator and the parameters used for the rail recording.

W1 covers frames 0 to 138, an opening interval of 139 samples. W2 is frame 430 and W3 is frame 534, again single transition samples across the lift. W4 covers frames 1083 to 1498, a closing interval of 416 samples, because the cube stays on the rail while the recording continues after the transfer. Table 7.2 reports the resulting lengths against the same physical reference, and Figure 7.2 shows the trajectory.

Segment	Physical length (cm)	Reconstructed length (cm)	Absolute error (cm)	Relative error (%)
W1 to W2	25.5	22.68	2.82	11.06
W2 to W3	3.5	4.35	0.85	24.18
W3 to W4	37.5	40.33	2.83	7.56

Table 7.2. Handover recording: reconstructed marker-centre segment lengths against the physical reference, using the columns of Table 7.1.



Handover recording, 1487 accepted samples

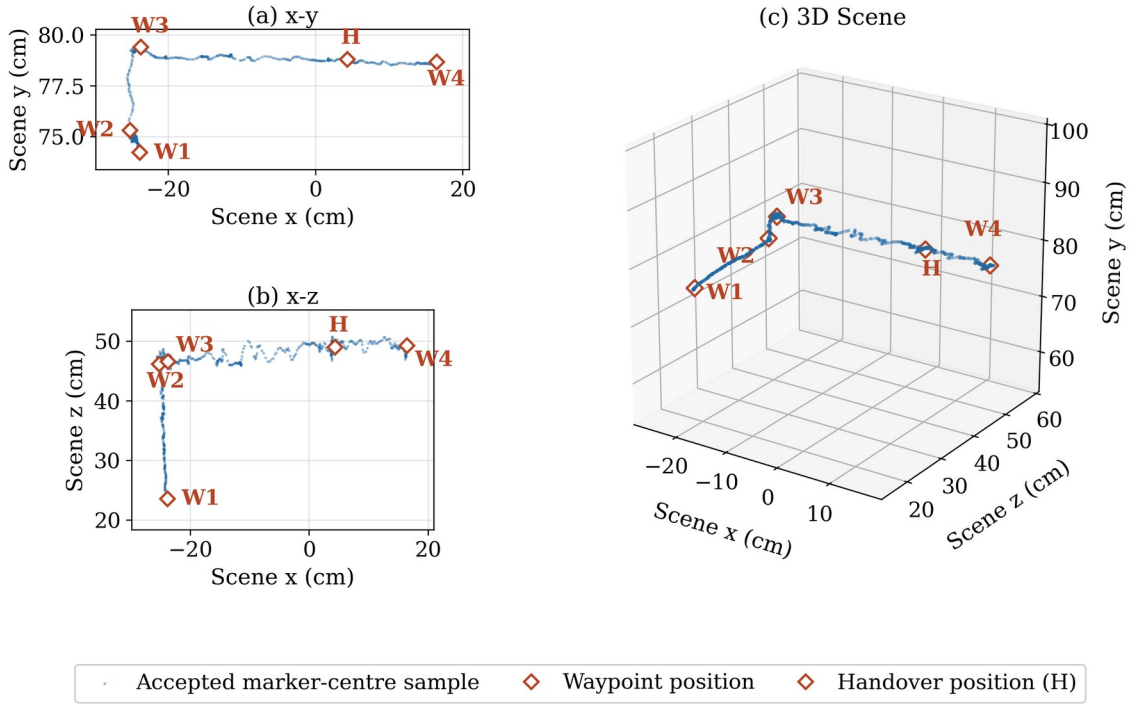


Figure 7.2. Handover recording: video frames, accepted marker-centre samples, the four reconstructed waypoints and handover position H in {Scene}, with this recording's floor translation and scale conventions of Figure 7.1.

The ranges of reconstructed lengths under the tested detection settings are 22.42 to 24.00 centimetres for W1 to W2, 4.28 to 4.76 centimetres for W2 to W3 and 38.78 to 40.60 centimetres for W3 to W4. W2 and W3 are again single accepted samples. Within 7

frames on either side the depth coordinate spreads 15.4 millimetres at W2 and 25.7 millimetres at W3, so the local depth variation at both transition endpoints is larger than in the rail recording.

Over the 539 samples between W3 and W4 the median perpendicular distance from the fitted line is 0.39 centimetres, the p95 is 1.44 centimetres and the maximum is 2.82 centimetres. The fitted line lies 0.6 degrees from horizontal. This remains a within-recording comparison and is reported separately from the length errors.

The two recordings agree on the sign of every error. The depth leg from W1 to W2 reconstructs short in both, while the two legs after the lift reconstruct long in both. The handover recording disagrees with the physical reference more than the rail recording on all three segments, in absolute and in relative terms. Every detection variant included in the ranges above keeps these signs, so the threshold changes the error magnitude within each range but does not produce the pattern. A wider sweep shifts a boundary by ten accepted samples. That sweep lies outside the set of variants above, and it reverses one sign in each recording: W3 to W4 in the rail recording and W2 to W3 in the handover recording.

The proper gravity-alignment rotation G and the common floor translation of Section 6.4 preserve Euclidean distances, including segment lengths and perpendicular distances from a trajectory to its fitted line. They therefore leave Tables 7.1 and 7.2 and the scalar scatter statistics unchanged. Absolute coordinates depend on the origin. Displacement components, fitted-line tilt and the checks used to confirm task phases depend on the axis orientation. Those checks identify which axis has the largest motion. The floor translation shifts the plotted positions without changing those displacements, tilts or checks. The trajectory panels of Figures 7.1 and 7.2 use the final Scene coordinates throughout.

7.3 Human Reconstruction Accuracy under Valid Landmark Measurements

For each arm, the reference contains raw MediaPipe elbow and wrist coordinates with valid aligned depth. A valid-landmark frame is a frame in which the shoulder, elbow, wrist and required torso inputs used by the reconstruction of this section satisfy the landmark-validity checks. Such a frame requires observed, finite, positive-depth

shoulder, elbow, wrist and torso inputs. No input may be filled or rejected after filtering. The per-frame and per-event landmark checks must also accept the frame. Frames outside the detector evaluation interval are excluded. These checks define the conditions under which a measurement counts as valid; they do not establish an anatomical reference.

The corresponding Unity joints are the forearm bone head for the elbow and the hand bone head for the wrist. Their saved world positions include display filtering and rig mapping. The calibrated camera mapping, gravity alignment and recording-specific floor translation carry the measured landmarks into {Scene}, where the logged rig joints are expressed. Frame identifiers match exactly, and timestamp and pelvis correspondence checks verify the saved capture. For joint k at frame t , positional error is

$$e_{k,t} = \left\| {}^{Scene}p_{k,t}^{rig} - {}^{Scene}p_{k,t}^{measured} \right\|. \quad (7.5)$$

Both positions in equation (7.5) are expressed in the Unity scene frame of Section 6.4. The left superscript names that frame, and the right superscript names the source of the position, either the rendered rig or the measured landmark. The statistics describe reconstruction error relative to measured 3D landmarks. Detector-accepted inputs can still produce constrained root or held twist output; the comparison retains those outputs as part of the rendered reconstruction. It does not isolate a kinematic solve whose output groups were all measured.

The chapter distinguishes two reconstruction quantities throughout, and neither is the other. The kinematic-model error compares the measured joint with the position that the kinematic model and forward kinematics place directly from the solved angles, before Unity rig mapping and display filtering. The rendered rig error compares the same measured joint with the final Unity rig joint, after the complete display and rendering path. Equation (7.5) applies to both, with the corresponding reconstructed position.

Both Unity captures ran the avatar with the arm and trunk lengths of the recording being replayed. The arm lengths are the landmark lengths the reconstruction of Chapter 5 solved on. The trunk length is the median distance from that reconstruction's pelvis to the measured mid-shoulder over the torso-clean frames of the same recording. A check on each capture compares the rig the receiver measured against the lengths requested for that

recording, so a capture whose rig carries another recording's lengths cannot be scored here. Tables 7.3 and 7.4 are therefore end-to-end rendered results whose rig reproduces the geometry the solver used. Display filtering, rig mapping and the remaining stages of the display path are not separated by this comparison, so no part of the rendered difference is attributed to any one of them.

7.3.1 Single-Hand Rail Task

The working right arm supplies 102 matched valid-landmark frames from the saved capture. Table 7.3 reports the elbow and wrist errors on this same frame set. The captured frames are not a complete or uniform sample of the task. Every retained frame has a constrained root output, so these results do not establish reconstruction performance without root constraints.

The right forearm twist output is held on every valid-landmark frame of this recording, alongside the constrained root output just noted. A kinematic-model wrist distance computed here would include the effects of the held twist and the constrained root. It would not isolate the agreement between the kinematic model and the measured landmarks. No such distance is reported for this recording or compared with the handover results. The quantitative kinematic-model accuracy comes from the handover recording.

Arm	Joint	n	Median (cm)	p95 (cm)	Maximum (cm)
Right	Elbow	102	6.4	7.5	19.5
Right	Wrist	102	11.4	15.4	22.9

Table 7.3. Single-hand task: rendered joint error relative to accepted measured 3D landmarks on the captured subset of valid-landmark frames.

Figure 7.3 shows the marker-centre and model-wrist trajectories through the single-hand rail task. These views provide spatial context for the captured reconstruction.

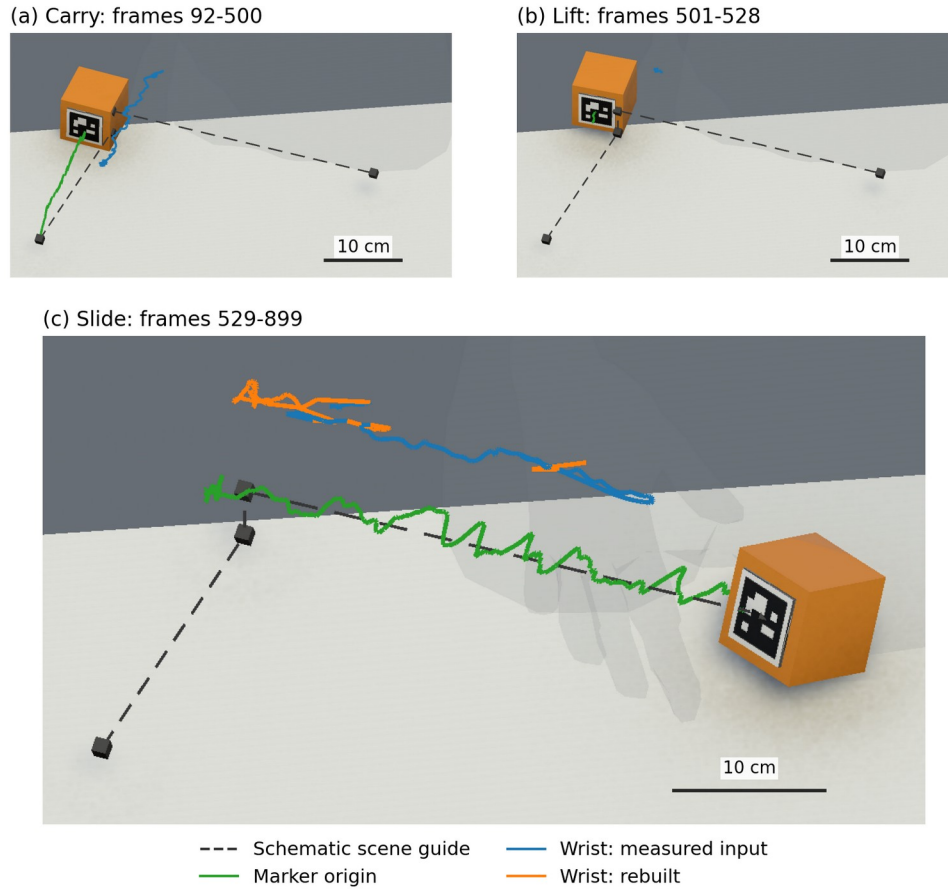


Figure 7.3. Single-hand rail task in Unity: marker-centre and model-wrist trajectories for carry, lift and slide. Wrist colours give the recorded state; avatar and cube show each interval's end.

7.3.2 Handover Task

Table 7.4 evaluates both arms separately. The elbow and wrist of each arm share a valid-landmark frame set; the two arms have different coverage because their measurement validity differs. Figure 7.4 shows the saved Unity trajectories through the handover for spatial context. The quantitative capture is incomplete and includes constrained root output on part of each valid-landmark set. Differences from Table 7.3 cannot be attributed solely to the handover condition because the recorded subsets and rig states also differ.

Arm	Joint	n	Median (cm)	p95 (cm)	Maximum (cm)
Right	Elbow	650	5.3	6.7	9.4
Right	Wrist	650	5.5	6.8	15.0
Left	Elbow	589	6.1	11.9	17.7
Left	Wrist	589	6.8	12.3	17.7

Table 7.4. Handover task: rendered joint error relative to accepted measured 3D landmarks, reported separately for each arm on its captured subset of valid-landmark frames.

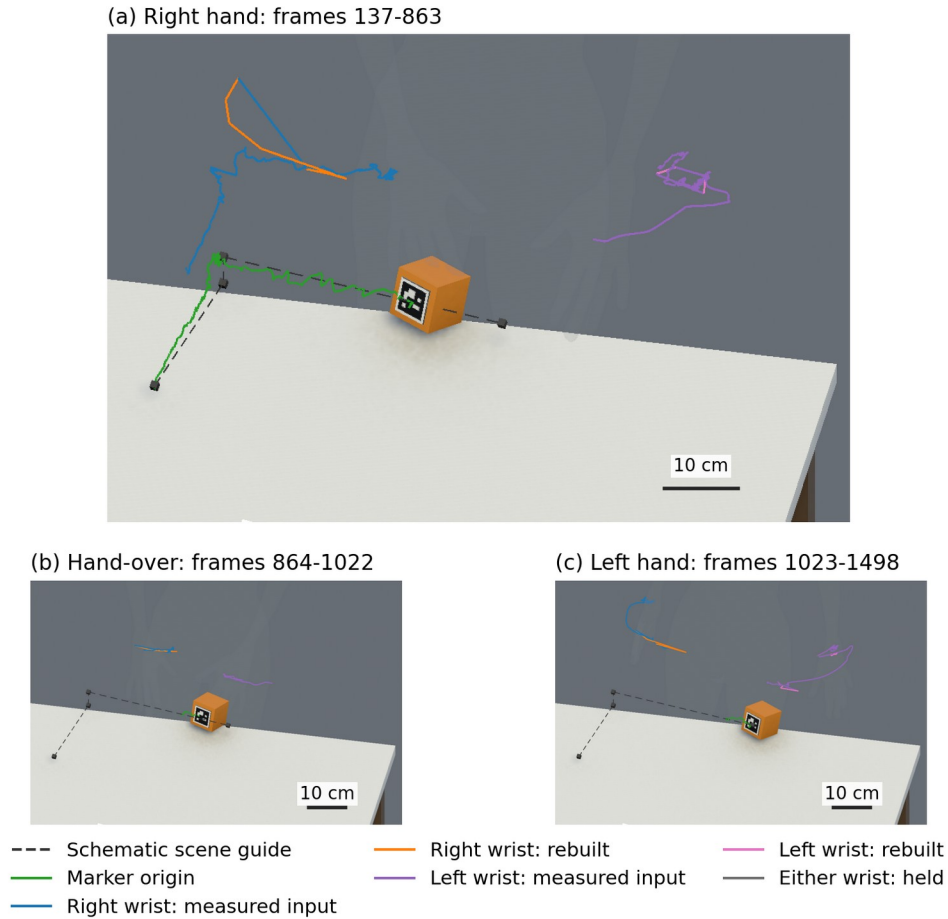


Figure 7.4. Handover task in Unity: marker-centre and both model-wrist trajectories for the right-hand, transfer and left-hand intervals. Wrist colours give the state; avatar and cube show each interval's end.

The kinematic-model wrist error separates the model from the display path. It uses the same 650 right-arm and 589 left-arm valid-landmark frames as Table 7.4. On those frames the kinematic-model wrist lies a median of 0.83 centimetres from the measured wrist on the right and 2.34 centimetres on the left, against the rendered rig medians in that

table. The additional discrepancy arises after the kinematic model, along the path described above.

That capture subset is not representative of the full valid-landmark frame population, and least so on the left arm. It over-represents frames on which the forearm twist output is held, at 55 per cent of the subset against 41 per cent of the valid-landmark frames of the whole recording. Across all valid-landmark frames the kinematic-model wrist median is 1.03 centimetres on the right, over 1156 frames, and 1.03 centimetres on the left, over 1095 frames. These recording-wide values are the representative performance of the kinematic model, and they are not set against Table 7.4, whose frame set is different.

The state of the forearm twist output separates two cases. Where the twist is measured the kinematic-model wrist median lies between 0.6 and 0.9 centimetres; where it is held it lies between 2.3 and 2.8 centimetres. The elbow is almost unaffected, because the twist rotates about the upper-arm axis: on the frame sets of Table 7.4 the kinematic-model elbow median is 0.36 centimetres on the right and 1.08 centimetres on the left.

7.3.3 Human-Object Spatial Check

The separation between the wrist and the held cube has one external physical reference. The author measured approximately 16 centimetres between the physical location corresponding to the MediaPipe wrist landmark and the object ArUco marker centre, for a normal single-hand grip. This is a scalar physical reference, not a per-frame constraint.

Figure 7.5 pairs recorded and rendered views of the single-hand grip during acquisition, lift and slide, captured on 14 September 2026. Red spheres identify joint groups that the stream does not report as measured. The views illustrate the task and retain the captured pose and object-placement discrepancies; the numerical tables use their own logged frame sets and capture configurations.

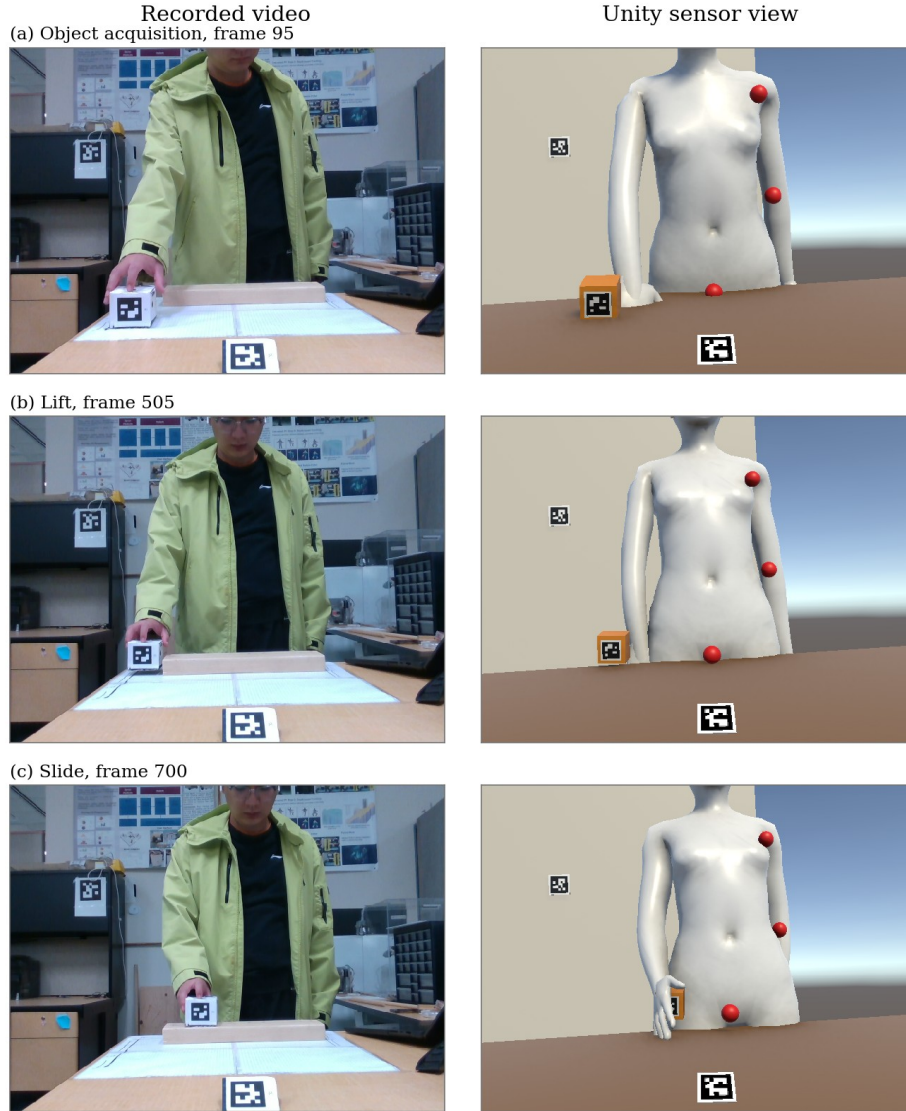


Figure 7.5. Rail recording: object acquisition, lift and slide, with recorded video on the left and the saved Unity sensor view on the right.

Reconstruction places the wrist a median of 15.77 centimetres from the marker centre while the right hand holds the cube alone, and 14.08 centimetres while the left hand holds it alone. Both reconstructed positions are expressed in $\{\text{Scene}\}$. The common floor translation cancels in their difference, and the proper gravity-alignment rotation preserves its norm, so these distances are unchanged. Both medians sit within about two centimetres of the physical reference, so the reconstructed human-object separation is of the correct magnitude.

The two medians recorded during the transfer itself, 17.91 centimetres for the right hand and 15.85 centimetres for the left, are not evaluated against the same reference. Grip geometry changes while both hands are on the cube, so the single-hand measurement does not describe that configuration. Those two values describe the reconstructed grip geometry during transfer.

Figure 7.6 shows the change of holding hand in the recorded scene and the corresponding saved Unity views.

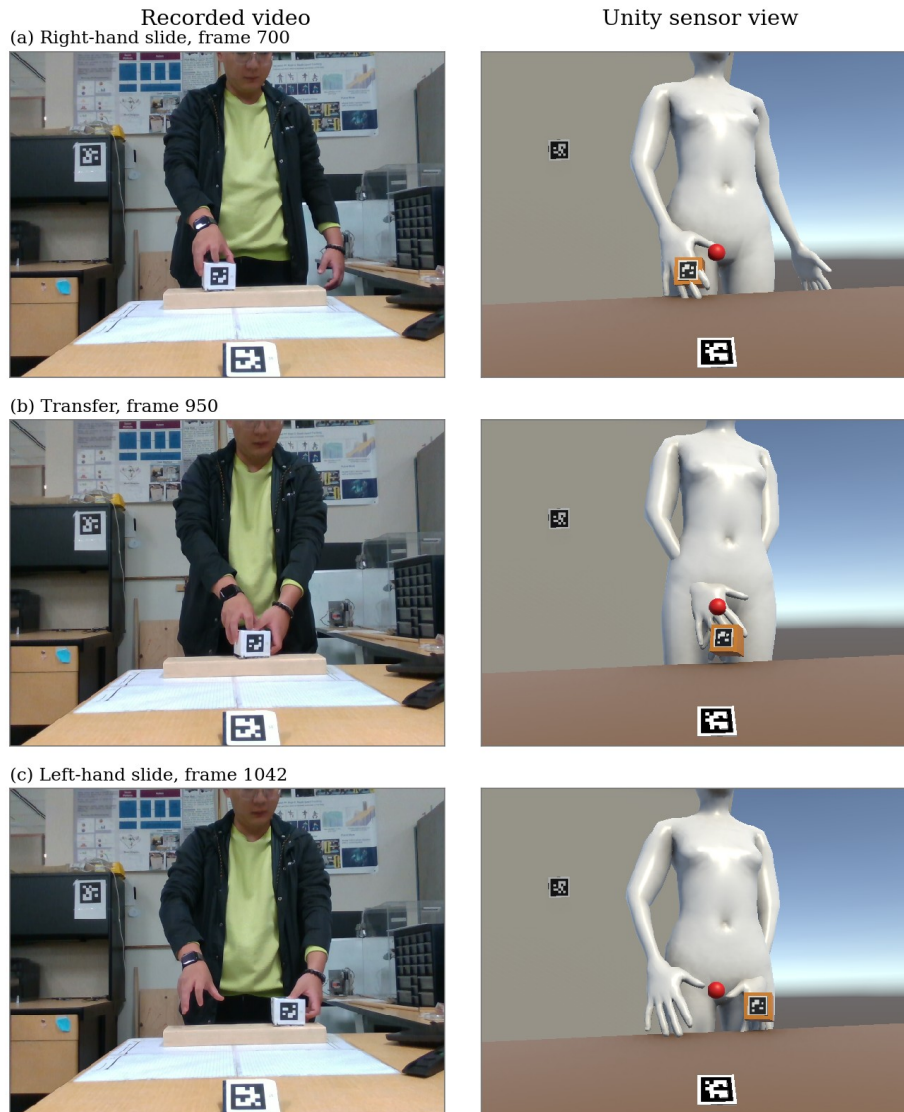


Figure 7.6. Handover recording: right-hand slide, transfer and left-hand slide, using the paired views of Figure 7.5. This visualization was captured on 9 September 2026 with a torso-length override. The visualization supplies qualitative examples; Table 7.4 uses a separate capture.

This is an approximate external physical check. It is not an anatomical reference for the wrist, because the physical endpoint is the location corresponding to a detector landmark rather than a joint centre. It also does not validate the three-dimensional hand-object offset that the recovery of Chapter 5 uses, because one scalar distance constrains neither the direction of that offset nor its behaviour from frame to frame.

7.4 Reconstruction During Landmark Failure

7.4.1 Synthetic Landmark Removal

The experiment removes the right elbow and wrist inputs after offline filtering, while retaining the measured shoulder and torso. The reference is the unmasked kinematic reconstruction of the same frames. Every selected reference frame has valid observed arm and torso inputs, accepted landmark checks, an established grip, and measured root, shoulder swing, shoulder twist and elbow output groups. The handover recording provides suitable windows; the single-hand recording does not provide a fully measured root-and-arm reference window under these conditions.

All eligible windows contain 45 frames, matching the recovery limit. Motion magnitude is the maximum distance of the unmasked wrist from its first position within a window. The windows were selected before any method was scored. The first window has the smallest excursion. The second has the largest excursion among windows that do not overlap the first. The third has the excursion closest to the median of all candidates and overlaps neither earlier selection. Table 7.5 lists the selections. These are contrasting windows from one recording, not independent experimental trials.

Window	Frame range	n	Reference wrist excursion (cm)
Lower motion	490 to 534	45	0.6
Intermediate motion	445 to 489	45	2.8
Higher motion	557 to 601	45	5.1

Table 7.5. Synthetic removal windows selected using valid unmasked wrist motion before method errors were calculated.

The methods are hold-last joint angles, direction-memory recovery, and object-assisted recovery. Each method replays the recording from a fresh initial state and agrees with the

reference immediately before removal. Masked landmarks do not update the hand-object offset, which is already established and remains frozen during the gap. Each method uses the same retained shoulder position and calibrated segment lengths to reconstruct the elbow and wrist. For method m , the deviation from the unmasked reference is

$$e_{m,k,t} = \left\| {}^{Camera'}p_{m,k,t}^{masked} - {}^{Camera'}p_{k,t}^{unmasked} \right\|. \quad (7.6)$$

Both positions in equation (7.6) are expressed in the y-up camera frame {Camera'}, the left-handed frame of Section 3.1 in which the kinematic chain is built, and no display frame enters this section. Tables 7.6 and 7.7 use identical frames and metrics for every method. This is a test of missing solver inputs with fixed offline preprocessing and calibration. It does not measure anatomical accuracy or the effects of removing observations before filtering.

Window	Method	n	Median (cm)	p95 (cm)	Maximum (cm)
Lower motion	Hold-last	45	0.4	0.8	1.0
Lower motion	Direction memory	45	0.3	0.7	0.7
Lower motion	Object-assisted	45	0.7	1.6	1.8
Intermediate motion	Hold-last	45	0.7	1.4	1.4
Intermediate motion	Direction memory	45	1.7	2.4	2.4
Intermediate motion	Object-assisted	45	1.6	2.7	3.0
Higher motion	Hold-last	45	0.6	1.3	1.4
Higher motion	Direction memory	45	0.6	1.5	1.6
Higher motion	Object-assisted	45	2.2	2.9	3.0

Table 7.6. Synthetic elbow reconstruction error relative to the valid unmasked model reconstruction.

Window	Method	n	Median (cm)	p95 (cm)	Maximum (cm)
Lower motion	Hold-last	45	0.7	1.4	1.6
Lower motion	Direction memory	45	0.4	0.7	0.8
Lower motion	Object-assisted	45	1.2	2.0	2.4
Intermediate motion	Hold-last	45	1.2	2.8	3.0
Intermediate motion	Direction memory	45	1.9	3.0	3.0
Intermediate motion	Object-assisted	45	1.2	2.0	2.2
Higher motion	Hold-last	45	1.5	2.5	3.1
Higher motion	Direction memory	45	2.5	4.2	4.6
Higher motion	Object-assisted	45	1.9	2.7	2.8

Table 7.7. Synthetic wrist reconstruction error relative to the valid unmasked model reconstruction.

No method has the smallest error across every window and joint. Object assistance reduces the largest wrist deviation in the intermediate- and higher-motion windows, but increases elbow deviation in the higher-motion window. The lower-motion window favours direction memory. The results therefore support a window-dependent comparison rather than a general recovery ranking.

7.4.2 Natural Occlusion

Natural occlusion is evaluated on the loop recording. A single annotator marked the bead bracelet on the left wrist and the watch on the right wrist. The same visible feature defines the manual wrist label on clean and occluded frames. Each click is deprojected using the aligned depth and colour intrinsics. Table 7.8 compares the label with a filtered wrist measurement in the camera frame {Camera}, and Table 7.9 compares it with the reconstructed wrist in the y-up camera frame {Camera'}. The two frames share an origin and differ by the fixed axis flip of Section 3.1, which preserves distance, so every reported separation is the same in either frame.

The loop recording holds 2099 frames, of which 2058 are detected frames for the object marker. Of the 41 frames without a detected marker, 40 lie in the two-hand transfer on the desk, where a hand crosses the border of the printed square, and the remaining frame

sits at its near edge. The object therefore supplies no geometric constraint across that stretch, while every frame labelled below carries a detected marker. A detected frame is not necessarily an accepted object sample or a reliable trajectory sample. Detection coverage records only the detector's return of an object observation, not the observation's survival through the later cleaning.

Figure 7.7 shows four retained video frames from the natural-failure sample, frames 1427, 1462, 1777 and 1890. The visible wrist features allow manual wrist labelling even when the corresponding landmark input fails.

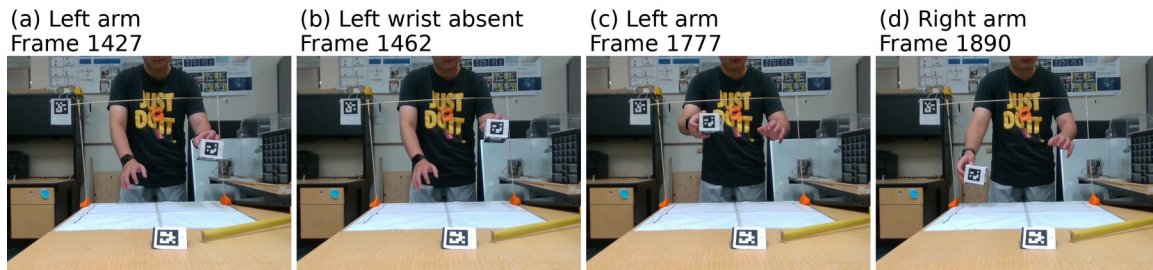


Figure 7.7. Loop recording: four natural-failure frames, with the affected arm named above each view. The bracelet or watch stays visible for manual wrist labelling.

Table 7.8 first measures the distance of the label from accepted, filtered MediaPipe and depth wrist measurements on clean wrist frames. That filtered measurement locates the detected landmark for this annotation comparison and is not the evaluation reference of Section 7.1; the table reports a label separation and not a reconstruction error. The visible feature is displaced from the landmark definition of the wrist; its separation is not an anatomical calibration.

Arm	Label	n	Median (cm)	p95 (cm)	Maximum (cm)
Left	Bracelet	3	4.5	5.4	5.5
Right	Watch	4	2.1	4.8	5.1

Table 7.8. Clean-frame distance between the manual wrist label and an accepted filtered wrist measurement.

During occlusion, the plain solve uses the reported landmarks, hold-last retains the last accepted joint angles, and object-assisted recovery reconstructs the missing inputs. Each method is evaluated against the same retained label points for that arm. The metric is the

Euclidean distance from the reconstructed wrist to the label, summarized in Table 7.9; the clean-frame scalar offset is not subtracted.

Figure 7.8 illustrates a left-wrist failure with no MediaPipe wrist observation. The object-derived wrist estimate and the wrist placed by the reconstructed arm are distinct points. On that frame the plain-solve and object-assisted distances to the label are 14.0 and 13.7 centimetres.

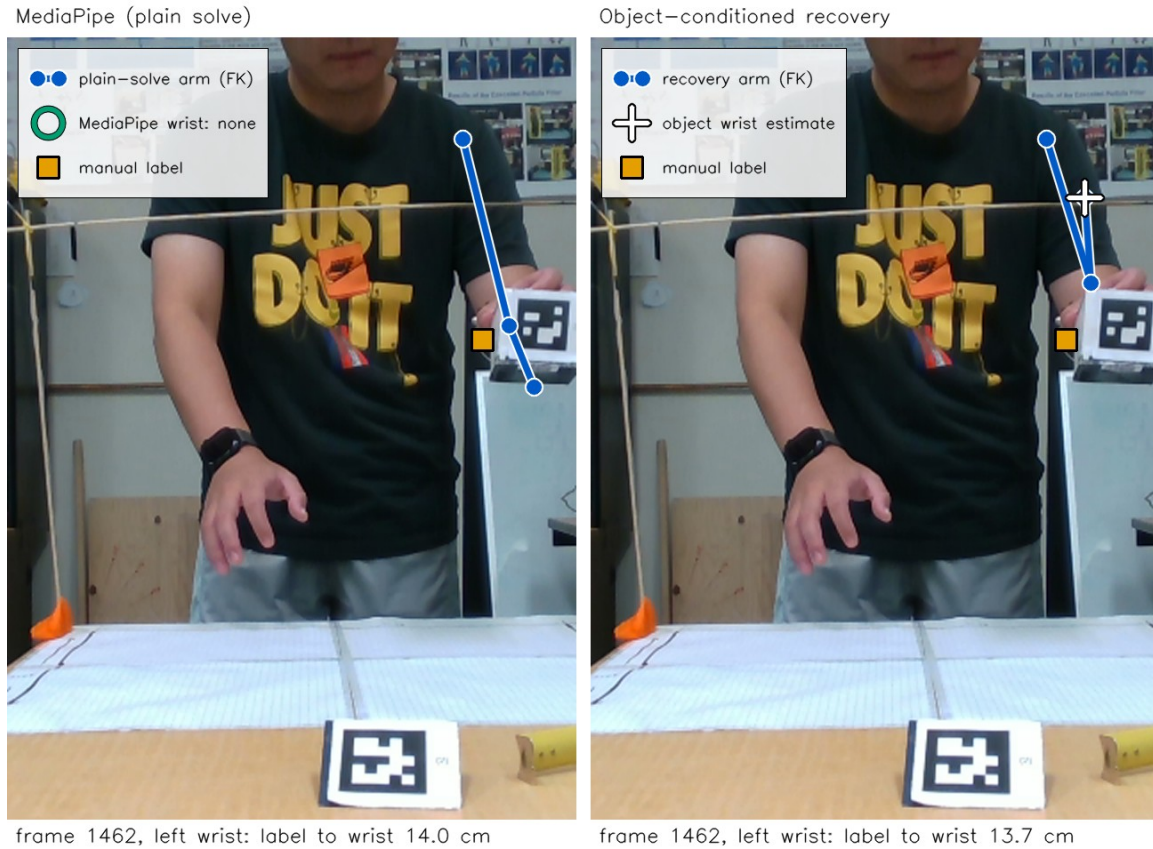


Figure 7.8. Loop frame 1462, left wrist: the square marks the manual wrist label, the blue chain is the reconstructed arm, and the cross is the object-derived wrist estimate.

Arm	Method	n	Median (cm)	p95 (cm)	Maximum (cm)
Left	Plain solve	16	12.3	14.1	14.2
Left	Hold-last	16	12.7	14.3	14.3
Left	Object-assisted	16	13.1	15.2	19.1
Right	Plain solve	4	11.4	13.6	14.0
Right	Hold-last	4	17.6	20.0	20.4
Right	Object-assisted	4	5.2	5.5	5.5

Table 7.9. Reconstructed wrist distance to the same manual wrist label on retained natural-occlusion frames, separately for each arm.

Figure 7.9 shows a right-wrist example in which object assistance places the reconstructed wrist closer to the manual wrist label. On loop frame 1890 the object-assisted wrist lies 5.0 centimetres from the label, against 14.0 centimetres for the plain solve. Together with Figure 7.8, it illustrates the reason the two arms are reported separately.

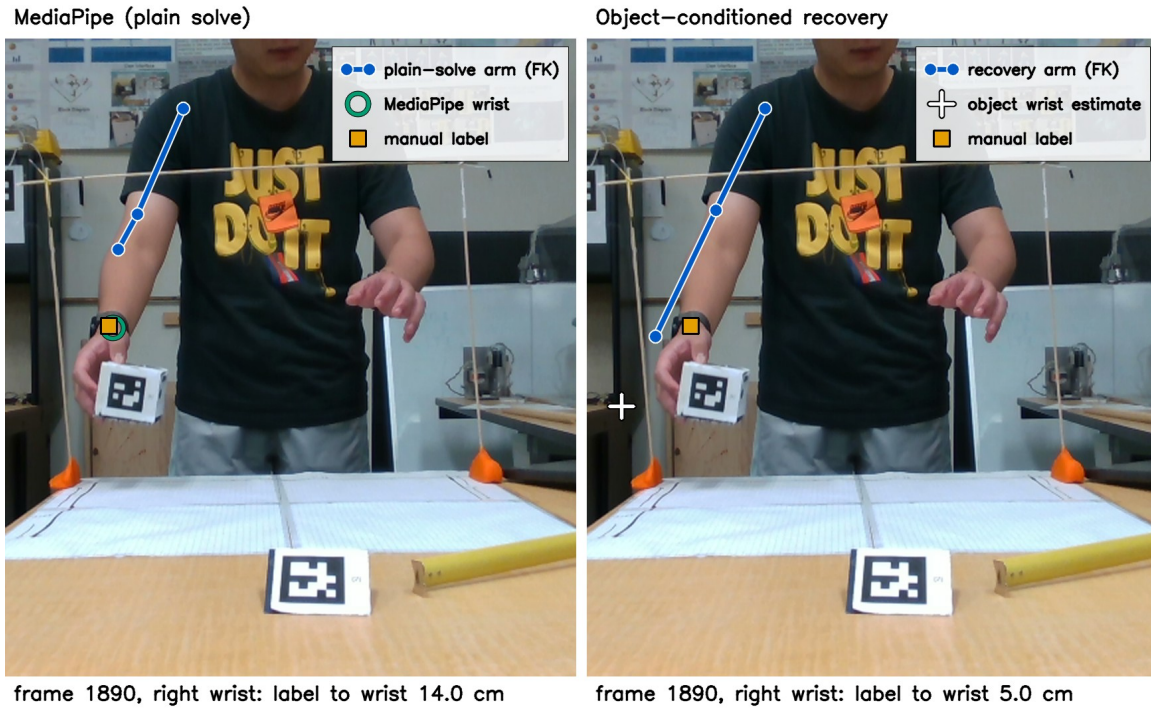


Figure 7.9. Loop frame 1890, right wrist, with the symbols of Figure 7.8. The cross marks the object-derived estimate, not the final reconstructed wrist.

Object assistance is closer to the right-wrist label on this small labelled set, while the left-wrist comparison shows no corresponding improvement. Label-to-wrist separation can change with arm orientation, clothing or jewellery motion, and depth selection. Repeated-

click and depth uncertainties were not quantified. Frames on which the marked feature was completely hidden were excluded, so the results describe selectively visible samples rather than all occlusion frames. These distances do not establish anatomical accuracy. Checking the reconstructed wrist against the hand-object offset used by recovery tests only the compliance of the wrist with the recovery constraint. The offset is not an independent accuracy reference.

Chapter 8: Real-Time Feasibility

Every result of Chapter 7 came from a finished recording, and a clinical or robotic use needs the reconstruction while the motion happens. This chapter tests the transfer of the architecture of Chapters 2 to 6 to causal processing, in which no sample from the future is available. It names the offline operations that read the future and gives the causal replacements.

8.1 From Offline to Causal Processing

A stage is causal when its output for a frame depends only on that frame and the frames before it [50]. The offline pass breaks that condition in six places, each one reading a sample a live system would not yet hold. The landmark smoother of Section 2.5 is a fourth-order Butterworth low-pass run forward and backward over the whole signal, and the two passes cancel the phase delay only because the backward pass starts at the end of the recording [50], [52]. The despiking test of the same section compares a sample with the median of a window centred on it, so half of that window lies ahead. Gap bridging needs the sample on the far side of a gap.

The hand-object offset of Chapter 5 warms up on the mean over a whole grip episode. The grip tracker of the same chapter confirms an episode exit over a run of five frames and then clears those frames, so the state of a frame changes after the frame has passed. The cleaning rule of Chapter 4 reads the object track as a complete time series, after it has been smoothed.

The causal structure must replace each of those operations with one that reads only the past, or do without it. This chapter tests the exchange by replaying the rail recording of Chapter 2 through the causal structure at its recorded pace and comparing the output with the offline pass frame by frame. The task phase names of this chapter, among them the lift and the slide, are those of Section 2.1. Replaying one recording keeps the scene fixed, so a difference between the two outputs follows from the processing path and not from the motion, the lighting or the camera placement.

8.2 Causal and Real-Time Architecture

Figure 8.1 sets the offline pass and the causal structure side by side. Offline, every script opens the recording itself and writes a table that the analysis tools of Chapter 7 read afterwards. Live, one capture process owns the sensor or the recording, aligns depth to colour once, stamps each frame with the session clock, and publishes the pair into a shared frame buffer (Section 6.1). The landmark branch and the marker branch read that buffer as separate processes and write their records. Every record carries the frame index and the session time of the frame it came from, and that pair is the only synchronization between the branches.

The merger pairs the two records at a render time that trails capture by a configured delay of two frames. It writes the combined record that the Unity receiver of Chapter 6 draws. The per-joint output states of Chapter 5 travel in the streamed packet, so a constrained joint group is distinguishable on screen from a held one.

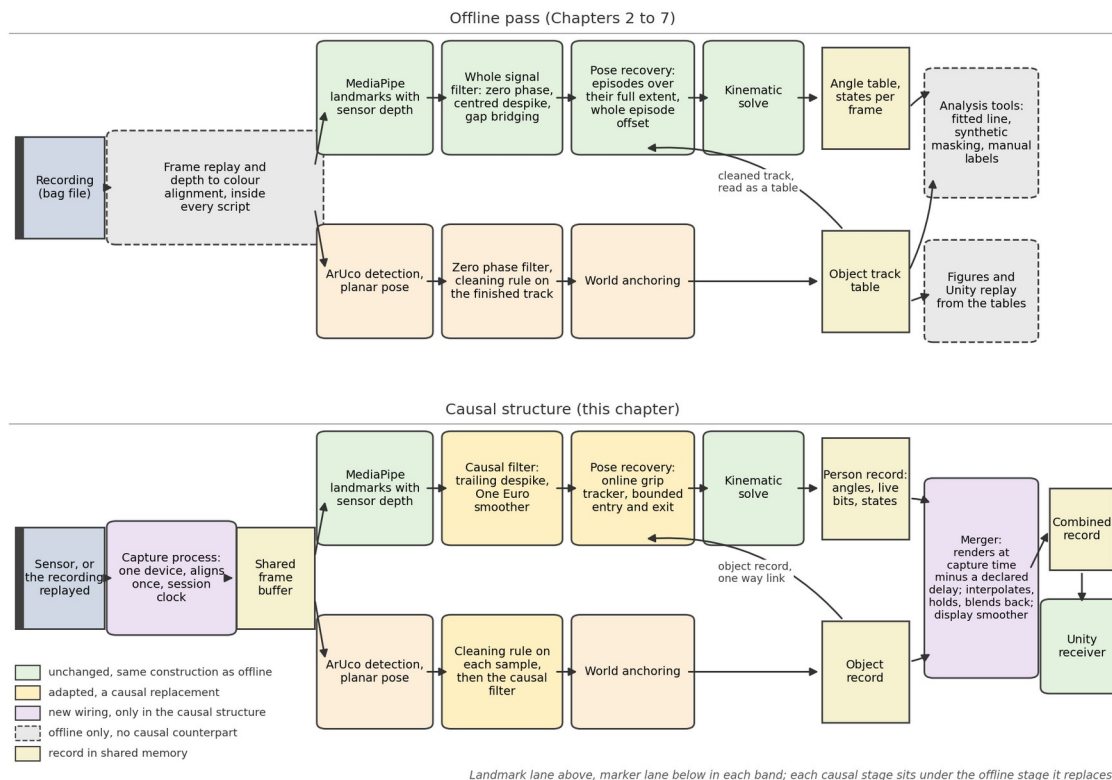


Figure 8.1. The offline pass (above) and the causal structure (below), each causal stage under the offline stage it replaces. The legend names the lane and role of each box.

The structure rests on three choices. A single process owns the device, because two processes cannot open the same camera. The merger renders at that delayed render time rather than at the newest sample. The delay is a buffering choice, distinct from the lag of the causal filter, which Section 8.3 shows and Section 8.4 separates from this delay. The display smoother of Section 6.3 acts on the output record only, so each branch's record holds only the values that branch computes.

Table 8.1 sorts the stages into three groups: unchanged, adapted, and analysis tools with no live form. The kinematic solve of Chapter 3 is closed form and holds no state across frames. The recovery layer of Chapter 5 does carry state: the direction memories, the offset estimate, the holding state, the twist hold and the rate limit. Every update reads the current frame and the state that earlier frames left behind, at a bounded cost per frame. The marker detector and the planar pose solver read a single image.

The despiking window trails the sample instead of surrounding it, and a rejected sample becomes a missing sample. After the filter rejects three samples in a row, it accepts the next one and rebuilds its window around it, so the trailing window does not go stale. Gap bridging moves into the merger, which bridges only within its delay and holds the last pose beyond it (Section 6.1.2). The zero-phase Butterworth smoother gives way to the One Euro filter on each landmark coordinate [53]. Its cutoff frequency increases as the coordinate changes faster, so it smooths a still pose and follows a moving one.

The marker branch keeps its mechanism and changes its order. The cleaning rule of Chapter 4 discards a sample farther from the last kept sample than the slow task can produce. Live, it runs on each detected sample ahead of the causal filter, whereas offline it runs on the smoothed track.

The grip tracker of Chapter 5 changes in two places. The hand-object offset keeps its slow update, and its warm-up becomes a running mean over the first clean frames of the episode, fitted in the frame the solver works in. The test for a carried object takes the first object samples as its rest reference and gates entry only, so a pause with the object held does not split an episode. Exit is confirmed only after five consecutive frames of a cleanly measured wrist beyond the release radius.

capability	offline form	live form	relation
Failure detectors, hip depth preparation, twist hold, angle rate limit	per frame	same construction	unchanged
Wrist from object pose; two-link elbow solve	per frame	same construction	unchanged
Object pose read by landmark branch	cleaned track, as a table	newest marker record, rebuilt in {Camera}	new data path
Per-joint output states to the reconstruction	in the record	in the streamed packet, through the merger's buffer	new data path
Landmark despiking	window centred on sample	trailing window; rejection becomes a missing sample	adapted
Interpolation across a gap	whole signal	only between samples the merger holds; beyond, last pose held	adapted
Landmark smoothing	forward-backward Butterworth	One Euro per coordinate	adapted
Cleaning rule on the object track	on the smoothed track	on the detected sample, before the causal filter	adapted
Hand-object offset and grip episodes	warm-up over whole episode, then slow update	warm-up over first clean frames, then same update; entry and exit decided over a few frames	adapted
Comparison against the line fitted to the slide; failure mask as reference; within-recording comparison under synthetic landmark removal	over a finished recording	none, beyond the solver's gates and the grip-plausibility detector	offline only

Table 8.1. The parts of the offline pass of Chapters 2 to 5 that the causal structure keeps; the replacements of this section are the adapted rows.

The two capabilities marked as a new data path needed no new mathematics. Live, the recovery reads the newest record the marker branch published and rebuilds the object

pose in the camera frame {Camera}. It undoes the axis swap of Section 6.2. It then composes two poses: the calibrated pose of the world frame in {Camera} of Section 2.2.2, and the object pose in the world frame. It rejects a record whose frame index sits more than three frames from the frame being solved. The offline-only group has no live form beyond the grip-plausibility detector of Section 5.2 and the solver's own gates, which do the work of the failure mask.

8.3 Offline-Causal Comparison

Setting the two outputs against each other is a within-recording comparison, one output of the pipeline beside another output of the same recording. Neither path is the reference for the other, and a difference between them is a processing-path difference rather than an error against physical truth. Chapter 7 evaluates the offline pass against physical references, measured landmarks and manual wrist labels, and against its own outputs where no independent reference exists. Section 7.2 sets the reconstructed object segment lengths against tape measurements of the physical route, and Appendix H verifies the kinematic solve against synthetic exact references.

Figure 8.2 draws the right elbow flexion and the right shoulder twist of both paths on the same frame axis in three windows of the recording. Figure 8.3 draws both chains on the colour frame at one moment of each window. Each chain is carried back into {Camera} by the fixed axis flip of Section 3.1, and the colour intrinsics then place it on the image. The angles of the recovery solve of Section 7.4 place the offline chain and those of the causal path place the causal chain, both built in the y-up camera frame {Camera'} on the measured shoulder of the frame.

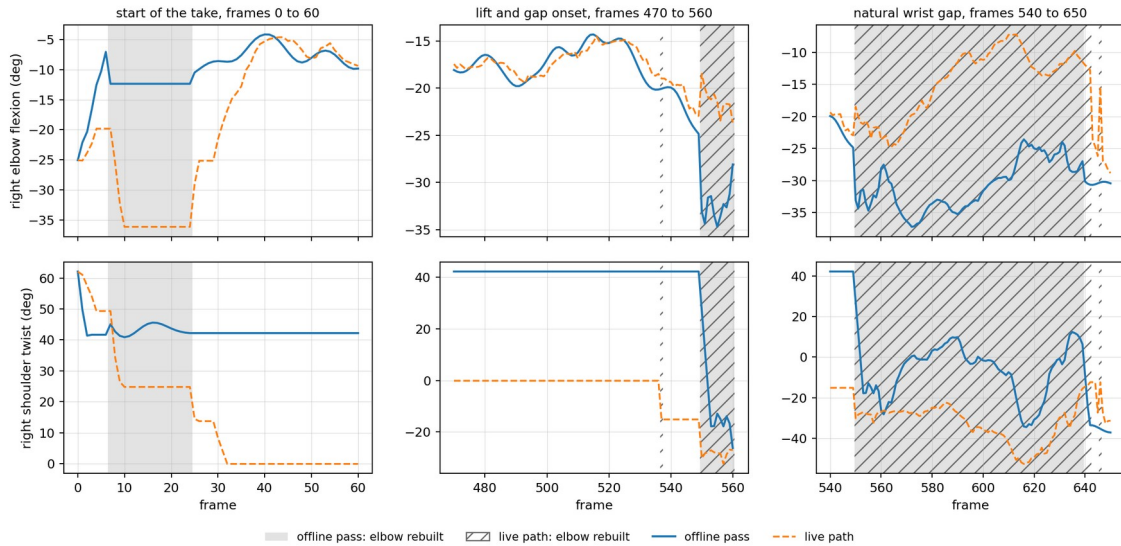


Figure 8.2. The two paths: the right elbow flexion above, the right shoulder twist below. The grey band marks frames the offline pass rebuilt, the striped band the causal path.

The smoother loses its view of the future, so the causal filter follows a moving joint late. In the lift onto the rail of Figure 8.2 the causal curve trails the offline one by a few frames and rounds its turns. The lag is small on this slow task and grows with the speed of the motion.

The states carried by the solver depend on the starting frames. The direction memories, the hand-object offset and the twist hold of Section 5.5, which acts while the elbow is nearly straight, all keep whatever the earliest frames gave them. On this recording the offline failure mask rebuilt the right arm over a window of frames 7 to 24 while the live gates held it. The two paths therefore froze shoulder twists about 40 degrees apart and kept them until the elbow bent far enough inside that window to release the twist hold (Figure 8.2, lower row). At frame 18 the offline pass rebuilt the arm and the causal path held it, and the wrists of the two forward-kinematics chains sit 12.0 centimetres apart (first panel of Figure 8.3). The frozen twists keep the two wrists apart on the later frames where every landmark is measured.

The hand-object offset differs as well, so the two paths rebuild the wrist onto different points of the object. Just before the natural wrist gap opens at frame 548, the offline pass still follows the tracker's wrist, which has already drifted onto the object, while the causal path has begun to rebuild from the object (second panel of Figure 8.3). Inside that gap the

rebuilt elbow-flexion angles differ by as much as 24.1 degrees (Figure 8.2, right), although at frame 595 the wrists nearly coincide (third panel of Figure 8.3). The recovery also starts and ends on different frames along the two paths, because the live gates and the offline mask do not fire on the same frames. The causal exit rule keeps the wrist anchored to the object during the frames that confirm a release, while the offline episode logic clears those frames once it sees the release.

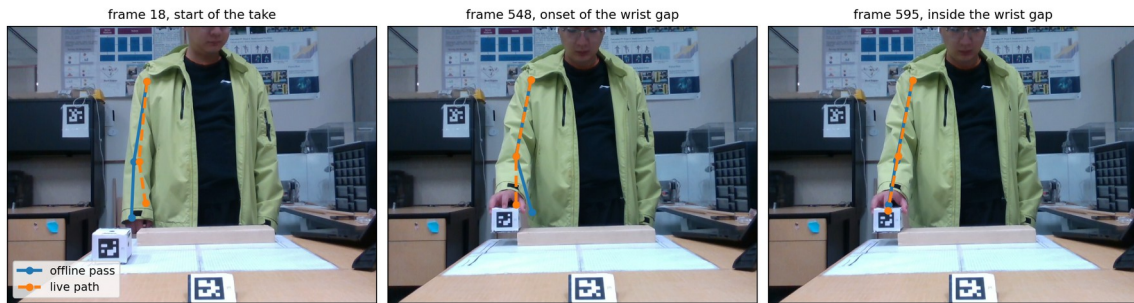


Figure 8.3. The offline chain (blue) and the causal chain (orange) drawn on the colour image at frames 18, 548 and 595, one moment from each window of Figure 8.2.

A rerun of the live validator on the archived records of the same replay puts numbers on the comparison. Each causal frame is paired with the offline frame the alignment places beside it, and the arm must be cleanly measured on both sides of the pair. The alignment that brings the two angle series closest together is five frames, and the pairing keeps 174 frames. At that alignment the elbow flexion of the two paths differs by a median of 0.4 degrees, the shoulder elevation by 1.4, and the shoulder twist by 7.0.

The shoulder azimuth differs by a median of 16.4 degrees and carries the shoulder group to a 95th percentile of 23.3 degrees. Section 3.4.2 gives the reason: the arm hangs close to straight down through the slide, near the configuration in which the azimuth is undefined. The comparison covers the arm alone, because the hip depth preparation of Section 2.6 rejects a measured hip on 858 of the 900 frames of this recording, so the root is rarely a measured group on both sides at once.

The merger blends a return from a long gap over two ticks instead of snapping to it (Section 6.1.2). On this recording the merger never had to hold a person record mid-run.

8.4 Runtime, Buffering, and Delay

On the workstation of Appendix B the causal structure consumed all 900 frames of the recording at its recorded pace, sustaining about 29 frames per second, with none dropped, and the merger emitted its output at a steady interval. Detection takes most of each frame's budget on both branches, and no stage this thesis adds exceeds 0.63 milliseconds at the 99th percentile. Each branch also finishes inside the frame budget of 33.3 milliseconds at the 99th percentile, the landmark branch within 13.0 milliseconds and the marker branch within 16.7. A lighter detector variant is the documented way to reduce the cost [54].

This chapter reports three timing quantities, and they are separate. The configured two-frame delay is a buffering and render delay: the merger renders the world two frame intervals behind capture so that it can interpolate between two samples it already holds. It is not a measured end-to-end latency from the sensor to the screen, which this thesis does not measure. The second quantity is the lag of the causal filter, which depends on the One Euro filter and the speed of the motion. The five-frame alignment of Section 8.3 is the third, an estimate of the shift that brings the two angle series into closest agreement on this recording. None of the three quantities bounds the others.

8.5 Feasibility and Remaining Validation Gap

Sections 8.2 to 8.4 show that the architecture transfers to causal processing on this recording. Causal forms replaced every operation that read the future, and the reconstruction still followed the task through the natural wrist gap of Section 8.3.

The experiment does not demonstrate a working live system. No complete live-camera session was validated, and the step from the recording to the device leaves several items untested. A scene calibration built from a live stream needs the printed size of each marker before the first detection, including the 45-millimetre desk and object markers of Table 2.1. The calibration of this thesis was computed from a recording. The rest reference of the grip tracker assumes that the object begins at rest and in view, which a recording guarantees and a live session does not. The capture process has been exercised on a file and not on the sensor. The synthetic landmark removal of Section 7.4.1 and the manual wrist labels of Section 7.4.2 both read a finished recording, so a live run can

report only the outcome its own gates decided. Section 7.4.1 evaluates the removal against the unmasked reconstruction.

The result is therefore feasibility evidence for the architecture under causal processing, and not a validated real-time system. Chapter 9 relates the differences of Section 8.3 and the remaining validation gap to the rest of the thesis.

Chapter 9: Discussion

This chapter reads the results of Chapters 7 and 8 as evidence about one integrated system. The system runs two branches from one RGB-D camera and three printed markers. The landmark branch gives a torso pose and four joint angles per arm, and the marker branch gives the world frame and the object pose. A recovery layer marks each joint group as measured, held or constrained, and the same chain also runs causally.

Each part was evaluated against a reference of its own. The points below state the conclusions those results support and the limits they reveal. The limits are of three kinds: the view of a single camera, the quality of the measurements, and the values the system carries over from earlier frames.

- Comparing the reconstructed object path with simple tape-measure distances shows a clear pattern in the errors: carry segments come out a bit shorter than they really are, while lift and slide segments come out a bit longer. Within the range of detector settings tested, adjusting them changes the size of the errors, not their direction. Lift points are the least reliable because each one depends on depth from a single video frame. Some extra uncertainty also comes from the wobble and rotation of the object, since those checks rely only on the recording itself, not an external reference. When the system loses the marker, it simply gives no position rather than a wrong one, and these losses happen only outside the labelled parts of the data, so the labelled comparisons remain unaffected. Figure 9.1 shows one such loss, and Figure 7.3 shows the three segments of the reconstructed path.

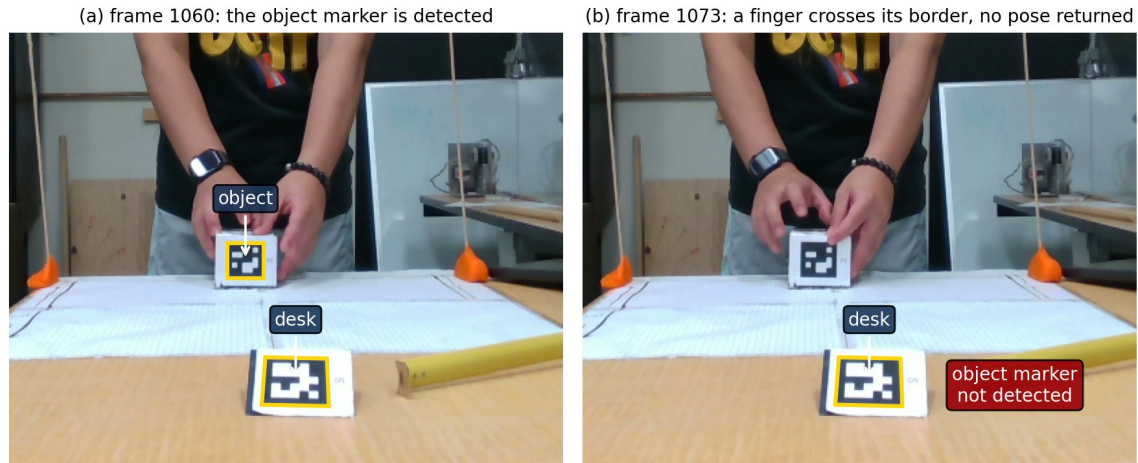


Figure 9.1. Object marker loss on the loop recording. (a) Both markers are detected, their corners outlined. (b) A finger crosses the printed square and no object pose is returned.

- The placement of the person and the object together in the scene has one outside check: the tape-measured distance from the wrist to the cube in the hand. The reconstruction comes close to it. The wrist the kinematic model places sits closer to the measured wrist than the wrist the drawn avatar shows, and the difference is not blamed on any single step of the path that draws the avatar. The distance also depends on the state of the forearm twist, measured or held; on the rail recording it is always held, so the avatar hand lands beside the cube. The two recordings also give different arm lengths, because the elbow landmark sits farther down the arm on one of them, and that is a property of the landmark rather than of the person. Figures 7.5 and 7.6 show the video beside the drawn avatar, and Figures 7.8 and 7.9 show a wrist label against the reconstructed arm.
- Using the object to place a hidden hand helped on one side and not on the other. The stored hand-to-object distance is learned while the hand is visible and updates slowly, so it lags behind a changing grip and freezes when tracking fails. An arm pointing straight at the camera leaves the wrist depth uncertain, so a small error there moves the elbow a long way, and a twist held through a gap is kept by the frames that follow. In the tests where landmarks were removed on purpose, no method was best on every window and joint. A closer wrist position on its own does not prove that the whole arm is right. Figure 5.3 shows the stored hand-to-object distance, Figure 5.5 the elbow reconstruction, and Figure 7.7 the natural failure frames.

- The torso frame that every arm angle is measured against depends on the two hip landmarks, and the desk can hide them. A hidden hip takes the depth of whatever is in front of it, so an upright person appears tilted forward, and because both hips shift together, no failure detector notices the shift. A landmark can also be wrong while the detector is confident about it: the detector once placed the elbow on the hand (Figure 9.2), and the check on the arm's length caught the mistake. When the hip depth preparation switches on or off, the torso frame moves with it. Holding the hips at their last good position while covered would look steady, but nothing measures the hips independently, so that rule remains untested. Figure 3.4 shows the torso frame built from the hips.

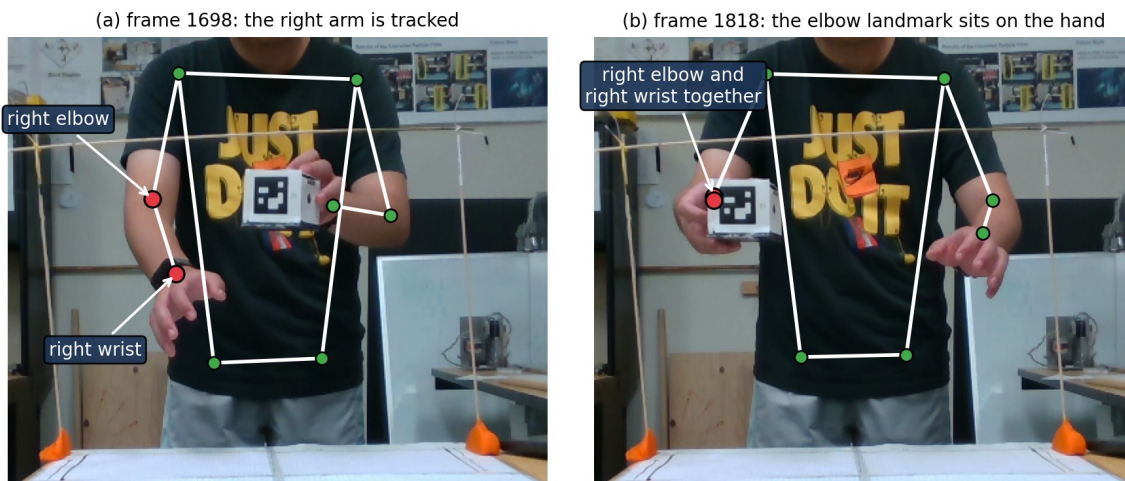


Figure 9.2. Landmark tracking failure on the loop recording. (a) The right elbow and wrist landmarks sit correctly. (b) The elbow landmark has moved onto the hand.

- The model itself solves four of the seven ways an arm can rotate, treats the wrist as a point with no orientation and the torso as one rigid plate, and does not track the legs. The other limits are of three kinds. A single camera cannot see a hand behind the cube or the desk, cannot judge depth well along the line of sight, and cannot tell the turn of a hanging arm or of a wrist. Measurements can be wrong: a landmark can sit off the body, a depth reading can come from the wrong surface, and marker corners can be noisy. The calibration's vertical direction also rests on the mounting angle of the wall marker, which was never measured. Finally, values carried between frames, such as a stored hand-to-object distance or a held angle, can be out of date. Figure 3.6 shows the arm chain and Figure 3.7 the swing and twist of a joint.

- Running the system live, frame by frame, gives a different result from processing the whole recording afterwards, because the live version cannot look ahead. In one gap the offline pass rebuilt the arm while the live pass held it, so the two froze the forearm twist at different angles, and their wrists ended up about 12 centimetres apart. The gap did not close once tracking returned. Live runs match each other only if both start from the same first frames, so running it for real needs a defined start, with the object at rest in view. The live version kept pace with the recording, but only as a replay of a file. Chapter 8 lists the points a real camera session still has to prove, and Figure 8.3 draws the offline and live chains on the same frames.
- The evaluation rests on one person doing a slow task in one controlled setup, recorded three times. Nothing here shows the accuracy of the system for other people, grips, clothing, tasks or rooms. The wrist labels used for checking are points clicked on a bracelet or a watch, not the exact wrist the detector defines, and the label sets are small, so a single frame can change a summary statistic. Some references are missing altogether. The physical route was never lined up in the same frame of reference as the reconstruction. The object's orientation has no reference of its own. The live result is a replay rather than a camera session, and the model stops at the upper body. Table 9.1 collects the six main limitations with the numbers behind them.

Limitation	Cause	Evidence on the recordings
Object lengths disagree with the physical reference	Not isolated by these recordings; the detector settings set the size of the disagreement, not its direction	Section 7.2: the carry reconstructs short on both recordings, by 1.14 and 2.82 cm, and the lift and the slide reconstruct long, by 0.53 to 2.83 cm
Physical route not registered	The measured route is a set of endpoint separations and is not expressed in the calibrated world frame	Section 7.2 compares endpoint separations and reports no distance from the reconstructed track to the physical path
Rendered output differs from the kinematic solve	The discrepancy appears downstream of the solve, in the rig and display path, whose stages these results cannot separate	Handover recording, captured frames: kinematic model wrist 0.83 and 2.34 cm from the measured wrist against 5.5 and 6.8 cm for the rendered rig joint (Section 7.3.2)
Segment lengths are effective, not anatomical	The calibrated upper arm and forearm are medians of the landmark track, and the elbow landmark need not sit at the anatomical elbow	Rail right arm 31.7 and 20.5 cm, loop right arm 25.6 and 25.2 cm, beside an independent measurement of about 25 cm for both segments on the subject; the rail elbow landmark sits 5.9 cm farther along the arm
Recovery helps on one occluded window and not on the other	Hand-object offset frozen with the estimator's lag; a nearly straight arm along the optical axis	Loop recording, right arm: object-assisted 5.2 cm from the manual wrist label against 11.4 and 17.6, on four retained frames (Section 7.4.2)
No live-sensor session	The capture process has not been run against the camera, and the online offset estimate needs the object at rest at the start	The real-time result comes from the recording replayed at its recorded pace

Table 9.1. The six principal limitations of the system, their causes, and the evidence for each on the recordings.

The results show that the approach works on the tasks that were recorded. They do not establish the accuracy of the system for other people or other tasks. Chapter 10 concludes and sets out future work.

Chapter 10: Conclusions and Future Work

10.1 Conclusions

This thesis develops and evaluates a single-view RGB-D framework for reconstructing upper-body motion together with the motion of a manipulated object in a shared metric scene. The framework combines RGB-D body landmarks, fiducial object tracking, an explicit upper-body kinematic model, failure detection, object-assisted pose recovery, and graphical reconstruction in Unity, with body and object motion expressed in one calibrated world frame. Under the tested conditions, the system reconstructed the torso, both arms, and the tracked object from one consumer-grade RGB-D camera, and the same architecture also operated causally at approximately the recorded frame rate.

The results support the following points.

- A single camera cannot see everything. Hands disappear behind the object or the desk, and distance along the line of sight is hard to judge, so some joints are hidden or poorly placed with any detector. The recovery layer that fills those gaps is therefore a needed part of the system, not a cosmetic filter.
- Using the object's position to place a hidden hand helped on one hand in one recording and did not help on the other hand. It works only under the right conditions: the stored hand-to-object distance must still be current and the arm must not point straight at the camera, and even then the evidence is a handful of frames rather than a general rule. A good wrist position on its own also does not prove that the whole arm is right.
- The errors come from the parts interacting, not from one weak part. A better detector alone would still leave the parts of the scene hidden from a single camera, and more cameras alone would still leave the errors that come from carrying old values forward, so no single upgrade removes them.

The contribution is a working integrated system together with a full account, in Chapter 9, of the motion it can and cannot recover from one camera. The evaluation covers one person and three controlled recordings. The physical route was never lined up in the same frame of reference as the reconstruction, so no point-to-path accuracy was established,

and the live version was tested by replaying a recording rather than with a camera in the room. The results show that the approach works; they are not a general accuracy claim.

10.2 Future Work

The current system struggles when the stored hand-to-object distance goes out of date or when several measurements fail at the same time. Future work can take three directions.

- The first is more cameras looking from different sides. Several cameras placed around the scene could see the parts hidden from one alone, so calibrated cameras watching the same wrist, elbow, hip or marker would leave far fewer gaps to fill, while everything would still be measured in the same shared scene.
- The second is a newer, more reliable pose detector. The current detector, MediaPipe, sometimes places a landmark on the wrong body part with high confidence. A stronger model such as RTMPose-m, using the COCO-17 keypoint layout and running on a graphics card through ONNX Runtime, could make such mistakes rarer without changing the rest of the system, provided the detector stays fast enough for live use.
- The third is a small learned model that judges which measurements to trust. It would look at the recent landmark history, the current pose, the detector's confidence, the failure flags and the object's position, and weigh them. The kinematic chain would keep the geometry consistent so that the arm can never take an impossible shape.

As continued improvements from hardware scaling become increasingly difficult and costly [59], system architecture and integration become more important.

Appendix A: RGB-D Data Acquisition

A.1 Acquisition Procedure

Prepare and record the session in the following order.

1. Draw the working area on grid paper on the desk. Use the RealSense D435 sensor [10] with the workstation of Appendix B for one continuous take.
2. Place the rail inside the working area and set it level.
3. Fix the three markers at the back wall, at the world origin on the desk and on one face of the cube.
4. Mount the camera on its support at the desk edge, at about chest height. Aim it to keep the subject, desk, rail and all three markers in view.
5. Request the colour stream in 8-bit-per-channel bgr8 format and the depth stream in 16-bit z16 format. Set both streams to 640 by 480 pixels at 30 frames per second.
6. Verify the negotiated stream profiles against the requested profiles before keeping any frame. Stop capture with a message naming the stream if they differ. A slower bus can cause the device to select a lower resolution or frame rate without warning.
7. Start recording with the streams running once the subject is at the desk. Perform the task once: lift the cube from the desk onto the rail, then slide it to the far end. Write everything the device delivers into the single session file described in Section A.2.
8. Stop recording after the subject steps back. Retain the complete take, including the approach and retreat spans, so later processing stages can segment it from the recorded data.

A.2 Structure of the Recorded Bag File

The sensor stores each session as a single bag file, a rosbag container in which named topics carry messages held in compressed chunks. The layout separates description from data, as Figure A.1 draws and Table A.1 lists. The file stores its version, the device and sensor descriptions, the value of every sensor option at record time, and the calibration of each stream once at the start. The camera intrinsics of Appendix D come from that calibration. The rest is a sequence of per-frame messages, an image message and a metadata message per captured frame.

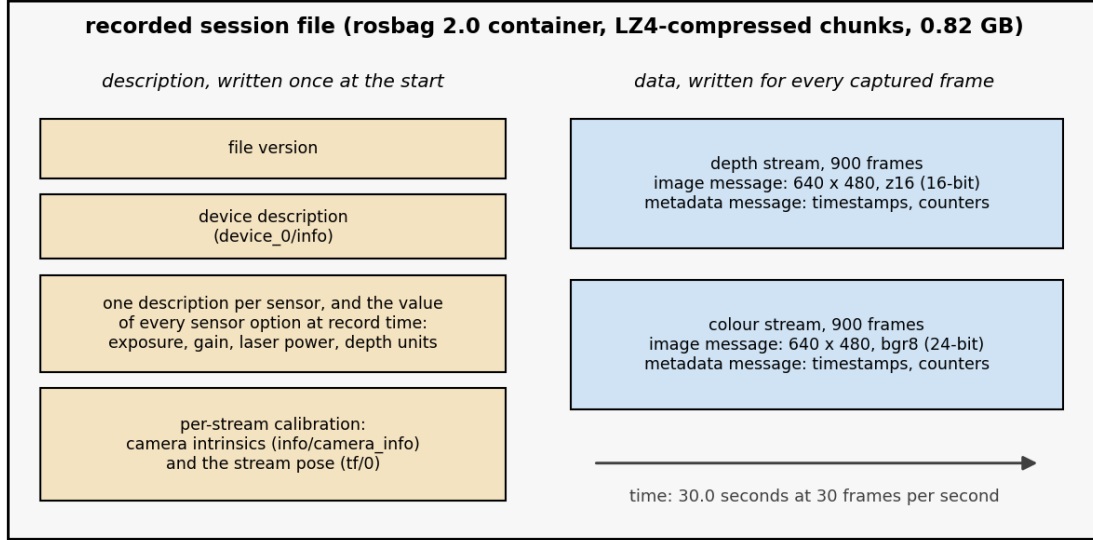


Figure A.1. Structure of the recorded session file: the one-time description topics on the left, the per-frame image and metadata messages on the right.

Replay reads the same structure back, so a recording carries its own calibration and needs no separate calibration file.

Property	Value, read from the recording
container	rosbag 2.0, compressed chunks
streams	depth (z16, 16-bit) and colour (bgr8, 24-bit)
resolution and rate	640 by 480 at 30 frames per second, both streams
duration	29.99 seconds, first to last frame
frames delivered on replay	900 depth and 900 colour
raw pixel payload	1.38 GB
file size on disk	0.82 GB
one-time topics	file version, device and sensor descriptions, sensor options, per-stream intrinsics and pose
per-frame topics	image data and frame metadata, per stream

Table A.1. Contents of the recording's session file.

A.3 Replay and Visual Inspection

The vendor's viewer application opens the same session file and replays it as though a live sensor were attached, which makes it the quickest way to audit a recording before

any processing runs. Figure A.2 places the camera's own view of one moment beside the point cloud the viewer deprojects from it.

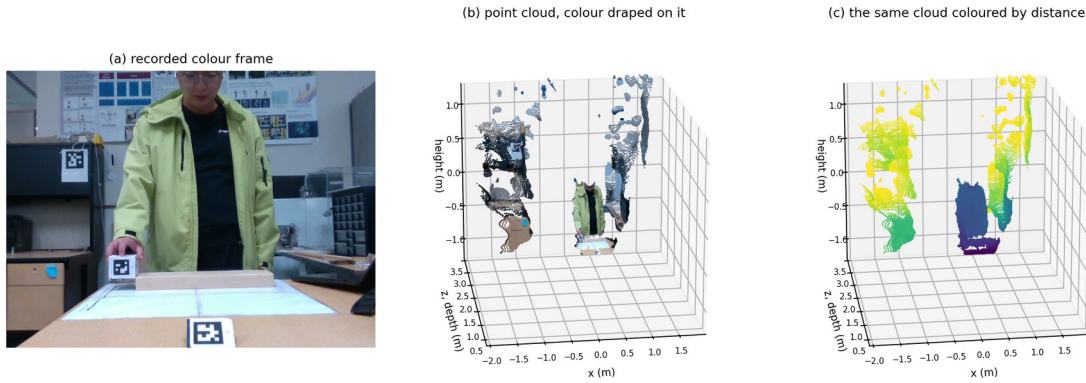


Figure A.2. (a) The recorded colour frame of the worked-example moment. (b) The same moment as a metric point cloud with the colour stream mapped onto it. (c) The same cloud coloured by distance.

A second, automated pass reads the raw outputs of the two extractors and reports the contents of the recording before any modelling starts. It lists the frames in which the subject is present, the segmentation into approach, manipulation and retreat, the frames on which each marker and each landmark is available, and an inventory of the occlusion episodes. The pass reports these conditions without rejecting the recording, with a single exception: because the desk marker defines the world origin, a frame on which it is missing stamps the recording as rejected and stops every stage after it.

Appendix B: Hardware and Software Platform

Every processing result and every timing in this thesis was produced on the one workstation of Table B.1, and the cost measurements of Chapter 8 should be read against this configuration. The division of labour is fixed throughout: the pose detector's neural network inference runs on the graphics card through the framework's GPU delegate. Everything else, including the depth sampling, the marker pose recovery, the kinematic mathematics and the filtering, runs on the processor in a single thread.

Component	Value
processor	AMD Ryzen 9 7950X, 16 cores, 32 threads
memory	64 gigabytes (GB)
graphics card	NVIDIA GeForce RTX 3080, 10 GB
sensor	Intel RealSense D435 RGB-D camera [10]
operating system	Ubuntu 24.04 long-term support (LTS)
Python	3.11.9
RealSense library	2.55.1
MediaPipe	0.10.15
OpenCV	4.10.0
NumPy	1.26.4
SciPy	1.13.1
Unity editor	6000.3.19f1

Table B.1. The workstation and the software versions behind every result in this thesis.

Appendix C: Pose Landmark Detection

C.1 MediaPipe BlazePose Overview

The detector this work uses is the BlazePose model [54], deployed through the MediaPipe framework [30], whose landmark network regresses 33 keypoints inside the region of interest that a person detector localizes. For each of them it returns normalized image coordinates, a depth estimate inferred from appearance, and a visibility score between 0 and 1 for its confidence that the landmark is present and unoccluded. The pipeline runs the heavy variant in video mode, which tracks the person across frames rather than detecting from scratch on each one. It expects a single person and leaves the detection, presence and tracking confidence thresholds at 0.30. The appearance-derived depth is not used, because it is unreliable during fast motion and object interaction [3], [13]. Metric depth comes instead from the aligned depth image of Appendix D, which is a measurement rather than an inference.

C.2 Landmark Coordinate Extraction

Of the 33 landmarks the model returns, this thesis uses the eight of Table C.1, covering the torso and the two arms.

ID	Landmark	ID	Landmark
L11	left shoulder	L12	right shoulder
L13	left elbow	L14	right elbow
L15	left wrist	L16	right wrist
L23	left hip	L24	right hip

Table C.1. The eight MediaPipe landmarks used in this work, with the model's indices.

The extractor takes three steps for each landmark on each frame. The normalized coordinates are scaled to the colour image and truncated to whole pixels, so the pixel u is the integer part of the horizontal coordinate times 640 and v the integer part of the vertical coordinate times 480. The aligned depth image is sampled there by the 5 by 5 median of Section D.3, and the pixel with its depth is back-projected by equation (D.2). Figures C.1 and C.2 show the eight landmarks of the worked-example frame, at the sampled pixels and as position vectors from the sensor origin.

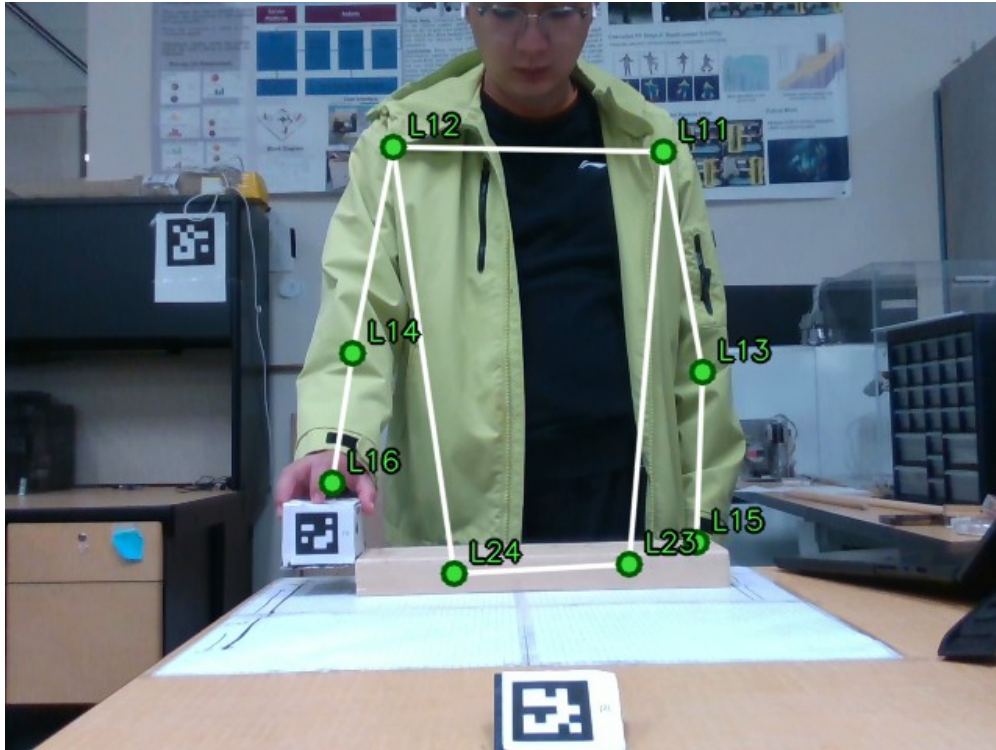


Figure C.1. The eight upper-body landmarks on the worked-example frame, drawn at the pixels the extractor sampled, with the segments between them.

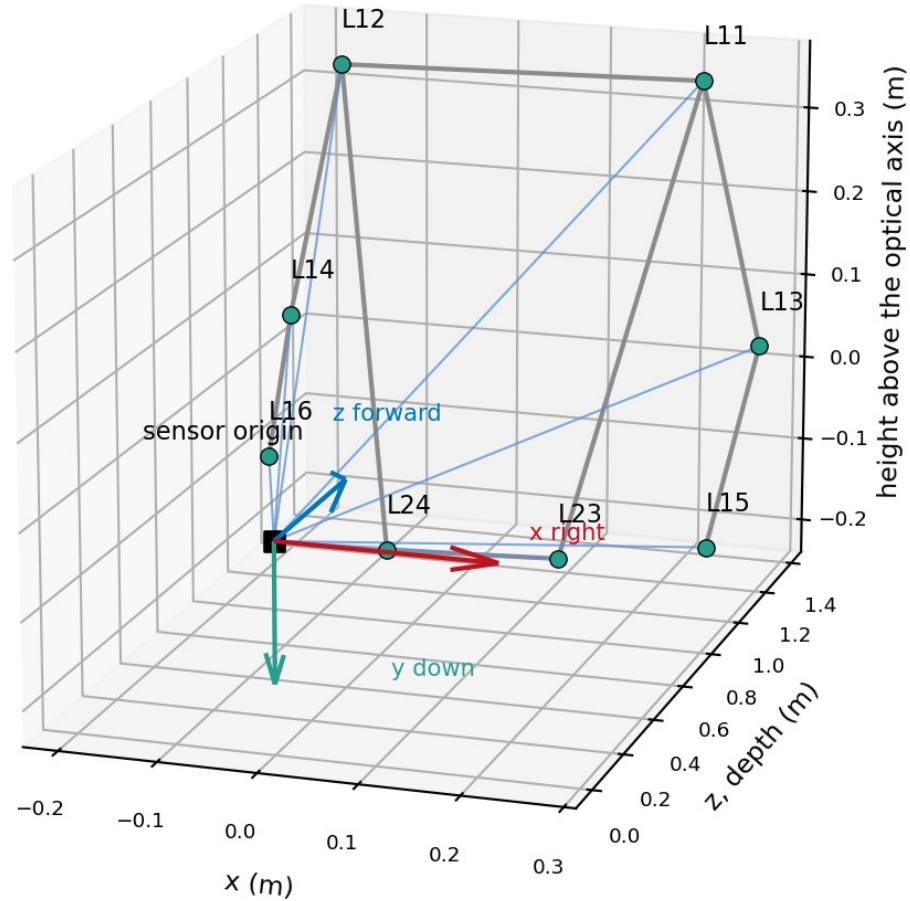


Figure C.2. The same eight landmarks as metric position vectors from the sensor origin, with the camera axes drawn at the origin.

The model returns coordinates for all 33 landmarks, visible or not, so an occluded wrist still receives a position. The visibility score is then the only signal that separates a measurement from a guess. A landmark whose visibility falls below 0.50 is therefore written as an empty record, and the same applies when the 5 by 5 depth window holds no valid return. Every exported landmark carries a code recording which of three cases produced it: a measured sample, a sample blocked by low visibility, or a sample blocked by missing depth. Later stages can therefore tell a gap from a value.

C.3 Worked Example

Take the right wrist, landmark L16, on frame 533 of the recording, the frame of Figure C.1. The detector reports it with a visibility of 0.58, above the 0.50 gate. The extractor stores the sampled pixel, not the normalized pair. Multiplying the normalized coordinates by 640 and 480 and truncating gives the pixel (209, 308).

The 5 by 5 median depth at pixel (209, 308) is 1.06 metres. Back-projection through equation (D.2), with the intrinsics of Table D.1, gives

$$x = 1.06 \cdot \frac{209 - 323.94}{607.56} = -0.20$$

$$y = 1.06 \cdot \frac{308 - 248.02}{607.02} = +0.10,$$

so the right wrist sits at (−0.20, 0.10, 1.06) metres in the camera frame. Table C.2 carries the same three steps through the other two landmarks of the right arm, and Section 3.5 continues from the filtered form of these numbers into the kinematic solve.

Landmark	Visibility	Pixel	Depth (m)	Camera-frame position (m)
L12 right shoulder	1.00	(248, 93)	1.31	(−0.16, −0.33, 1.31)
L14 right elbow	0.70	(222, 225)	1.20	(−0.20, −0.05, 1.20)
L16 right wrist	0.58	(209, 308)	1.06	(−0.20, 0.10, 1.06)

Table C.2. The right-arm landmarks of the worked-example frame, from the visibility score through the sampled pixel and depth to the back-projected position. These are raw extractor positions, before the filtering of Section 2.5; the filtered values are in Table 3.1.

Appendix D: Depth Sensing and Deprojection

The Intel RealSense D435 is a stereo-depth camera that produces a per-pixel depth measurement alongside a colour image [10]. Depth comes from triangulating between two infrared imagers separated by a known baseline, assisted by an infrared projector that adds texture to surfaces too uniform for stereo matching. The scene spans depths from about 0.53 metres at the desk marker to about 3.2 metres at the wall marker, inside the sensor's usable range. Stereo depth noise grows roughly with the square of the distance [40], [41].

D.1 RGB Camera Calibration

Calibration establishes the relationship between a 3D point in the scene and the pixel it lands on. The pinhole model describes it through four numbers: the focal lengths in pixels along the two image axes, and the principal point, the pixel where the optical axis meets the image [60]. A point p with coordinates x , y and z in the camera frame projects, as shown in Figure D.1(a), to the pixel

$$u = f_x \frac{x}{z} + c_x, v = f_y \frac{y}{z} + c_y. \quad (\text{D.1})$$

This work performs no manual calibration. The vendor's library exposes the factory calibration held in each device, and the recording carries those values in its own calibration topic, so the intrinsics used are the ones the sensor reports for the colour stream at 640 by 480, listed in Table D.1.

Parameter	Value	Meaning
fx	607.56 px	focal length along the horizontal image axis
fy	607.02 px	focal length along the vertical image axis
cx	323.94 px	principal point, horizontal (image centre 320)
cy	248.02 px	principal point, vertical (image centre 240)
distortion model	inverse Brown-Conrady	the model the stream declares
distortion coefficients	all zero	the recorded values for this stream

Table D.1. Factory colour-camera intrinsics, read from the calibration topic of the recording.

The table connects back to the model in three ways. The two focal lengths differ by less than 0.1 percent, so the pixels are square to that precision. The principal point sits a few pixels from the geometric image centre, so the deprojection of Section D.3 measures every pixel offset from that point rather than from the centre. The distortion coefficients are all zero, so the vendor's general deprojection routine reduces to the closed form of equation (D.2) for this data. Ignoring the small principal-point offset, the focal lengths and image size give a nominal field of view of 55.6 by 43.1 degrees, which guided the camera placement in Chapter 2.

D.2 Depth-to-RGB Alignment

The depth imager and the colour camera sit at a small offset inside the sensor housing, and each has its own intrinsics [10], so a raw depth pixel does not observe the same scene point as the colour pixel at the same location. The alignment step drawn in Figure D.1(b) resolves this. Each depth pixel is lifted to a 3D point through the depth intrinsics, moved into the colour camera's frame through the known transformation between the two imagers, and projected back onto the colour image grid through the colour intrinsics of Table D.1. That transformation is held inside the vendor library and is never applied by this pipeline, so only the aligned colour optical frame, the camera frame, is used downstream. The registration is geometric: a colour pixel and its aligned depth now share

one ray, but the depth belongs to the nearest surface along that ray. A landmark on a body part behind the cube therefore takes the cube's depth, the case Section 5.2 deals with.

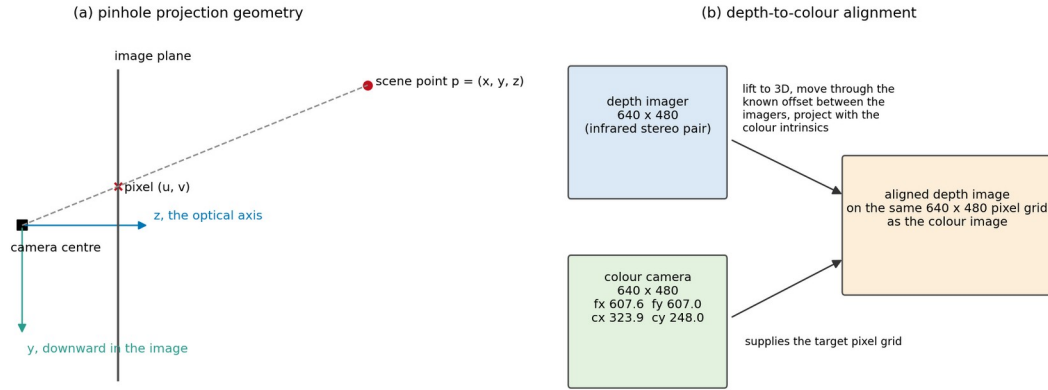


Figure D.1. (a) The pinhole geometry the intrinsics describe: a scene point, the ray to the camera centre, and the pixel where it crosses the image plane. (b) Depth-to-colour alignment: the depth image is reprojected into the colour camera's geometry so that both buffers share one pixel grid.

Both buffers are 640 by 480, so the aligned depth image shares the pixel grid of the colour image and the two values at one pixel describe the same physical point. That aligned pair is the only input the rest of the pipeline sees, and both detectors work in colour pixel coordinates that index the depth image directly.

D.3 Pixel-to-Camera Mapping

A pixel with a valid depth is lifted to a 3D point in the camera frame by running the projection backwards:

$$x = z \frac{u - c_x}{f_x}, y = z \frac{v - c_y}{f_y}, \text{Camera } p = (x, y, z). \quad (\text{D.2})$$

The camera frame is right-handed, with x to image right, y downward and z forward along the optical axis, and the depth measurement is the z coordinate itself. The depth image stores integer counts, one count being one millimetre for this device. The fixed y -axis flip of Section 3.1 expresses the same point in the left-handed y -up camera frame $\{\text{Camera}'\}$, used for body kinematics. Its origin is still the camera origin; it is not person-centred. Chapter 6 gives the further mapping into Unity.

Reading depth at a single pixel fails in two ways. The pixel can give a zero return, which is common at object edges and on dark or reflective surfaces, or it can carry single-pixel speckle. The pipeline therefore takes the median of the nonzero depths in a 5 by 5 window centred on the target pixel, counting a zero return as an absent sample rather than as zero metres. When the window holds no valid return, the sample is reported as missing.

D.4 Worked Example

Take the object marker on frame 533, the frame of Appendix C. The marker detector of Appendix E places its centre at pixel (200, 341), where the 5 by 5 median depth is 0.99 metres, and equation (D.2) with the intrinsics of Table D.1 gives

$$x = 0.99 \cdot \frac{200 - 323.94}{607.56} = -0.20$$

$$y = 0.99 \cdot \frac{341 - 248.02}{607.02} = +0.15,$$

so the marker centre sits at (−0.20, 0.15, 0.99) metres in the camera frame. Running the result forward through equation (D.1) returns the pixel the detection started from, so the two equations are inverses.

Appendix E: Fiducial Marker Detection and Pose Recovery

The landmark detections of Appendix C measure the subject. The objectives of Chapter 1 also require a measurement of the manipulated object in a frame that does not move with the camera, and a scene geometry against which the reconstruction can be read. Fiducial markers supply both, because they are printed patterns whose geometry is known exactly, so their pose follows from a single image with no training data and no appearance model. Table 2.1 gives the three markers and their roles. This work uses ArUco markers because they detect reliably, cost little and encode their own identity [31]; the size and distance relationship of [32] set their printed sizes.

E.1 Detection and Sub-Pixel Corners

Detection runs in four stages. Adaptive thresholding turns the image into binary regions by comparing each pixel with the mean of its own neighbourhood, which keeps the detector working under uneven lighting. Contour extraction finds candidate quadrilaterals. The interior of each candidate is perspective-corrected and matched against the dictionary, which either rejects it or returns a marker identity, and the four corner positions are refined below the pixel grid. Figure E.1 shows the result on the worked-example frame.

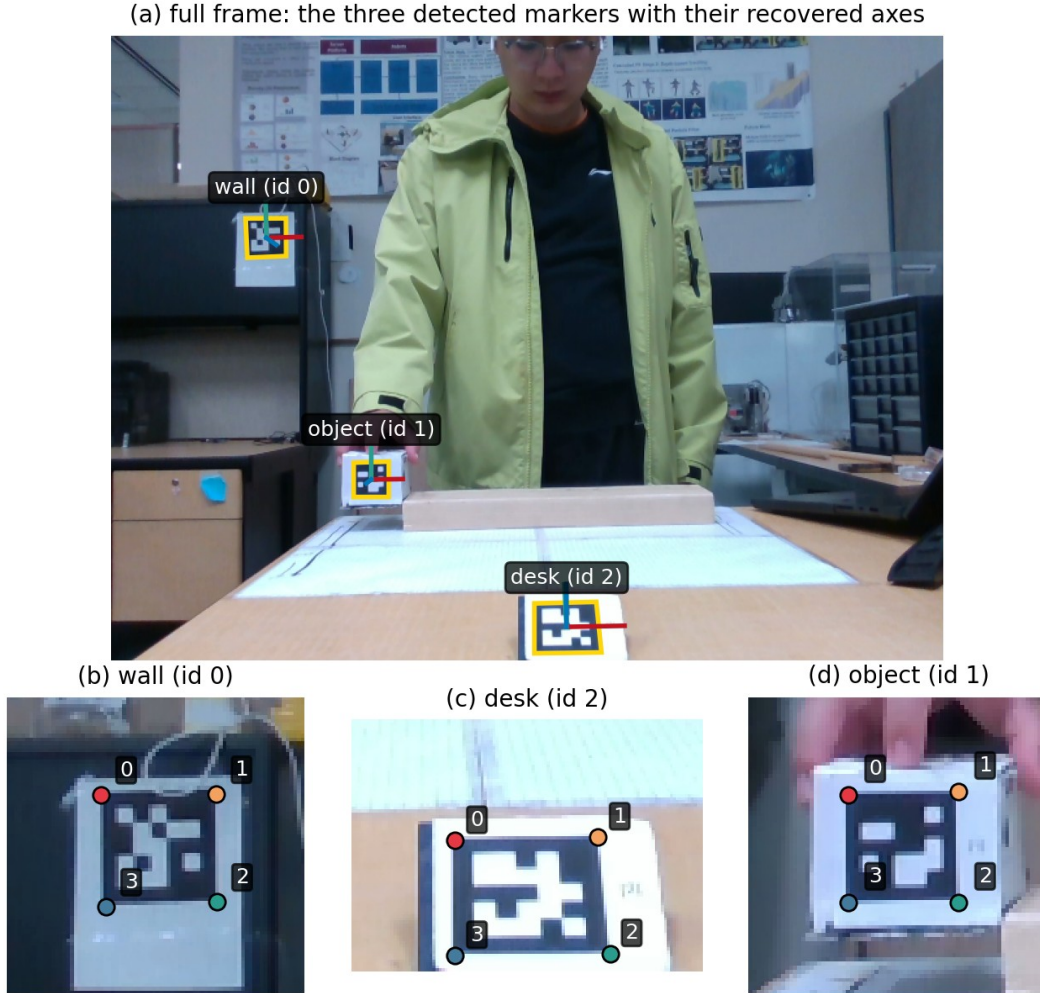


Figure E.1. Marker detection on the worked-example frame. (a) The full frame with the three detected markers outlined and the recovered axes at each centre. (b) to (d) Each marker enlarged, with its four refined corners numbered in detection order.

Detection itself is standard and is not a contribution of this work [31]. The corner refinement that follows it is a design decision, because corner precision limits pose precision. With the detector's default settings the coordinates land on whole pixels, so a marker that sits still is reported at identical corners frame after frame, which is the detector returning to the same pixel grid, not a stable measurement. The pipeline therefore enables the refinement based on the AprilTag method [61], which takes each corner as the intersection of straight lines fitted to the two marker edges that meet there. Every frame of the recording uses the refined corners.

The marker branch also records, at detection time, the median of a 5 by 5 patch of aligned depth values at each marker's centre pixel. The pose recovery never reads it. The value comes from the other measurement principle, the stereo depth stream rather than the marker geometry, and it remains in the recorded data, although no analysis of this thesis reads it. Table E.1 lists it beside the recovered translation for the worked-example frame. A frame on which a marker is not detected stays an empty record.

E.2 Planar Pose Recovery and the Two-Solution Ambiguity

A detected marker gives four corner points in the image. The printed marker gives the same four corners in its own frame, at plus and minus half the printed side length L in the marker plane where the third coordinate is zero, in the order the detector returns them. Recovering the pose means finding the orientation of the marker's own frame relative to the camera frame, and the position t of the marker origin in the camera frame. Together they carry these four known corners onto the four detected pixels through the projection of equation (D.1). The marker's own frame follows the rule of Section 2.2.1, so the pose recovered here is that of the world frame for the desk marker, of the wall marker frame, or of the object frame, each relative to the camera frame. The corners carry noise, so the solver finds the orientation and position that minimize the summed squared image distance between the detected and projected corners. The four corners provide eight equations for the six unknowns.

A planar target has a structural weakness that a target with depth does not have. Because the four corners lie in one plane, a plane seen nearly face-on can be tilted toward the camera or away from it by the same small angle and produce the same four pixels to within the corner noise. The two readings are mirror images of the marker normal about the camera ray. The solver used here, the infinitesimal plane-based method [47], returns both locally optimal poses, called lobes below, rather than choosing one of them. Figure E.2 draws the geometry and the pair returned for the wall marker on the worked-example frame. The branch records its near-degenerate flag there, as on almost every frame. The two normals differ by 36.4 degrees, and both readings reproject with a largest corner gap of 0.22 pixels on sides averaging 30.03 pixels.

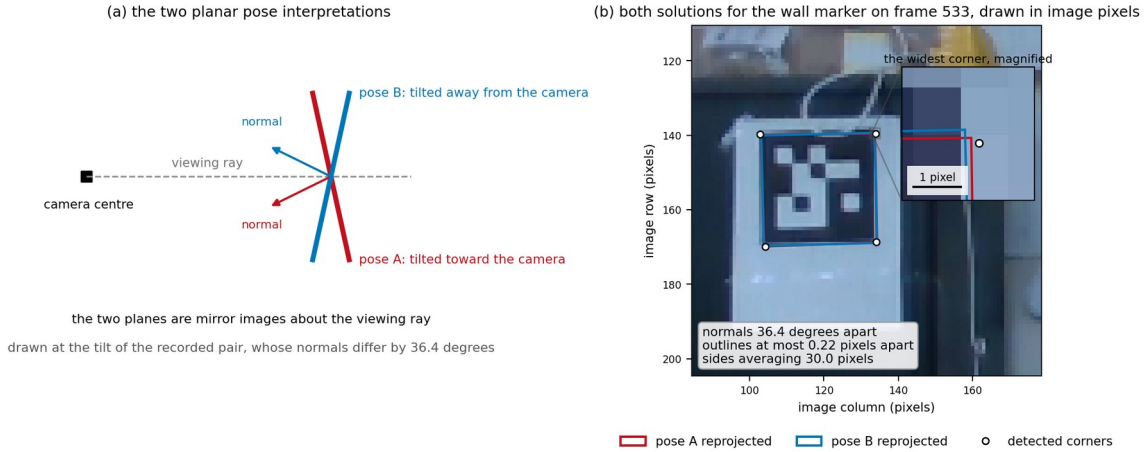


Figure E.2. (a) A nearly face-on plane admits two pose readings, tilted toward the camera or away from it, that are mirror images about the camera ray. (b) Both readings the solver returns for the wall marker, reprojected over the detection they come from, with the corner where they separate most magnified in the inset.

The pipeline selects its policy by the angular separation of the two lobes, not by the reprojection accuracy of either lobe. That choice is a heuristic of the implementation. A marker close to the camera shows visible perspective across its own width. The lobes then sit far apart, and the wrong one places its corners visibly away from the detection, so the closer-fitting lobe is taken. A marker whose projection is nearly an affine copy of the printed square gives almost no perspective cue. The two lobes can then stay far apart in angle while their reprojections coincide, since a plane tilted toward the camera and one tilted away by the same angle project to the same affine image. Angular separation therefore does not indicate the distinguishability of the two readings on the image. The threshold between the two cases is a lobe separation of 40 degrees, and the pair of Figure E.2 sits just below it, so the discriminators below decide instead.

Below that separation the discriminator becomes physical and temporal instead. On the first such frame the plumb wall decides: the lobe whose in-plane up axis better agrees with an estimate of the gravity direction in the camera frame is taken. That estimate is seeded from the plane fitted to the depth returns around the desk marker and needs to be right only to within about 10 degrees. On every later frame the lobe closest to a reference pose is kept, and the reference updates only when the chosen pose lies within 30 degrees of it, so one corrupted detection cannot corrupt the frames that follow. After five

consecutive frames far from the reference the tracker re-seeds. A per-frame flag records the method used to resolve each pose.

E.3 From the Rodrigues Form to a Rotation Matrix

The pose solver returns its rotation in the compact Rodrigues form, a single 3-vector whose direction is the rotation axis and whose length is the rotation angle in radians. The rest of the system works in rotation matrices, so the vector is converted on receipt, and the same formula reappears in the object-track cleaning of Chapter 4.

Write w for the unit axis and θ for the angle, so the Rodrigues vector is θ times w . The skew-symmetric matrix of w , written with the cross subscript, has rows $(0, -w_3, w_2)$, $(w_3, 0, -w_1)$ and $(-w_2, w_1, 0)$, and multiplying it by a vector v gives w cross v . The conversion is Rodrigues' rotation formula [44]:

$$R(w, \theta) = I + \sin(\theta)[w]_{\times} + (1 - \cos(\theta))[w]_{\times}^2. \quad (\text{E.1})$$

Equation (E.1) is a rotation operator. It builds a rotation from an axis and an angle, so it carries no frame labels, and a frame is named only where the rotation it returns is used. The formula holds for every angle and produces a proper rotation matrix by construction. Its inverse, recovering the axis and the angle from a matrix, is the matrix logarithm the gap bridging of Chapter 4 uses, so the pipeline moves between the two forms without introducing quaternions.

E.4 Worked Example: Frame 533 Detections

Frame 533 of the recording, shown in Figure E.1, carries detections of all three markers. Table E.1 collects the values the marker branch records for each of them: the printed side length, the centre pixel, the recovered marker origin in the camera frame, the range it implies, and the independent 5 by 5 median depth at the same pixel. The wall marker is printed at 150 millimetres because it sits farthest from the camera, and the object marker occupies one face of the 70 millimetre cube. The desk and object translations listed rest on the 45 millimetre size of Table 2.1, and the wall translation on the 150 millimetre size of the same table.

Marker	Role	Black square (mm)	Centre pixel	Marker origin t, camera frame (m)	Range (m)	Depth sample (m)
id 0	wall	150	(119, 155)	(−1.03, −0.47, 3.05)	3.26	3.19
id 2	desk	45	(350, 455)	(0.02, 0.19, 0.55)	0.58	0.53
id 1	object	45	(200, 341)	(−0.21, 0.16, 1.02)	1.06	0.99

Table E.1. The three marker detections of the worked-example frame, in the order of the panels of Figure E.1. The marker origin and the range come from the corner geometry alone and carry the printed-size scaling. The depth sample comes from the stereo depth stream and does not enter the pose recovery. The range is the length of that position vector, and the depth sample is the coordinate along the optical axis, so the two columns are different quantities.

Take the object marker through the rotation conversion. The pose recovery returns the rotation of the object frame relative to the camera frame

$${}_{Object}^{Camera}R = \begin{bmatrix} 1.00 & -0.01 & 0.06 \\ -0.01 & -1.00 & -0.04 \\ 0.06 & 0.04 & -1.00 \end{bmatrix}.$$

The entries are printed to two decimals and the angle to a tenth of a degree. The angle follows from the trace:

$$\theta = \arccos\left(\frac{\text{tr}\left({}_{Object}^{Camera}R\right) - 1}{2}\right) = 177.7^\circ,$$

and the axis from the skew-symmetric part, each component being one off-diagonal difference divided by twice the sine of the angle:

$${}_{Camera}w = (1.00, 0.00, 0.03).$$

The marked face of the cube is turned almost 180 degrees about an axis close to the camera's x axis, the cube sitting on the rail with its face square to the camera. The skew-symmetric matrix of those components is

$$[w]_{\times} = \begin{bmatrix} 0.00 & -0.03 & 0.00 \\ 0.03 & 0.00 & -1.00 \\ 0.00 & 1.00 & 0.00 \end{bmatrix},$$

and its square is

$$[w]_{\times}^2 = \begin{bmatrix} 0.00 & 0.00 & 0.03 \\ 0.00 & -1.00 & 0.00 \\ 0.03 & 0.00 & -1.00 \end{bmatrix}.$$

The two scalar factors of equation (E.1) at this angle are the sine, 0.04, and one minus the cosine, 2.00, so the three terms assemble to

$$\begin{matrix} \text{Camera} \\ \text{Object} \end{matrix} R = I + 0.04[w]_{\times} + 2.00[w]_{\times}^2 = \begin{bmatrix} 1.00 & -0.01 & 0.06 \\ -0.01 & -1.00 & -0.04 \\ 0.06 & 0.04 & -1.00 \end{bmatrix}.$$

The recomposed matrix agrees with R to the displayed rounding. This is a numerical consistency check, not a measurement of pose accuracy. The third column is the object frame's own outward normal, its third axis expressed in the camera frame, (0.06, -0.04, -1.00), pointing almost straight back at the camera. That pose is the input the world anchoring of Chapter 4 carries forward.

Appendix F: Characteristics of the Candidate Filters

The filtering stage of Section 2.5 can select any of four smoothers. This appendix describes the computation and behaviour of each smoother. This provides a common basis for the choice in Chapter 2 and the real-time comparison in Chapter 8. Every parameter quoted is the one the filter run of the recording used, at the recorded rate of 29.98 samples per second, the nominal 30 frames per second.

Section 2.5 gives the two stages ahead of the smoothers. The rolling Hampel test comes first, with an implemented robust scale based on rolling absolute residuals [48], [49]. Each sample is first compared with its own seven-frame rolling median; a second rolling median of those absolute residuals is multiplied by about 1.48. This differs from taking every deviation about the current window's single median. Linear interpolation then fills gaps of up to 5 frames. Their rules matter here in two respects: leading and trailing gaps are not backfilled, and each smoother runs on each unbroken run of valid samples separately. No filter therefore smooths across a gap it cannot see into.

The zero-phase Butterworth low-pass is the candidate the results use, chosen for a passband that is flat rather than rippled, so the motion band passes without ripple [50], [51]. A fourth-order filter with a 3 hertz cutoff runs twice over each run, forward and backward [52], and the second pass cancels the phase shift of the first. Cancelling phase delay does not undo attenuation: the two passes multiply their amplitude gains, so a movement with components near 3 hertz comes out smaller. A peak whose components are attenuated unevenly can shift by a sample. Running a filter backward needs the samples that follow the current one, so this candidate exists only offline.

The Savitzky-Golay smoother fits a second-order polynomial by least squares to a sliding window of 9 samples and takes its value at the centre [57]. Fitting a curve rather than averaging preserves a peak that a moving average would flatten, which Figure F.1(a) shows, but the fit gives every sample in the window a weight, so a single far-off sample still pulls the result.

The rolling median replaces each sample by the median of a 9-sample window. Because the median takes the middle value of the window, an isolated extreme sample is

discarded rather than averaged in, although it can still move the median by one rank. A steady ramp passes through a centred window unchanged, since the middle value of a monotone run is the sample at its centre. The cost falls on a peak, where the median takes a sample below the top and flattens it.

The One Euro filter is the causal candidate, the one the real-time pipeline of Chapter 8 can run, because it needs only the samples already received [53]. It is a first-order low-pass whose cutoff is not fixed. The filter keeps a smoothed estimate of the rate of change of the signal and raises its cutoff in proportion to it, through a minimum cutoff of 0.05 hertz and a coefficient of 1.0. Jitter is therefore suppressed while the signal holds still, and the filter follows while it changes. Being causal, it cannot avoid lag, which Figure F.1(a) shows as the late arrival of its output at the end of the move.

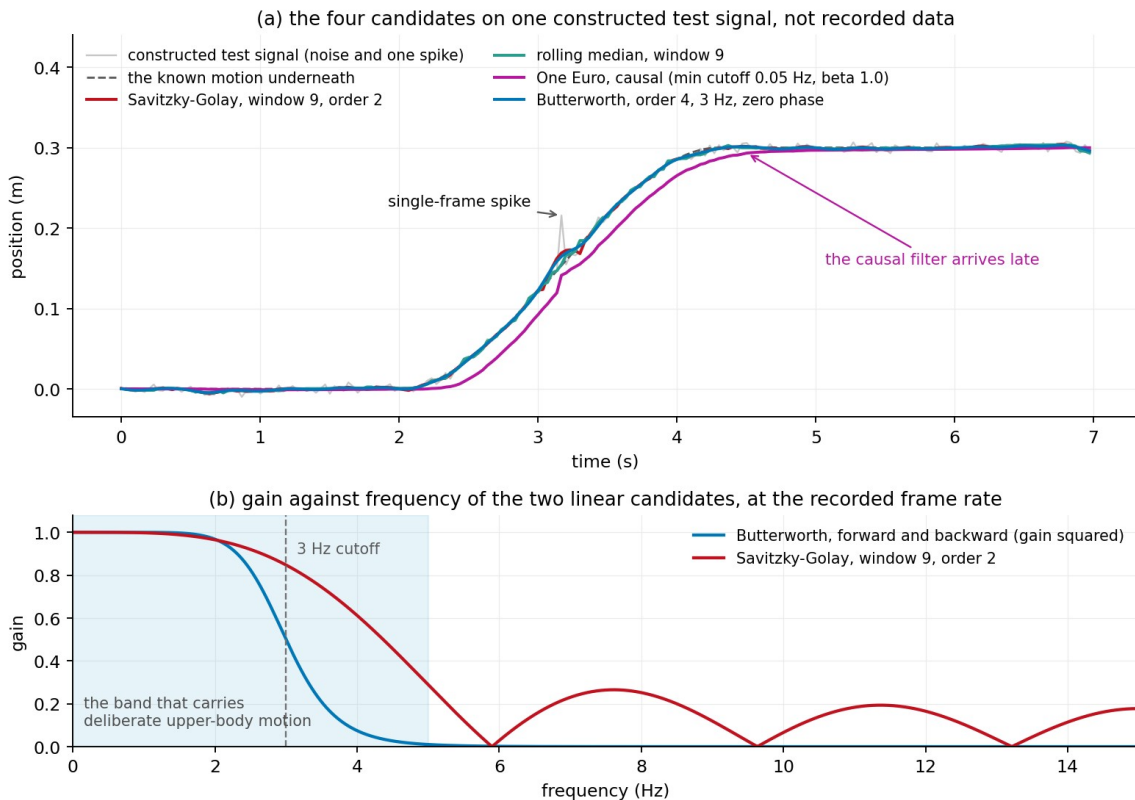


Figure F.1. (a) The four candidates on one constructed test signal, not recorded data: a profile that holds still, moves and holds again, with broadband noise and one single-frame spike added. (b) The amplitude gain of the two linear candidates against frequency, including both passes of the Butterworth filter, with the 3 hertz cutoff marked and the band of deliberate upper-body motion shaded.

Table F.1 compares phase and response to spikes. Centred offline filtering avoids a systematic causal delay, but smoothing can change peaks and event boundaries. A spike missed by the Hampel stage is rejected by the median or spread across neighbouring samples by the other candidates, so despiking runs first. Offline availability and lag determine which smoother the pipeline can use.

Candidate	Computation	Phase	Single-frame spike	Parameters used
Butterworth, zero phase	low-pass, applied forward and backward	none	spread over the filter length	order 4, cutoff 3 Hz
Savitzky-Golay	least-squares polynomial over a sliding window	none	spread over the window	window 9, order 2
rolling median	median of a sliding window	none	removed	window 9
One Euro	first-order low-pass with a rate-dependent cutoff	lag	partly passed through	minimum cutoff 0.05 Hz, coefficient 1.0

Table F.1. The four candidate smoothers, with the parameters used in Figure F.1(a). The Butterworth, Savitzky-Golay and rolling median smoothers need the whole recording or a window around each sample; only the One Euro filter runs on the samples already received.

Appendix G: Hip Depth Preparation

Section 2.6 introduces the preparation used before the body solve. This appendix gives its rules for the hips and shoulders, whose depth can come from an object in front of the body even when the landmark pixel remains usable. The rules operate in the y-up camera frame {Camera'}; flipping the camera y axis leaves the camera-depth coordinate z unchanged.

G.1 Inputs and Initial State

The state contains calibrated hip and shoulder widths, distances from each torso landmark to the opposite pair's midpoint, and remembered pair directions and depths. Offline evaluation supplies the arm segment lengths and the shoulder width. Every remaining length is fixed at the median of its first 60 available observations, offline and live alike. These lengths include the hip width and the four distances to the opposite pair midpoint. Direction and depth memories start empty. Accepted pair directions are normalized after blending 70 percent of the previous direction with 30 percent of the current one.

For a point p with camera-depth coordinate p_z greater than 5 centimetres, its retained ray is p divided by p_z . If z_{star} is the chosen replacement depth, the repaired point p_{star} is

$$p_{star} = z_{star} \frac{p}{p_z}. \quad (G.1)$$

This scaling preserves the direction from the camera. A missing point or unusable ray cannot receive this repair.

G.2 Rejection Order

The preparation first checks each pair's width. A deviation greater than 30 percent from its calibrated width triggers an endpoint check. Each endpoint's distance to the opposite pair midpoint is compared with its own calibrated distance, using the same tolerance. If exactly one endpoint fails, that endpoint is rejected and its ray retained. An ambiguous width violation alone leaves both endpoints in place.

The next check compares the directed hip and shoulder lines when both remain available. With both direction memories seeded, disagreement above 20 degrees activates the check. The pair farther from its remembered direction is selected, with a tie assigned to the hips. That pair stays selected until disagreement falls to 10 degrees or below. The preparation applies the endpoint-distance check described for width deviations and rejects a single point when it can. Otherwise, a known pair width permits rejection of both endpoints, and the remembered pair direction is moved toward the surviving line. The solver's 30 percent width tolerance and 20 degree line trigger are not the thresholds of the offline detectors of Section 5.2, whose torso width ratio detector fires at 20 percent and whose torso line angle detector fires above 22 degrees.

The depth check next rejects each remaining torso point whose depth differs from its memory by more than 10 centimetres. If both endpoints fail, the width and direction are known, and no whole-pair repair has already been scheduled, the pair is rebuilt together. Otherwise, each rejected endpoint is considered separately.

The final check rejects a remaining hip more than 15 centimetres nearer the camera than the input shoulder midpoint. That midpoint is saved before any checks and is not recomputed after shoulder rejection. This check can operate before depth memory exists, but assumes the input shoulders have usable depths and the trunk does not lean substantially. The earlier depth checks take priority because this check only sees hips they left available.

G.3 Memory Updates and Replacement

After rejection and before replacement, each surviving torso sample updates its depth memory. Its first accepted depth seeds the memory; later updates blend 70 percent previous depth with 30 percent new depth. Repaired points never update this memory. It has no expiry, so a persistent rejected interval can continue to use an old accepted depth.

A single point rejected by the width, line or depth checks uses its remembered depth in equation (G.1). A hip rejected by the last check instead uses the saved shoulder-midpoint depth. Neither individual placement enforces pair width. The latter placement assumes an approximately upright trunk in camera depth.

For a scheduled whole-pair repair, both rays are first placed at their remembered depths. Their average gives the repaired midpoint. Their line direction is blended into the remembered pair direction with the same 70/30 weighting and normalized; following a line-angle rejection, this is an additional update after steering toward the other pair. The endpoints are then placed half the calibrated width on either side of that midpoint. The final endpoints therefore need not lie on their original rays. If either ray or depth memory is unavailable, or the rebuilt pair has zero width, the fallback keeps the input midpoint and rebuilds around it using the remembered direction and calibrated width. That fallback cannot remove a common depth error in the midpoint.

If one endpoint is still missing, a separate fallback places it one calibrated width from its available partner along the remembered pair direction. This fallback is limited to 45 consecutive unmeasured frames for that endpoint. The limit does not expire depth memory or the whole-pair repair. When the required geometry remains unavailable, the dependent joint group keeps its last solved value.

G.4 Output States and Scope

A successfully solved joint group that consumes a repaired point is tagged constrained and loses its live bit (Section 6.1). This includes the root when a hip, or the shoulder used as its tilt reference, was repaired. A group that cannot be solved is held. The plain, direction-memory and object-assisted solves of Chapter 7 all rest on this preparation. The hold-last solve keeps the last accepted joint angles and does not use it. Offline pelvis translation uses the repaired hip midpoint. The live path runs the same preparation before its solve but still publishes the original hip midpoint as the pelvis translation.

The preparation depends on correct landmark rays and accepted calibration samples. A desk-depth sample can seed memory if it passes the checks, and genuine lean or body translation can trigger a depth rejection. Starting with hidden hips is not evaluated: the shoulder-depth check is available, but success still depends on usable hip pixels and shoulder depths. Chapter 9 discusses the observed effect without treating it as independent anatomical validation.

Appendix H: Kinematic Solver Verification

Chapter 3 derives the solve that turns metric landmarks into a root pose and four angles for each arm. This appendix is the record of the checks made on that mathematics, first against injected angle trajectories and then on measured input. It verifies the equations of Chapter 3 and nothing else, so it sits here rather than among the experimental results of Chapter 7.

H.1 Synthetic Datasets

Before any recording ran through the kinematic solver of Chapter 3, the six synthetic datasets of Table H.1 checked it. Each dataset injects a known angle trajectory, synthesizes landmark positions in the camera frame {Camera}, pushes them through the complete pipeline, and compares the decoded angles with the injected ones. The injected trajectory is a synthetic exact reference, known before the landmarks exist.

Synthetic dataset	Motion
Root sweeps	torso yaw, pitch, roll, then combined
Shoulder only	shoulder swing and twist, torso still
Shoulder with torso	the same shoulder motion, torso moving
Elbow only	flexion, then a rotating flexion plane, then both
Full arm	root, shoulder, and elbow together
Both arms	torso and both arms, eleven angles nonzero

Table H.1. The six synthetic datasets. On each, the decoded angles agree with the injected ones over every frame and every angle, to the precision of the arithmetic.

The residual is floating-point rounding, so the solver inverts the geometric construction exactly. The gimbal-lock case behaves as derived: a pose at the singularity decodes without error, as a different set of numbers for the same physical pose.

H.2 Measured Input

On measured input, where no injected trajectory exists, the solver ran on the 77 frames of the rail recording, out of its 900, in which all eight landmarks are measured, all of them inside the slide. On the worst of those frames the angles it returns reconstruct the measured upper-arm direction on either arm to the precision of the arithmetic. Elbow

flexion stays inside its anatomical range throughout, between -30.8 and -14.8 degrees on the right arm and between -40.4 and -15.5 on the left.

H.3 Scope of the Checks

These checks verify the kinematic mathematics that was tested, on the trajectories that were tested. They do not validate the landmark detector, the depth the sensor returns, the marker pose, or the anatomical accuracy of the reconstructed pose. They also say nothing about the recovery of a missing observation, which Section 7.4 grades separately.

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